INCH-POUND

MIL-P-24764/1
29 November 1991

MILITARY SPECIFICATION

POWER SUPPLIES, SHIPBOARD, ELECTRONIC

5 VOLTS, KEY CODE AA3A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this associated detail specification and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation: MIL-P-24764.

1. SCOPE

1.1 Scope. This document, along with MIL-P-24764, establishes the requirements and test procedures for an electronic switching mode power supply with single or dual 5 volt outputs.

1.2 Classification. This specification covers family S1 power supplies in accordance with MIL-P-24764.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 6130
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards, and handbook. The following specification, standards, and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-P-24764 - Power Supplies, Shipboard, Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-241 - Design Guide for Electromagnetic Interference (EMI) Reduction in Power Supplies.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

(except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General requirements. Power supplies shall comply with the requirements of MIL-P-24764 except as specified herein.

3.2 Correlation requirements. Before any production power supplies are tested, all testers and test equipment shall be correlated in accordance with MIL-P-24764.

3.3 Reliability requirements. Power supplies shall have a minimum mean time between failures (MTBF) of 100,000 hours when predicted in accordance with MIL-P-24764. The prediction shall be made with a mounting surface temperature of 50 degrees Celsius (°C).

3.4 Performance requirements. Power supplies shall conform to the electrical requirements specified in table I and shall operate within the environment specified in MIL-P-24764.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A1	IV	Isolation-input to chassis and output, 500 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A2	IV	Isolation-output to chassis, 50 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A3	IV	Isolation-output to output, 20 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A4	IV	Isolation-excessive temperature, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A5	IV	Isolation-unsafe temperature, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A6	IV	Isolation-output status, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A7	IV	Isolation-input connector used to unused contacts, 100 VDC	X	100	INF	100	INF	100	INF	100	INF	kOhm
A8	IV	Isolation-input to chassis and output, 500 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A9	IV	Isolation-output to chassis, 50 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A10	IV	Isolation-excessive temperature, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A11	IV	Isolation-unsafe temperature, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A12	IV	Isolation-output status, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
A13	IV	Isolation-input connector used to unused contacts, 100 VDC	S	10	INF	1	INF	10	INF	1	INF	MOhm
B1	V	Chassis contacts to chassis continuity	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B2	V	5V sense return to remote ON/OFF return continuity	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B3	V	5V sense return to voltage program-1 return continuity	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B4	V	Out 2 (-) to voltage program-2 return continuity	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ohm

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
B5	V	5V sense return to output program return continuity	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B6-B58	V	Parallel connector contacts continuity	X	0.0	1.0	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B58-B63	V	Chassis contacts to chassis continuity	Q	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C1	VI	Load regulation, 5.05V, 15A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
C2	VI	Load regulation, 5.05V, 150A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
C3	VI	Load regulation, 5.25V, 15A	X	5.20	5.30	5.17	5.33	5.14	5.36	5.12	5.38	VDC
C4	VI	Load regulation, 5.25V, 150A	X	5.20	5.30	5.17	5.33	5.14	5.36	5.12	5.38	VDC
C5, C6	VI	Load regulation, 5.05V, 0A -5.05V, 7A and -5.05V, 70A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
C7, C8	VI	Load regulation, 5.05V, 00A -5.05V, 7A and -5.05V, 70A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
C9, C10	VI	Load regulation, -5.05V, 7A +5.05V, 0A and +5.05V, 00A	X	-5.10	-5.00	-5.13	-4.97	-5.15	-4.95	-5.18	-4.92	VDC
C11, C12	VI	Load regulation, -5.05V, 70A +5.05V, 0A and +5.05V, 00A	X	-5.10	-5.00	-5.13	-4.97	-5.15	-4.95	-5.18	-4.92	VDC
C13	VI	Load regulation, 5.25V, 00A -5.25V, 70A	X	5.20	5.30	5.17	5.33	5.14	5.36	5.12	5.38	VDC
C14	VI	Load regulation, -5.25V, 70A 5.25V, 00A	X	-5.30	-5.20	-5.33	-5.17	-5.36	-5.14	-5.38	-5.12	VDC
C15	VI	Load regulation, sgl output, no load	X	4.85	5.15	4.92	5.18	4.90	5.20	4.87	5.23	VDC
C16	VI	Load regulation, out-1 dual output, no load	X	4.85	5.15	4.92	5.18	4.90	5.20	4.87	5.23	VDC
C17	VI	Load regulation, out-2 dual output, no load	X	-5.15	-4.85	-5.18	-4.82	-5.20	-4.80	-5.23	-4.87	VDC
C18	VI	Load regulation, OS dual output, no load	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	VDC

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-18°C to 61°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
C18	VI	Load regulation, output-1 remote ON/OFF - OFF	X	0.0	0.5	0.0	0.5	0.0	0.5	0.0	0.5	VDC
C28	VI	Load regulation, output-2 remote ON/OFF - OFF	X	-5.10	-5.00	-5.13	-4.97	-5.15	-4.95	-5.18	-4.82	VDC
01	VII	Line regulation, 86 VRMS, 15A	S	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
02	VII	Line regulation, 86 VRMS, 150A	S	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
03	VII	Line regulation, 144 VRMS, 15A	S	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
04	VII	Line regulation, 144 VRMS, 150A	S	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
05	VII	Line regulation, 86 VRMS 5.05V, 0A, -5.05V, 7A	S	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
06	VII	Line regulation, 86 VRMS 5.05V, 0A, -5.05V, 7A	S	-5.10	-5.00	-5.13	-4.97	-5.15	-4.95	-5.18	-4.82	VDC
07	VII	Line regulation, 86 VRMS 5.05V, 80A, -5.05V, 70A	S	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
08	VII	Line regulation, 86 VRMS 5.05V, 80A, -5.05V, 70A	S	-5.10	-5.00	-5.13	-4.97	-5.15	-4.95	-5.18	-4.82	VDC
09	VII	Line regulation, 144 VRMS 5.05V, 0A, -5.05V, 7A	S	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
010	VII	Line regulation, 144 VRMS 5.05V, 0A, -5.05V, 7A	S	-5.10	-5.00	-5.13	-4.97	-5.15	-4.95	-5.18	-4.82	VDC
011	VII	Line regulation, 144 VRMS 5.05V, 80A, -5.05V, 70A	S	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
012	VII	Line regulation, 144 VRMS 5.05V, 80A, -5.05V, 70A	S	-5.10	-5.00	-5.13	-4.97	-5.15	-4.95	-5.18	-4.82	VDC
013	VII	Line regulation, 180 VDC, 15A	X	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
014	VII	Line regulation, 180 VDC, 150A	X	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
015	VII	Line regulation, 202 VDC, 15A	X	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
016	VII	Line regulation, 202 VDC, 150A	X	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
017	VII	Line regulation, Auxiliary Input 180 VDC, 150A	X	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC
018	VII	Line regulation, Auxiliary Input 202 VDC, 150A	X	5.00	5.10	4.87	5.13	4.95	5.15	4.92	5.18	VDC

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 202640-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 3°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
D19	VII	Line regulation, 150A 47 Hz, 115 VRMS	0	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D20	VII	Line regulation, 150A 63 Hz, 115 VRMS	0	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D21	VII	Line Regulation, 100 VDC 5.05V, 0A, -5.05V, 7A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D22	VII	Line Regulation, 100 VDC 5.05V, 0A, -5.05V, 7A	X	-5.10	-5.00	-5.13	-4.97	-5.15	-4.95	-5.18	-4.92	VDC
D23	VII	Line Regulation, 100 VDC 5.05V, 80A, -5.05V, 70A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D24	VII	Line Regulation, 100 VDC 5.05V, 80A, -5.05V, 70A	X	-5.10	-5.00	-5.13	-4.97	-5.15	-4.95	-5.18	-4.92	VDC
D25	VII	Line Regulation, 202 VDC 5.05V, 0A, -5.05V, 7A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D26	VII	Line Regulation, 202 VDC 5.05V, 0A, -5.05V, 7A	X	-5.10	-5.00	-5.13	-4.97	-5.15	-4.95	-5.18	-4.92	VDC
D27	VII	Line Regulation, 202 VDC 5.05V, 80A, -5.05V, 70A	X	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
D28	VII	Line Regulation, 202 VDC 5.05V, 80A, -5.05V, 70A	X	-5.10	-5.00	-5.13	-4.97	-5.15	-4.95	-5.18	-4.92	VDC
E	VIII	Low input voltage protection	0	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
F1-F3	VIII	Loss of phase protection	0	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
G1	IX	Efficiency, 5.05V, 150A	X	70	100	70	100	70	100	70	100	x
G2	IX	Efficiency, 5.05V, 15A	0	60	100	60	100	60	100	60	100	x
G3	IX	Efficiency, 5.25V, 150A	0	70	100	70	100	70	100	70	100	x
G4	IX	Efficiency, 5.05V, 80A -5.05V, 70A	X	70	100	70	100	70	100	70	100	x
H	IX	Power factor	0	0.8	1.0	0.8	1.0	0.8	1.0	0.8	1.0	-

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
J1-J3	IX	Input current balance	Q	-5	5	-5	5	-5	5	-5	5	%
K1, K2	X	Inrush current, 150A	Q	-	70	-	70	-	70	-	70	Apk
L1	XI	Remote sense compensation, 5.05V	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
L2	XI	Remote sense compensation, 5.25V	Q	5.20	5.30	5.17	5.33	5.14	5.36	5.12	5.38	VDC
M1	XII	Open + sense lead transient amplitude	Q	-400	100	-400	100	-400	100	-400	100	mV
M2	XII	Open + sense lead transient time	Q	0	100	0	100	0	100	0	100	ms
M3	XII	Open + sense lead voltage	S	4.80	5.56	4.80	5.56	4.80	5.56	4.80	5.56	VDC
M4	XII	Open - sense lead transient amplitude	Q	-400	100	-400	100	-400	100	-400	100	mV
M5	XII	Open - sense lead transient time	Q	0	100	0	100	0	100	0	100	ms
M6	XII	Open - sense lead voltage	S	4.80	5.56	4.80	5.56	4.80	5.56	4.80	5.56	VDC
M7	XII	Open + and - sense lead voltage	S	4.80	5.56	4.80	5.56	4.80	5.56	4.80	5.56	VDC
M8	XII	Sense lead reversal protection	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
N1	XIII	Overload protection, 5V, 150A	X	150	187.5	150	190.5	150	187.5	150	190.5	ADC
N2	XIII	Overload protection, 5V, 80A	X	80	100	80	101.6	80	100	80	101.6	ADC
N3	XIII	Overload protection, -5V, 70A	X	70	87.5	70	88.9	70	87.5	70	88.9	ADC
N4	XIII	Short circuit recovery, 5V, 150A	Q	0	100	0	100	0	100	0	100	ms
N5	XIII	Short circuit overshoot, 5V, 150A	Q	-253	253	-253	253	-253	253	-253	253	mV
N6	XIII	Short circuit recovery, 5V, 80A	Q	0	100	0	100	0	100	0	100	ms
N7	XIII	Short circuit overshoot, 5V, 80A	Q	-253	253	-253	253	-253	253	-253	253	mV
N8	XIII	Short circuit recovery, -5V, 70A	Q	0	100	0	100	0	100	0	100	ms
N9	XIII	Short circuit overshoot, -5V, 70A	Q	-253	253	-253	253	-253	253	-253	253	mV
N10	XIII	Short circuit current, 150A	S	0	187.5	0	190.5	0	187.5	0	190.5	ADC
N11	XIII	Short circuit current, 80A	S	0	100	0	101.6	0	100	0	101.6	ADC

-5See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N12	XIII	Short circuit current, 78A	S	0	87.5	0	88.8	0	87.5	0	88.8	ADC
P1	XIV	Turn-on transient, 5V, 150A	Q	-253	253	-253	253	-253	253	-253	253	mV
P2	XIV	Turn-on time, 5V, 150A	S	15	250	15	250	15	250	15	250	ms
P3	XIV	Turn-off transient, 5V, 150A	Q	-253	253	-253	253	-253	253	-253	253	mV
P4	XIV	Control turn-on transient	Q	-253	253	-253	253	-253	253	-253	253	mV
P5	XIV	Control turn-on time	S	15	250	15	250	15	250	15	250	ms
P6	XIV	Control turn-off transient	Q	-253	253	-253	253	-253	253	-253	253	mV
Q1-Q3	XV	Ripple, total, (115 VAC, 10 Hz - 1 MHz) 150A,88A,78A	X	0	65	0	75	0	65	0	75	mVp-p
Q4-Q6	XV	Ripple, low frequency (115 VAC, 10 Hz - 10 kHz) 150A,88A,78A	S	0	50	0	60	0	50	0	60	mVp-p
Q7-Q9	XV	Ripple, high frequency (115 VAC, 10 kHz - 1 MHz) 150A,88A,78A	S	0	65	0	75	0	65	0	75	mVp-p
Q10-Q12	XV	Ripple, total (115 VAC, 10 Hz - 20 MHz) 150A,88A,78A	Q	0	300	0	300	0	300	0	300	mVp-p
Q13	XV	Ripple, total, overload, >150A (115 VAC, 10 Hz - 5 MHz)	Q	0	200	0	200	0	200	0	200	mVp-p
Q14	XV	Ripple, total, no load (115 VAC, 10 Hz - 5 MHz)	Q	0	130	0	150	0	130	0	150	mVp-p
Q15	XV	Ripple, total, overload, >80A (115 VAC, 10 Hz - 5 MHz)	Q	0	200	0	200	0	200	0	200	mVp-p
Q16	XV	Ripple, total, no load (115 VAC, 10 Hz - 5 MHz)	Q	0	130	0	150	0	130	0	150	mVp-p
Q17	XV	Ripple, total, overload, >78A (115 VAC, 10 Hz - 5 MHz)	Q	0	200	0	200	0	200	0	200	mVp-p

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Q18	XV	Ripple, total, no load (115 VAC, 10 Hz - 5 MHz)	Q	0	130	0	150	0	130	0	150	mVp-p
Q18-Q21	XV	Ripple, total (85 VAC, 10 Hz - 5 MHz) 150A, 80A,70A	Q	0	130	0	150	0	130	0	150	mVp-p
Q22-Q24	XV	Ripple, total (144 VAC, 10 Hz - 5 MHz) 150A, 80A, 70A	Q	0	130	0	150	0	130	0	150	mVp-p
R1	XVI	Dynamic line regulation, +21 percent	Q	0	152	0	152	0	152	0	152	mV
R2	XVI	Dynamic line ringing, +21 percent (2nd half cycle)	Q	0	76	0	76	0	76	0	76	mV
R3	XVI	Dynamic line regulation, -21 percent	Q	0	152	0	152	0	152	0	152	mV
R4	XVI	Dynamic line ringing, -21 percent (2nd half cycle)	Q	0	76	0	76	0	76	0	76	mV
S1	XVII	Dynamic load regulation, 80% to 90%, 5.05V, sgl. out	X	0	152	0	152	0	152	0	152	mV
S2	XVII	Dynamic load ringing (2nd half cycle)	S	0	76	0	76	0	76	0	76	mV
S3	XVII	Dynamic load regulation, 80% to 90%, 5.05V, sgl. out	X	0	152	0	152	0	152	0	152	mV
S4	XVII	Dynamic load ringing (2nd half cycle)	S	0	76	0	76	0	76	0	76	mV
S5	XVII	Dynamic load regulation, 80% to 90%, full capacitor load, 5.05V, sgl. out	Q	0	152	0	152	0	152	0	152	mV
S6	XVII	Dynamic load ringing (2nd half cycle)	Q	0	76	0	76	0	76	0	76	mV

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 3°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S7	XVII	Dynamic load regulation, 90% to 60%, Full capacitor load, 5.05V, sgl. out	Q	0	152	0	152	0	152	0	152	mV
S8	XVII	Dynamic load ringing (2nd half cycle)	Q	0	76	0	76	0	76	0	76	mV
S9	XVII	Dynamic load regulation, 60% to 90%, 5.25V, sgl. out (max sense load)	X	0	162	0	162	0	162	0	162	mV
S10	XVII	Dynamic load ringing (2nd half cycle)	S	0	81	0	81	0	81	0	81	mV
S11	XVII	Dynamic load regulation, 90% to 60%, 5.25V, sgl. out (max sense load)	X	0	162	0	162	0	162	0	162	mV
S12	XVII	Dynamic load ringing (2nd half cycle)	S	0	81	0	81	0	81	0	81	mV
S13	XVII	Dynamic load regulation, 60% to 90%, 5.25V, dual out (max sense load)	X	0	162	0	162	0	162	0	162	mV
S14	XVII	Dynamic load ringing (2nd half cycle)	S	0	81	0	81	0	81	0	81	mV
S15	XVII	Dynamic load regulation, 90% to 60%, 5.25V, dual out (max sense load)	X	0	162	0	162	0	162	0	162	mV
S16	XVII	Dynamic load ringing (2nd half cycle)	S	0	81	0	81	0	81	0	81	mV
S17	XVII	Dynamic load regulation, 60% to 90%, 5.05V, dual out	S	0	152	0	152	0	152	0	152	mV
S18	XVII	Dynamic load ringing (2nd half cycle)	S	0	76	0	76	0	76	0	76	mV
S19	XVII	Dynamic load regulation, 90% to 60%, 5.05V, dual out	S	0	152	0	152	0	152	0	152	mV

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
S20	XVII	Dynamic load ringing (2nd half cycle)	S	0	75	0	75	0	75	0	75	mV
S21	XVII	Dynamic load regulation, 80% to 90%, -5.05V, dual out	S	0	152	0	152	0	152	0	152	mV
S22	XVII	Dynamic load ringing (2nd half cycle)	S	0	75	0	75	0	75	0	75	mV
S23	XVII	Dynamic load regulation, 90% to 80%, -5.05V, dual out	S	0	152	0	152	0	152	0	152	mV
S24	XVII	Dynamic load ringing (2nd half cycle)	S	0	75	0	75	0	75	0	75	mV
S25	XVII	Dynamic load regulation , 60% to 90%, 5.05V, sgl. out	S	0	500	0	500	0	500	0	500	mV
S26	XVII	Dynamic load ringing (2nd half cycle)	S	0	250	0	250	0	250	0	250	mV
S27	XVII	Dynamic load regulation , 90% to 60%, 5.05V, sgl. out	S	0	500	0	500	0	500	0	500	mV
S28	XVII	Dynamic load ringing (2nd half cycle)	S	0	250	0	250	0	250	0	250	mV
T	XVIII	Turn-on time into capacitive load	Q	15	250	15	250	15	250	15	250	ms
U1	XIX	Undervoltage - output status LED off	X	4.39	4.80	4.39	4.80	4.39	4.80	4.39	4.80	VDC
U2	XIX	Overvoltage - output status LED off	X(4/)	5.57	6.45	5.57	6.55	5.57	6.45	5.57	6.55	VDC
U3	XIX	Status LED check	X	----------			Status LED on				----------	-
V1	XIX	Output status signal, logic low (5 mA)	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	VDC
V2	XIX	Output status signal, logic high	X	29.0	30.5	29.0	30.5	29.0	30.5	29.0	30.5	VDC

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
V3, V4	XIX	Output status signal, undervoltage	X	4.39	4.80	4.39	4.80	4.39	4.80	4.39	4.80	VDC
V5	XIX	Output status signal, undervoltage	X	4.39	4.80	4.39	4.80	4.39	4.80	4.39	4.80	VDC
V6, V7	XIX	Output status signal, overvoltage	X (4/)	5.57	6.45	5.57	6.55	5.57	6.45	5.57	6.55	VDC
V8	XIX	Output status signal, overvoltage	X (4/)	5.57	6.45	5.57	6.55	5.57	6.45	5.57	6.55	VDC
V9	XIX	Output status signal leakage current	Q	0.0	4.4	0.0	4.4	0.0	50	0.0	200	uADC
V10	XIX	Output status signal settling time (high to low)	Q	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	ns
V11	XIX	Output status signal settling time (low to high)	Q	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	ns
V12	XIX	Output status signal, logic low (0.5 mA)	Q	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	VDC
H1	XX	Excessive temp indication (5 mA current)	Q	-	-	-	-	0.0	0.4	-	-	VDC
H2	XX	Excessive temp indication (500 mA current)	Q	-	-	-	-	0.0	0.4	-	-	VDC
H3	XX	Unsafe temp indication (5 mA current)	Q	-	-	-	-	0.0	0.4	-	-	VDC
H4	XX	Unsafe temp indication (500 mA current)	Q	-	-	-	-	0.0	0.4	-	-	VDC
H5	XX	Unsafe - excessive differential (5 mA)	Q	-	-	-	-	1.0	25	-	-	°C
H6	XX	Indicator reset (5 mA)	Q	-	-	-	-	60	-	-	-	°C
H7	XX	Unsafe - excessive differential (500 mA)	Q	-	-	-	-	1.0	25	-	-	°C
H8	XX	Indicator reset (500 mA)	Q	-	-	-	-	60	-	-	-	°C

-See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
X	XXI	Output voltage hold-up time, 100 VDC Input	Q	50	-	50	-	50	-	50	-	us
Y1	XXII	AC input to chassis capacitance	5	0.0	0.03	0.0	0.03	0.0	0.03	0.0	0.03	uF
Y2	XXII	Auxiliary input to chassis capacitance	5	0.0	0.05	0.0	0.05	0.0	0.05	0.0	0.05	uF
Z1-Z3	XXIII	Output impedance	Q	0	30	0	30	0	30	0	30	mOhm
RR1	XXIV	Open 5.05V load lead protection	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
RR2	XXIV	Open 5.05V rtn load lead protection	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
BB1	XXV	Spike voltage protection, AC input, initial transient	Q	-253	253	-253	253	-253	253	-253	253	mV
BB2	XXV	Spike voltage protection, AC input, secondary transient	Q	-126	126	-126	126	-126	126	-126	126	mV

-See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-18°C to 60°C mounting surface temperature				5/
				Initial		End of life		Initial		End of life		Units
				Min	Max	Min	Max	Min	Max	Min	Max	
BB3	XXV	Spike voltage protection, auxiliary input, initial transient	Q	-253	253	-253	253	-253	253	-253	253	mV
BB4	XXV	Spike voltage protection, auxiliary input, secondary transient	Q	-126	126	-126	126	-126	126	-126	126	mV
CC	XXVI	Reverse voltage protection on auxiliary input	Q	5.00	5.10	4.87	5.13	4.85	5.15	4.92	5.18	VDC
001	XXVI	Reverse voltage protection on output, sgl.	Q	-1.0	0.0	-1.0	0.0	-1.0	0.0	-1.0	0.0	VDC
002	XXVI	Reverse voltage bias, turn-on	Q	5.00	5.10	4.87	5.13	4.85	5.15	4.92	5.18	VDC
003	XXVI	Reverse voltage protection, out-1	Q	-1.0	0.0	-1.0	0.0	-1.0	0.0	-1.0	0.0	VDC
004	XXVI	Reverse voltage bias, turn-on	Q	5.00	5.10	4.87	5.13	4.85	5.15	4.92	5.18	VDC
005	XXVI	Reverse voltage protection, out-2	Q	-1.0	0.0	-1.0	0.0	-1.0	0.0	-1.0	0.0	VDC
006	XXVI	Reverse voltage bias, turn-on	Q	5.00	5.10	4.87	5.13	4.85	5.15	4.92	5.18	VDC
007	XXVI	ON/OFF parallel outputs, sgl, off	Q	5.00	5.10	4.87	5.13	4.85	5.15	4.92	5.18	VDC
008	XXVI	ON/OFF parallel outputs, OS high	Q	2.4	5.5	2.4	5.5	2.4	5.5	2.4	5.5	VDC
009	XXVI	ON/OFF parallel outputs, OS LED off	Q		Output	status	LED	OFF				-

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-10°C to 60°C mounting surface temperature				5/ Units
				Initial		End of life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
DD10	XXVI	ON/OFF parallel outputs, dual, output-1, off	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
DD11-DD13	XXVI	Positive voltage bias, turn-on, sgl, output-1, output-2	Q	4.85	5.15	4.92	5.18	4.90	5.20	4.87	5.23	VDC
EE1-EE6	XXVII	Common mode current, broadband	Q	0	200	0	200	0	200	0	200	nAp-p
EE7	XXVII	Common mode current (50 Hz - 10 kHz)	Q	INF	-60	INF	-60	INF	-60	INF	-60	dBA / √Hz
EE8	XXVII	Common mode current (10 kHz - 10 MHz)	Q	INF	-36.5	INF	-36.5	INF	-36.5	INF	-36.5	dBA / √Hz
EE9	XXVII	Common mode current, largest amplitude signal	Q	INF	-26	INF	-26	INF	-26	INF	-26	dBA / √Hz
FF1 - FF6	XXVIII	Voltage programming contacts, signal rejection	Q	15	-	15	-	15	-	15	-	dB
GG1	XXIX	Input voltage turn-on threshold	S	100	135	100	135	100	135	100	135	VDC
GG2	XXIX	Input voltage turn-off threshold	S	-	100	-	100	-	100	-	100	VDC
GG3	XXIX	Input voltage turn-on/turn-off hysteresis	S	10	-	10	-	10	-	10	-	VDC
GG4	XXIX	Input voltage turn-on threshold, auxiliary input	S	100	135	100	135	100	135	100	135	VDC
GG5	XXIX	Input voltage turn-off threshold, auxiliary input	S	-	100	-	100	-	100	-	100	VDC

- See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE I. Power supply electrical requirements - Continued.

Test cond	Ref test table	Parameter	Test type 1/-3/	Acceptance requirements 25 ± 5°C mounting surface temperature				-18°C to 80°C mounting surface temperature				5/ Units
				Initial		End of Life		Initial		End of life		
				Min	Max	Min	Max	Min	Max	Min	Max	
GG6	XXIX	Input voltage turn-on/turn-off hysteresis, auxiliary input	S	18	-	18	-	18	-	18	-	VDC
GG7	XXIX	Maximum steady state input current	Q	-	14.7	-	14.7	-	14.9	-	14.9	ADC
GG8	XXIX	Slowly rising input peak current	Q	-	14.7	-	14.7	-	14.9	-	14.9	Apk
GG9	XXIX	Slowly rising input rate of current rise	Q	-	11.4	-	11.4	-	11.5	-	11.5	A/ms
GG10	XXIX	Input peak current, remote turn-on	Q	-	14.7	-	14.7	-	14.9	-	14.9	Apk
GG11	XXIX	Input rate of current rise, remote turn-on	Q	-	11.4	-	11.4	-	11.5	-	11.5	A/ms
HH1-HH5	XXX	Incorrect output programming protection	Q	0.0	6.45	0.0	6.45	0.0	6.45	0.0	6.45	VDC
HH6, HH7	XXX	Incorrect output programming protection	Q	5.00	5.10	4.97	5.13	4.95	5.15	4.92	5.18	VDC
JJ1	XXXI	Emergency condition input voltage protection	Q	4.99	5.51	4.98	5.51	4.99	5.51	4.98	5.51	VDC
JJ2	XXXI	Emergency condition input voltage protection	Q	5.20	5.30	5.17	5.33	5.14	5.36	5.12	5.38	VDC
KK1	XXXII	Output status noise, positive	Q	0.0	500	0.0	500	0.0	500	0.0	588	mVp
KK2	XXXII	Output status noise, negative	Q	0.0	-500	0.0	-500	0.0	-500	0.0	-588	mVp

- See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

1/ "X" parameters shall be 100 percent tested during power supply quality conformance testing.

2/ "S" parameters shall be sample tested (one percent or two power supplies from each inspection lot, whichever is greater) during quality conformance testing.

3/ "Q" parameters shall be tested during qualification testing. It is not required that these parameters be tested during quality conformance testing, however, the vendor is responsible for ensuring that the power supplies meet these parameters.

4/ Proper operation of the overvoltage protection circuitry shall be verified as an "X" test by the power supply vendor during manufacture of the supplies.

5/ The symbols used in this table and not defined elsewhere are as follows:

A - amperes
ADC - amperes direct current
Apk - amperes peak
cond - condition
INF - infinite
mAp-p - milliamperes peak-to-peak
max - maximum
min - minimum
mOhm - milliohm
MOhm - megohm
mV - millivolts
mVpk - millivolts peak
mVp-p - millivolts peak-to-peak
% - percent
sgl - single
uADC - microamperes direct current
V - volts

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

3.5 Power supply configuration. The configuration of the power supplies shall be in accordance with MIL-P-24764 and as specified herein.

3.5.1 Size. Power supplies shall be style 3.

3.5.2 Weight. Power supplies shall have a maximum weight of 14 pounds.

3.5.3 Visual indicator. The visual indicator shall be located as specified on figure 1. A green light emitting diode (LED) shall indicate status of output voltage.

3.5.4 Marking. The power supply marking shall contain the information and shall be located in the relative location and orientation shown on figure 1. The marking "OUTPUT STATUS" shall be immediately adjacent to the output status LED. When authorization has been granted by the Shipboard Power Supply Products Quality Assurance Activity (SPSP-QAA), the qualification certification mark "QUAL" shall be marked as shown. The size of the letters, information per line, and spacing shall, within the limits of MIL-P-24764, be allowed to vary.

3.5.5 Keying. Keying shall be AA3A.

3.5.6 Mounting fastener torque. The power supply shall operate as specified when it is mounted to a cooling plate with the mounting fasteners torqued to 32 $\pm$ 3 inch-pounds.

3.6 Function diagram and contact assignments. The power supplies shall conform to the function diagram shown on figure 2 and contact assignments shall be as shown in table II. When multiple connector contacts are specified for an interface function (such as output) for current capacity, redundancy, and so forth, all multiple contacts for each function shall be connected together before being used to interface the power supply to external sources and loads. A summary of power supply functions is shown in table III.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

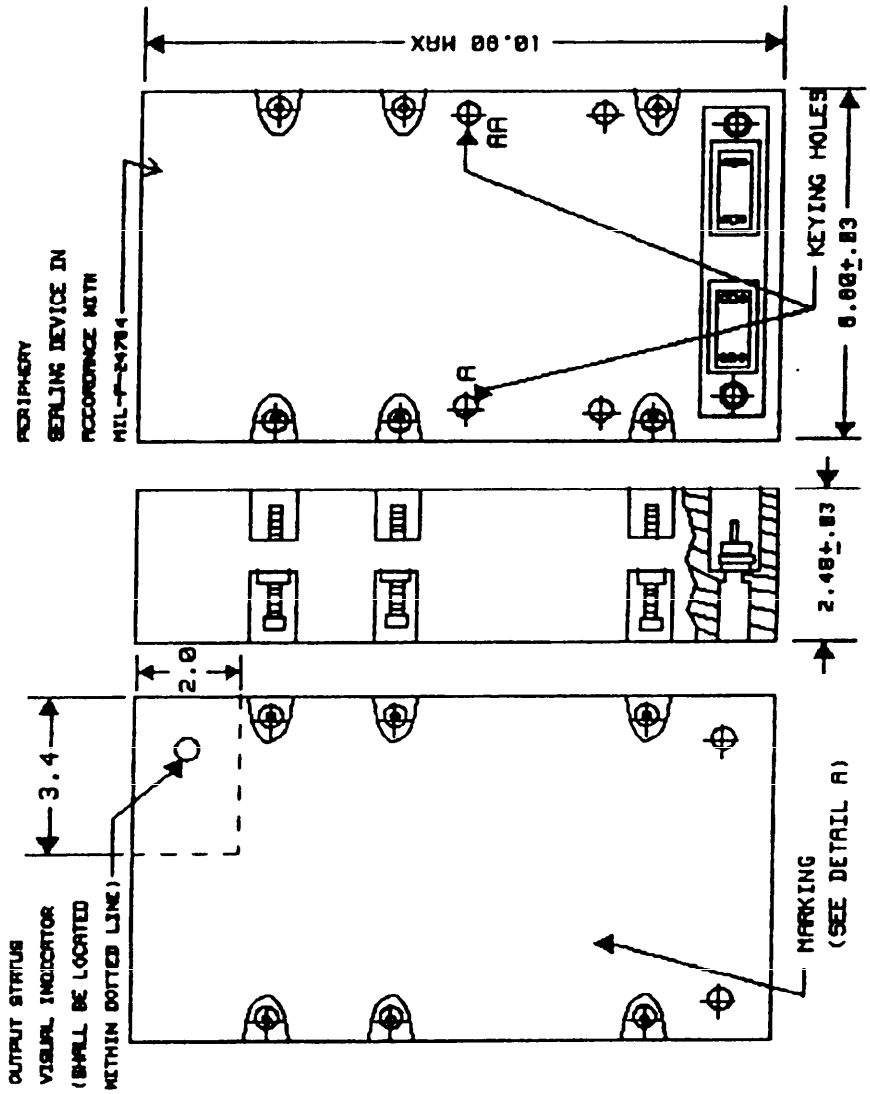
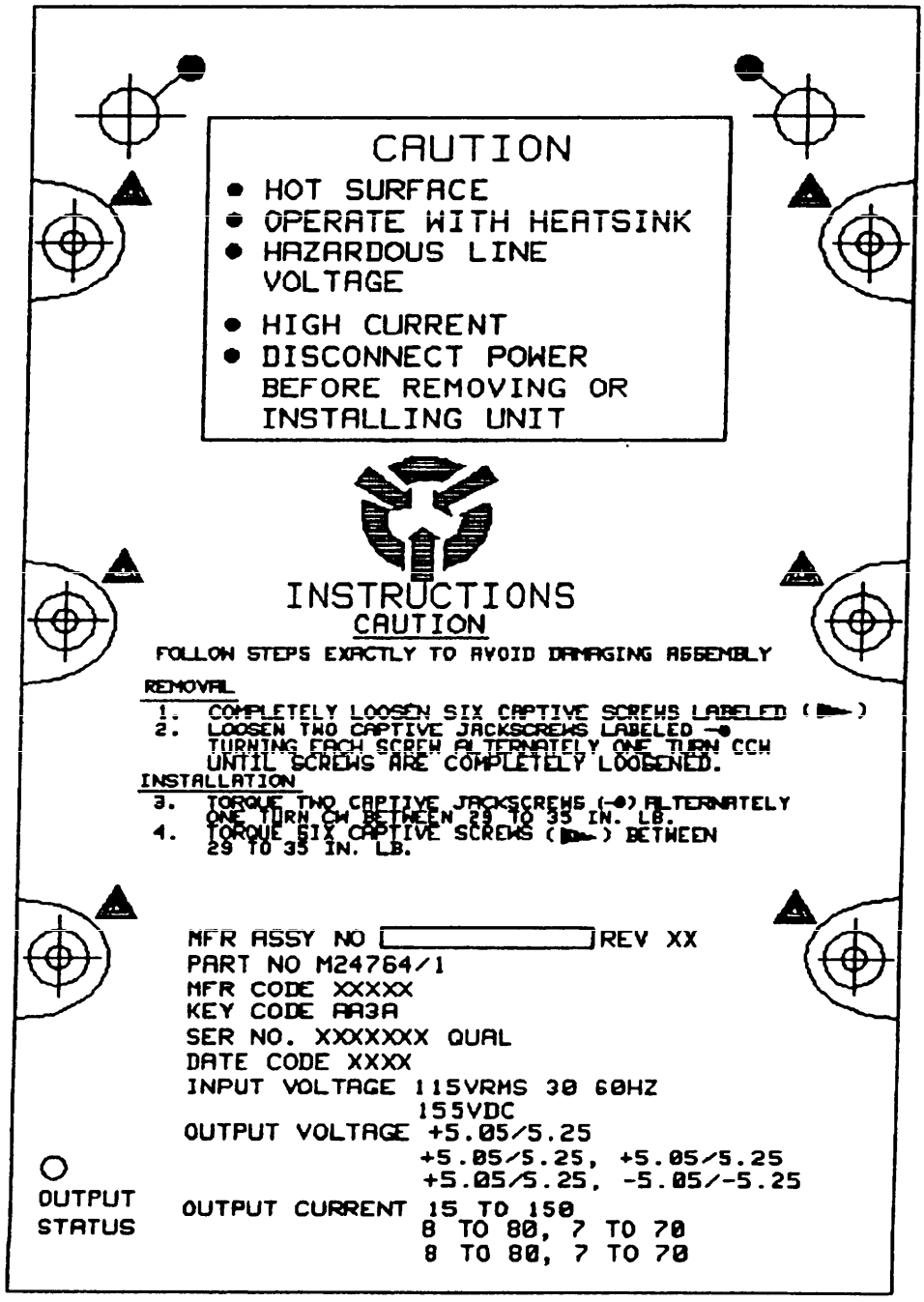


FIGURE 1. Power supply marking and visual indicator location.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.



DETAIL A

FIGURE 1. Power supply marking and visual indicator location - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

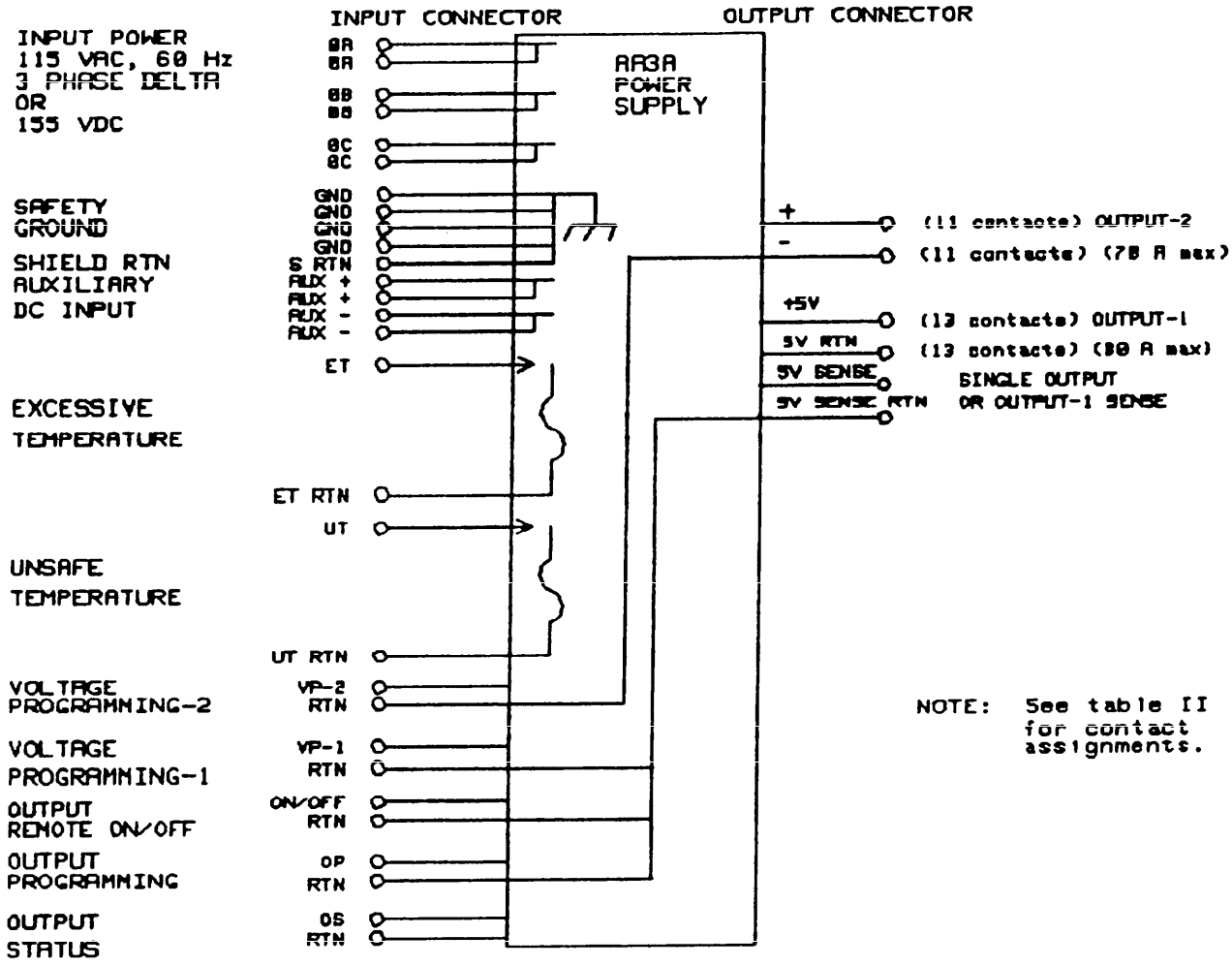


FIGURE 2. Power supply function diagram.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE II. Connector contact assignment.

INPUT CONNECTOR		OUTPUT CONNECTOR	
A	Phase A	A	5 V Sense
B	Unused	B	5 V RTN Sense
C	Phase C	C	Output 1 (+5 V)
D	Phase B	D	Output 1 RTN
E	Phase A	E	Output 1 (+5 V)
F	Unused	F	Output 1 RTN
H	Phase C	H	Output 1 (+5 V)
J	Phase B	J	Output 1 RTN
K	Auxiliary DC (+)	K	Output 1 (+5 V)
L	Unused	L	Output 1 RTN
M	Auxiliary DC (-)	M	Output 1 (+5 V)
N	Shield RTN	N	Output 1 RTN
P	Auxiliary DC (+)	P	Output 1 (+5 V)
R	Safety Ground	R	Output 1 RTN
S	Auxiliary DC (-)	S	Output 1 (+5 V)
T	Safety Ground	T	Output 1 RTN
U	Safety Ground	U	Output 1 (+5 V)
V	Unused	V	Output 1 RTN
W	Safety Ground	W	Output 1 (+5 V)
X	Unused	X	Output 1 RTN
Y	Excessive Temperature RTN	Y	Output 1 (+5 V)
Z	Remote ON/OFF	Z	Output 1 RTN
a	Excessive Temperature	a	Output 1 (+5 V)
b	Remote ON/OFF RTN	b	Output 1 RTN
c	Unsafe Temperature RTN	c	Output 1 (+5 V)
d	Voltage Programming-1	d	Output 1 RTN
f	Unsafe Temperature	e	Output 1 (+5 V)
g	Voltage Programming-1 RTN	f	Output 1 RTN
h	Output Status RTN	h	Output 2 (+)
i	Output Programming	j	Output 2 (-)
j	Output Status	k	Output 2 (+)
k	Output Programming RTN	m	Output 2 (-)
m	Voltage Programming-2 RTN	n	Output 2 (+)
n	Voltage Programming-2	p	Output 2 (-)
		r	Output 2 (+)
		s	Output 2 (-)
		t	Output 2 (+)
		u	Output 2 (-)
		v	Output 2 (+)
		w	Output 2 (-)
		x	Output 2 (+)
		y	Output 2 (-)
		z	Output 2 (+)

Source: https://assist.dla.mil -- Downloaded: 20260410-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE II. Connector contact assignment - Continued.

INPUT CONNECTOR	OUTPUT CONNECTOR	
	AA	Output 2 (-)
	BB	Output 2 (+)
	CC	Output 2 (-)
	DD	Output 2 (+)
	EE	Output 2 (-)
	FF	Output 2 (+)
	HH	Output 2 (-)

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE III. Power supply functions.

Function	Logic "0" or closed circuit	Logic "1" or open circuit	Reference
Remote ON/OFF single output (Input contacts Z and b)	150 A output ON	150 A output OFF	MIL-P-24764
Remote ON/OFF dual output (Input contacts Z and b)	Output-1 and Output-2 ON	Output-1 OFF Output-2 ON	MIL-P-24764

Function	Open Circuit	Closed circuit	Reference
Output programming (Input contacts i and k)	Single output	Dual output	3.11
Voltage programming-1 dual output (Input contacts d and g)	5.05 V (Output-1)	5.25 V (Output-1)	MIL-P-24764
Voltage programming-2 (Input contacts m and n)	5.05 V (Output-2)	5.25 V (Output-2)	MIL-P-24764
Voltage programming-1 single output (Input contacts d and g)	5.05 V (150A Output)	5.25 V (150A Output)	MIL-P-24764

-See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE III. Power supply functions - Continued.

Function	Logic "0"	Logic "1"	Reference
Output status signal (input contacts h and j)	Voltage within rating and remote ON/OFF = ON	Voltage out of tolerance or remote ON/OFF = OFF	3.12

Function	On	Off	Reference
Output status LED	Voltage within rating and remote ON/OFF = ON	Voltage out of tolerance or remote ON/OFF = OFF	MIL-P-24764

Function	Open 1/	Closure 1/	Reference
Excessive temperature (input contacts Y and a)	Any Tj prior to an excessive closure	When Tj is from 1 to 25°C prior to the unsafe closure	MIL-P-24764
Unsafe temperature (input contacts c and f)	Any Tj prior to an unsafe closure	When Tj exceeds the 80°C Tj by a minimum of 5°C	MIL-P-24764

1/ The symbols used in this table and not defined elsewhere are as follows:

Tj - semiconductor junction temperature

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

3.7 Quality conformance tests. All production power supplies shall be tested in accordance with 4.2 and footnotes 1 and 2 in table I at a mounting surface temperature of 25°C ± 5°C.

3.8 Qualification. Initial qualification and periodic check shall be in accordance with MIL-P-24764 (see 4.5).

3.9 End-of-life limits. Power supplies which have been plugged into an operating system, or exceed 2 years from the date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.3).

3.10 Output voltage polarity. The 150 amperes output and output-1 (80 amperes) of the power supply shall only be required to operate as positive outputs (output positive with respect to output return). However, the supply shall not be damaged by connecting the positive contact to chassis with the return floating. The output-2 (70 amperes) shall be capable of being utilized as a positive supply or as a negative supply.

3.11 Output programming. The power supply shall be programmable for either single or dual outputs by shorting or opening the output programming contacts. An open circuit shall select the single output configuration and a closed circuit shall select the dual output configuration. Power supply performance is not specified and no damage shall occur to a supply when it is programmed for dual outputs and operated as a single output or when it is programmed for single positive output and operated as dual positive outputs. However, it may not be protected when programmed for a single output and operated as a dual output with output-2 connected as a negative output.

3.12 Output status signal. The output status signal shall operate as defined in MIL-P-24764 for output voltage status. Any output out of tolerance shall cause an appropriate indication on the status signal output. The output status signal shall be at a logic "1" whenever the remote ON/OFF equals OFF regardless if programmed for single output or dual output.

3.13 Remote sense leads. The power supply shall have remote sense capability for the 150 amperes output (single output) and the output-1 (dual output mode) with sense and load leads up to 60 inches in length. The voltage at the +5V/5V return output contacts shall be capable of reaching a value of at least 5.408 volts, regardless of the voltage programming level, in order to compensate for load lead voltage drop. The power supply shall not be operated without the sense leads connected to the appropriate output.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

3.14 Parallel power supplies. The single output (150 amperes) or output-1 (80 amperes) of an AA3A power supply shall be capable of being connected in parallel with the single output or output-1, respectively, of one additional AA3A power supply. Two parallel configurations are possible:

(a) ON/OFF paralleling. One power supply output shall be energized and the other unenergized. The remote ON/OFF control on the unenergized power supply output shall be programmed for the OFF condition as long as the other power supply is operating. ORing diodes are not required.

(b) ON/ON paralleling. The output of both power supplies are energized at the same time. One power supply shall be programmed for an output voltage of 5.25 volts and the other for 5.05 volts. The power supply with the highest output voltage would be supplying all of the current to the load (no current sharing). If ORing diodes are used between the output and the load of each power supply, the voltage to the load would be reduced by the diode voltage drop (the remote voltage sensing feature may not be adequate to compensate for this voltage drop).

3.14.1 Output-2. Output-2 (in the dual output mode) of a supply shall not be controlled by the remote ON/OFF control. Power supply output-2 shall not be paralleled with another power supply output-2 when the power supplies are programmed for dual operation.

3.14.2 Output capacitive loading. When two AA3A power supplies are paralleled as in 3.14(a), the energized output shall provide specified output voltage and current into the maximum allowable capacitance of the external load and the output capacitance of the paralleled, unenergized power supply. The capacitance of the external load shall not exceed 30 microfarads (uF) per rated output ampere. The output capacitance of the paralleled power supply shall not exceed 65 uF per rated output ampere. Therefore, the energized power supply shall drive a maximum capacitance of 95 uF per rated output ampere.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

3.15 Electromagnetic interference requirements. The power supply shall meet the following electromagnetic interference (EMI) requirements of MIL-STD-461, class A4 and class A5, with modifications or extensions to the limits where indicated (see 4.5.7):

(a) CE01 direct current (DC) power leads - see figure 3.

(b) CE01 alternating current (AC) power leads - see figure 3.

(c) CE03 narrowband, DC power leads - see figure 4.

(d) CE03 broadband, DC power leads - see figure 5.

(e) CE03 narrowband, AC power leads - see figure 6.

(f) CE03 broadband, AC power leads - see figure 7.

(g) CS01.

(h) CS02.

(i) RE01 - see figure 8.

(j) RE02, installed configuration, see 4.5.7.2. The maximum test frequency shall be 1 gigahertz (GHz).

(k) RE02, connector opening, see 4.5.7.2. The maximum test frequency shall be 1 GHz.

(l) RS01.

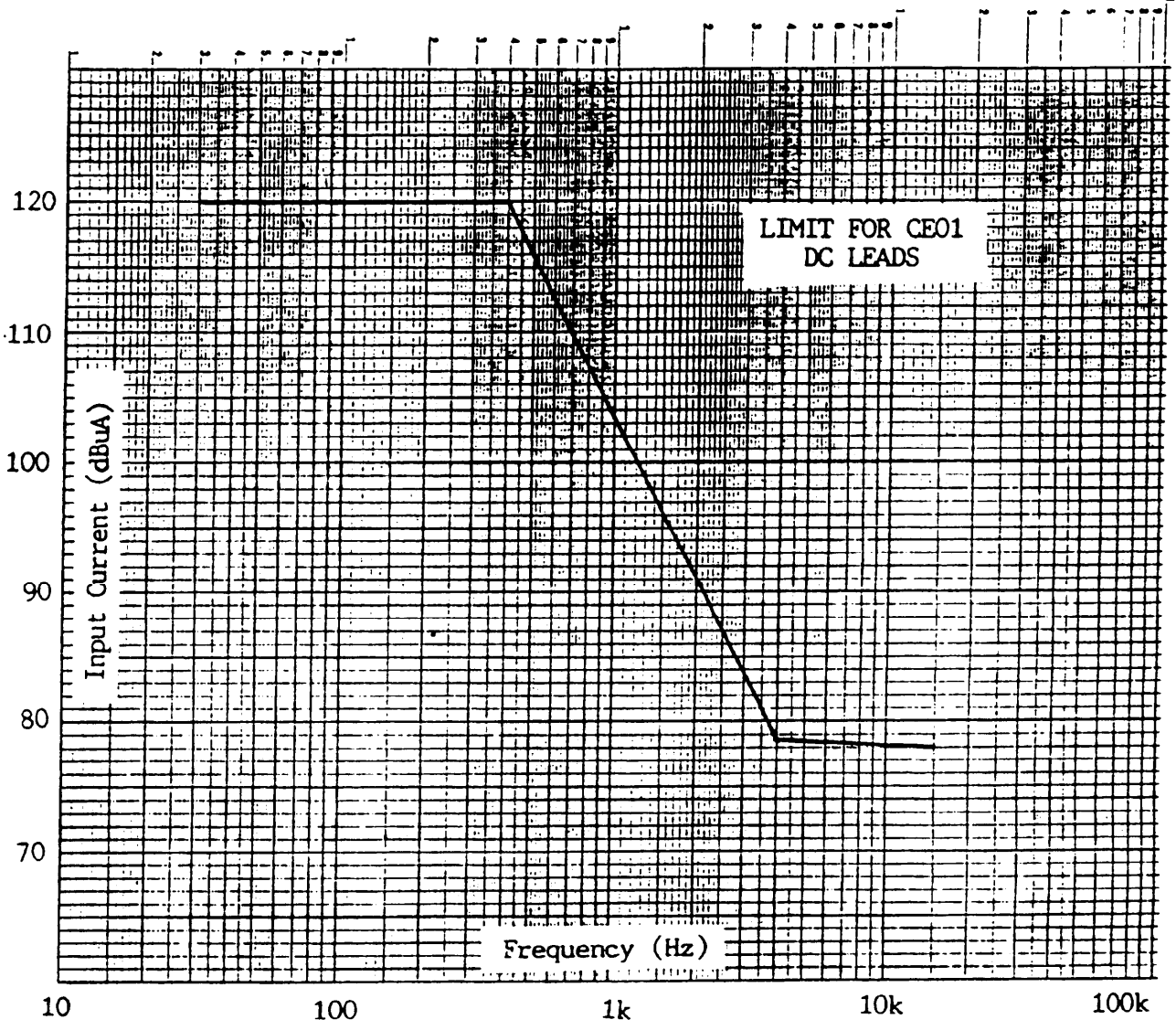
(m) RS02.

(n) RS03 - class A4 for below deck metallic hull.

3.15.1 Criteria for susceptibility. When the power supply is being tested for susceptibility to a conducted or radiated signal, the output voltage shall be monitored for compliance to the table I output regulation limits.

3.16 EMI filter design. The power supply is not required to have an internal input EMI filter. Therefore, for a power system to meet MIL-STD-461 conducted emissions requirement CE03, the system must have an EMI filter between the power source

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the currentversion before use.



NOTE:

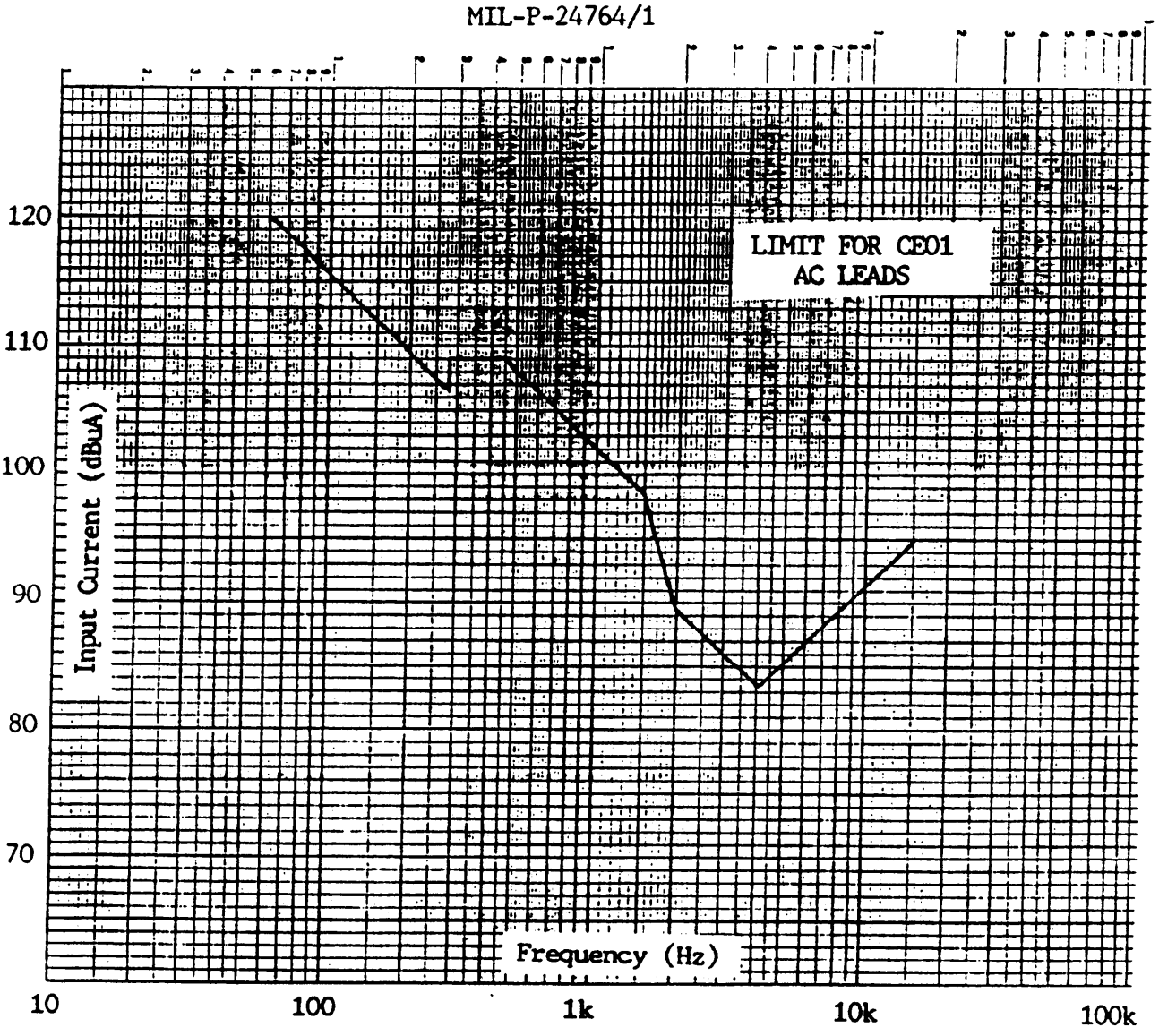
For power supplies with input current less than 1 ampere, the limit shall be as shown. For power supplies with input current greater than 1 ampere, the limit shall be relaxed as follows:

dB relaxation = 20 log(input current),

where input current is in amperes.
The unit dBuA is decibels with respect to 1 microampere.

FIGURE 3. Powerline harmonic current limitation (CE01).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.



NOTE:

For power supplies with input current less than 1 ampere, the limit shall be as shown. For power supplies with input current greater than 1 ampere, the limit shall be relaxed as follows:

dB relaxation = 20 log(input current),

where input current is the wideband, RMS value in amperes.

FIGURE 3. Powerline harmonic current limitation (CE01) - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

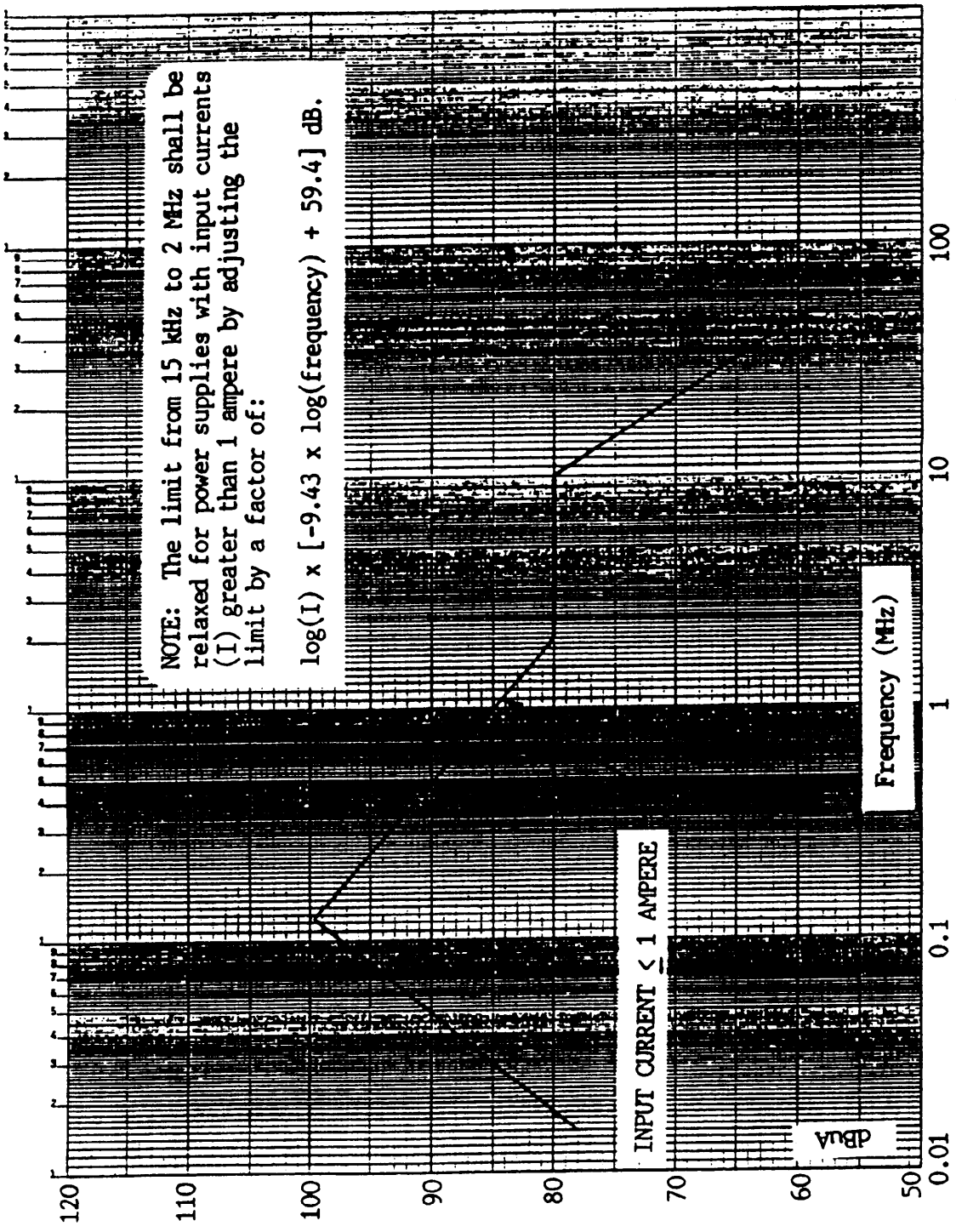


FIGURE 4. Narrowband conducted emissions on DC power leads (CE03).

Source: https://assist.dla.mil -- Downloaded: 20261O-20T01:52Z
Check the source to verify that this is the current version before use.

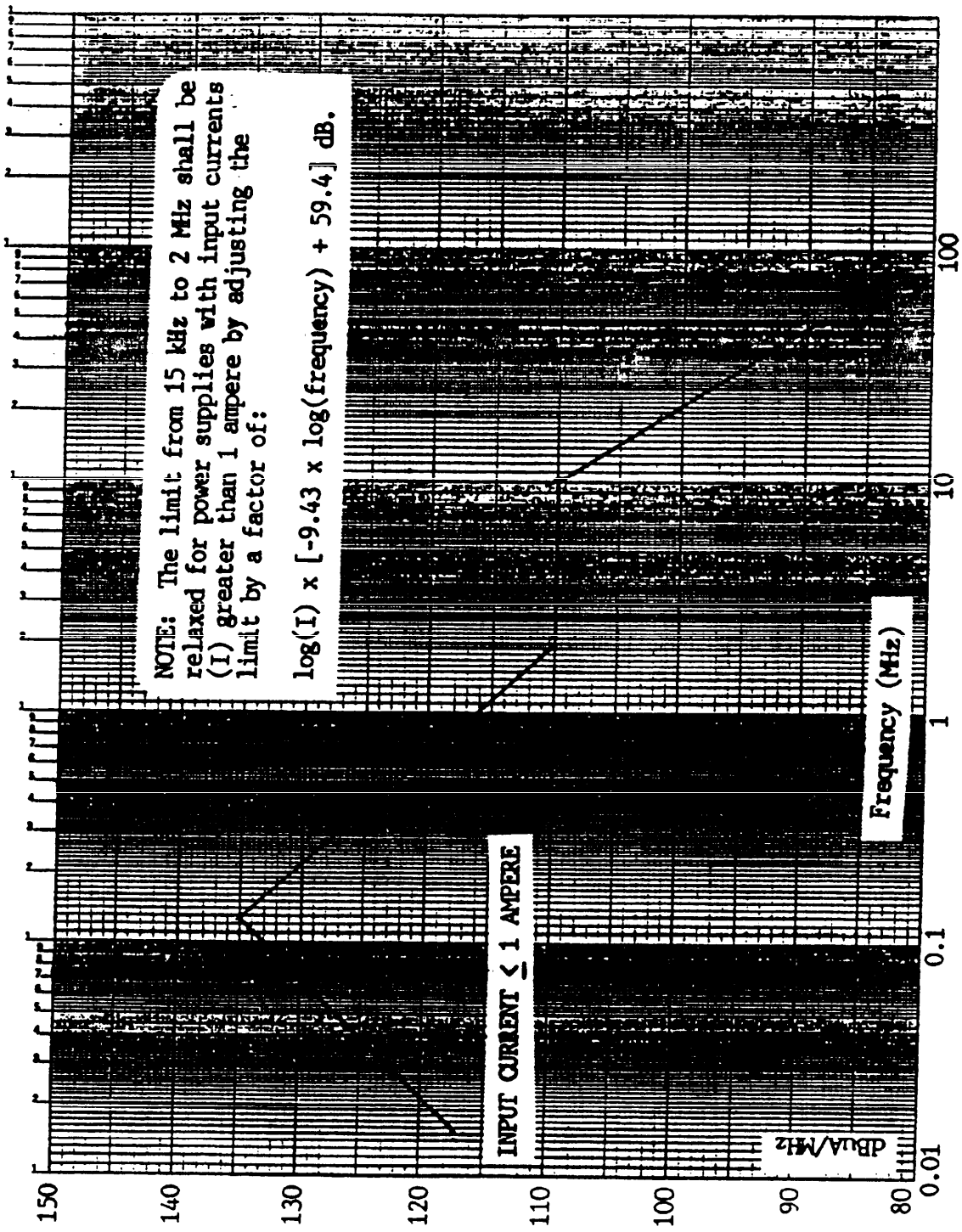


FIGURE 5. Broadband conducted emissions on DC power leads (CE03).

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

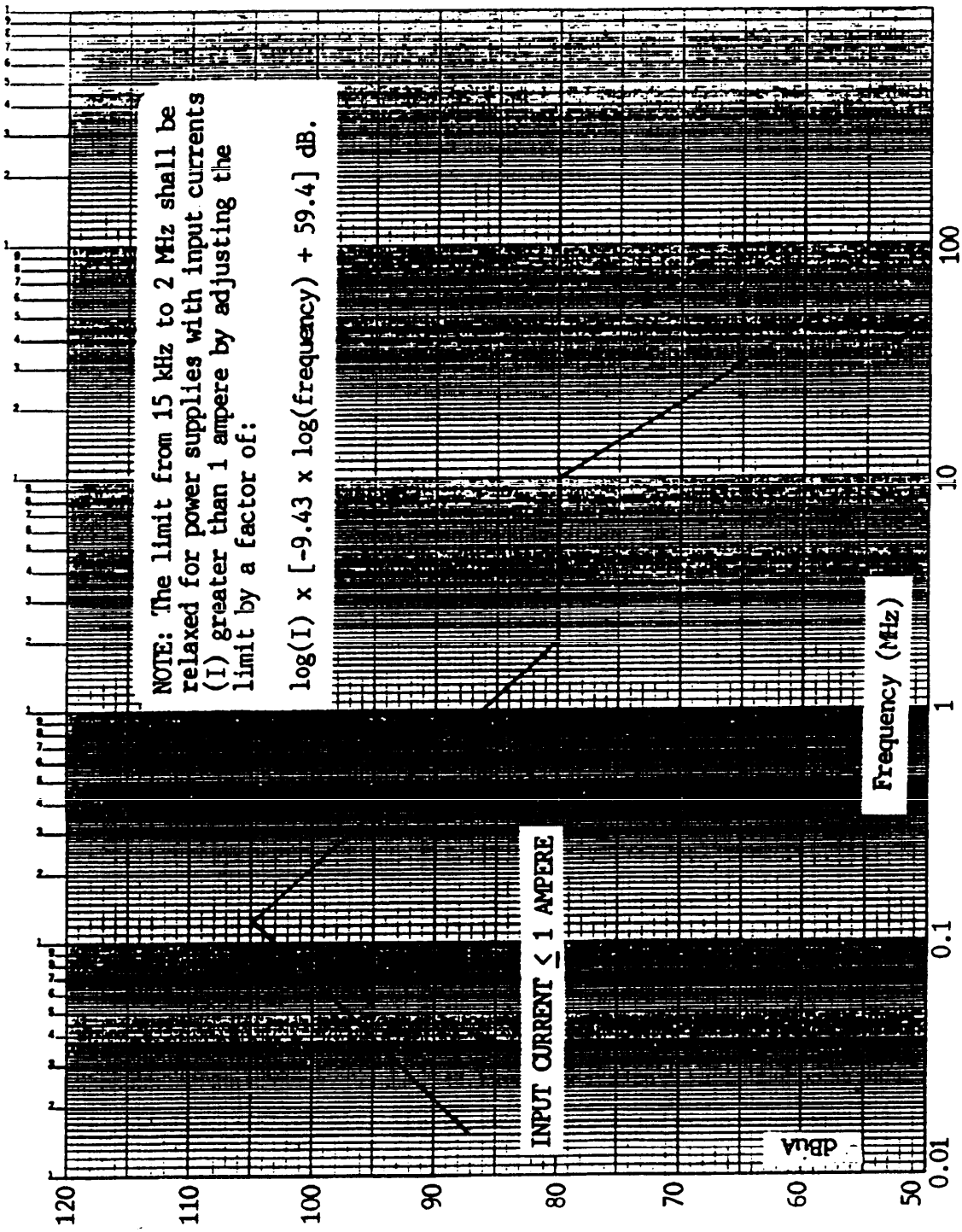


FIGURE 6. Narrowband conducted emissions on AC power leads (CE03).

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

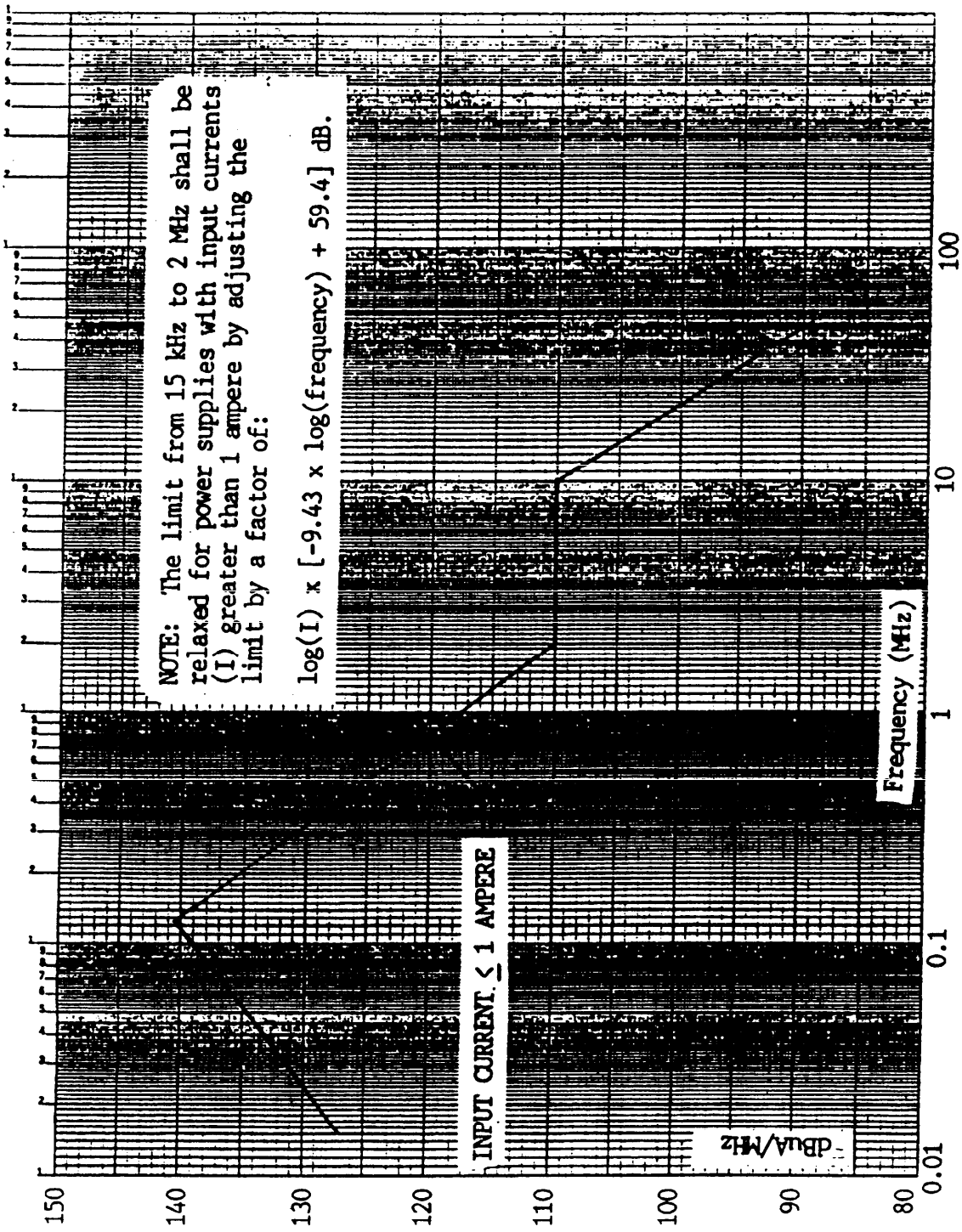


FIGURE 7. Broadband conducted emissions on AC power leads (CE03).

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the currentversion before use.

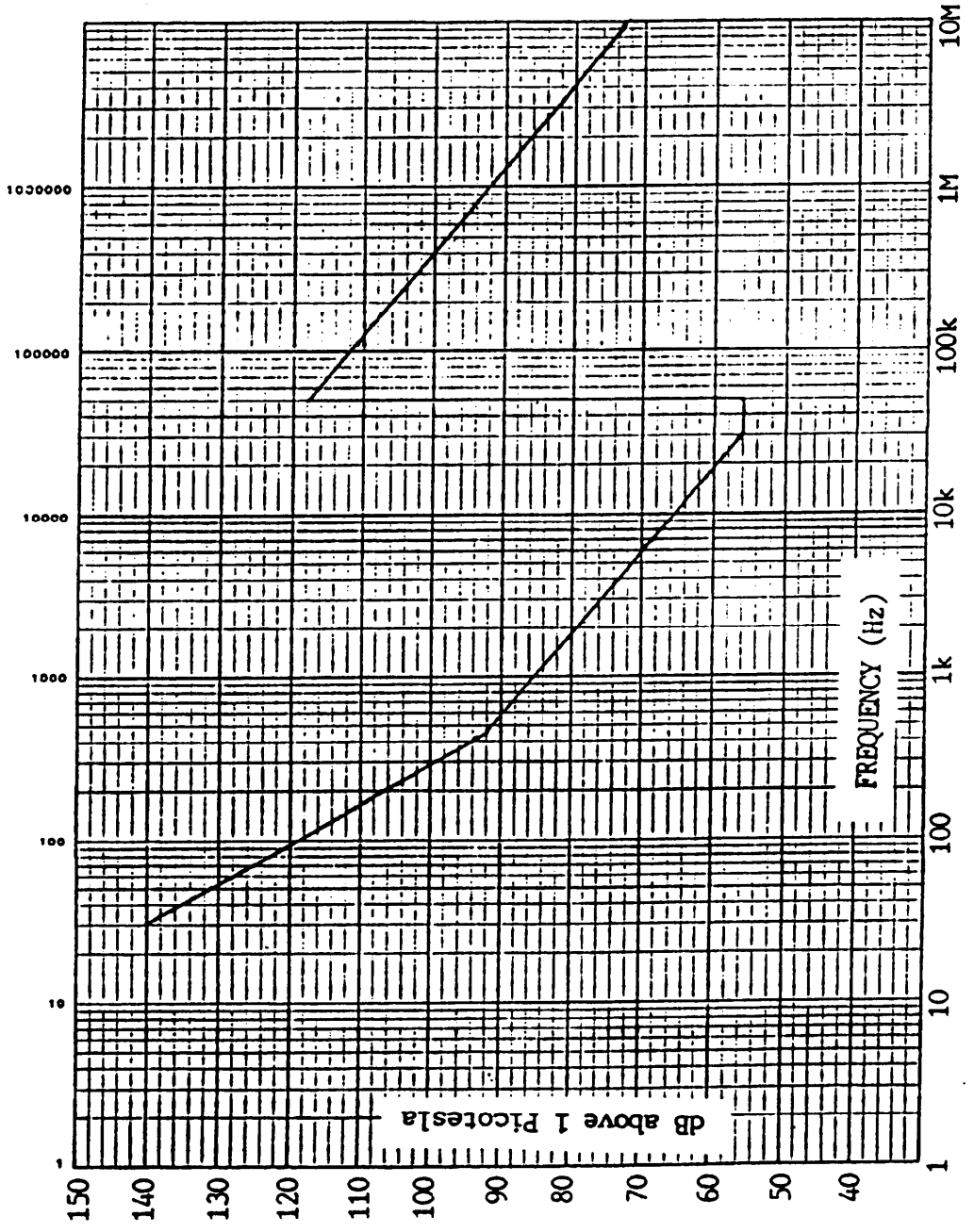


FIGURE 8. Magnetic field emissions (RE01).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

output and the power supply input. The necessary EMI filter attenuation can be found by taking the difference between the allowed applicable power supply conducted emissions curves (see figures 3 through 7) and the system required conducted emissions. The power supply input impedance characteristics as specified in 3.16.1 are also needed in order to design an EMI filter. MIL-HDBK-241 should be referenced for precautions to be taken in the design of an EMI filter to prevent the filter/power supply combination from forming a negative impedance oscillator. The design of the power source system must ensure that damage cannot occur under the system's worst case minimum input voltage including brownout conditions. The power supply output voltage may continue within regulation as the input voltage drops to half of nominal 155 volts direct current (VDC) (115 volts alternating current (VAC)); however, the power source voltage may not be specified to go this low.

3.16.1 Input impedance requirement. The power supply input impedance magnitude, at full load output, shall be greater than the figure 9 curve for the input voltages specified. The input impedance phase shall be more positive than the figure 10 curve for the input voltages specified. These curves are composites of the minimum input impedances from several different power supply designs. The actual model for each vendor's design may be obtained from the SPSP-QAA. During initial qualification testing, the SPSP-QAA will verify that each vendor's design meets the above impedance requirements.

3.16.2 Input impedance computer model. Each manufacturer of power supplies shall generate a computer model of the power supply input impedance (Zi) as observed from the input power contacts. The model shall diagram the input network and show both the AC and auxiliary inputs. It shall include elements simulating the negative impedance of the converter. Nominal component values together with their tolerances shall be given. To verify the accuracy of the model, each manufacturer shall measure the input impedance (magnitude and phase) of their power supply under the following conditions:

(a) From any two of the AC, phase A, B, or C input contacts. Also from the auxiliary input contacts if the data taken would differ significantly from that taken at the AC input contacts.

(b) Over the frequency range of 100 hertz (Hz) to 100 kilohertz (kHz) or up to the voltage feedback loop gain crossover frequency, whichever is greater.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

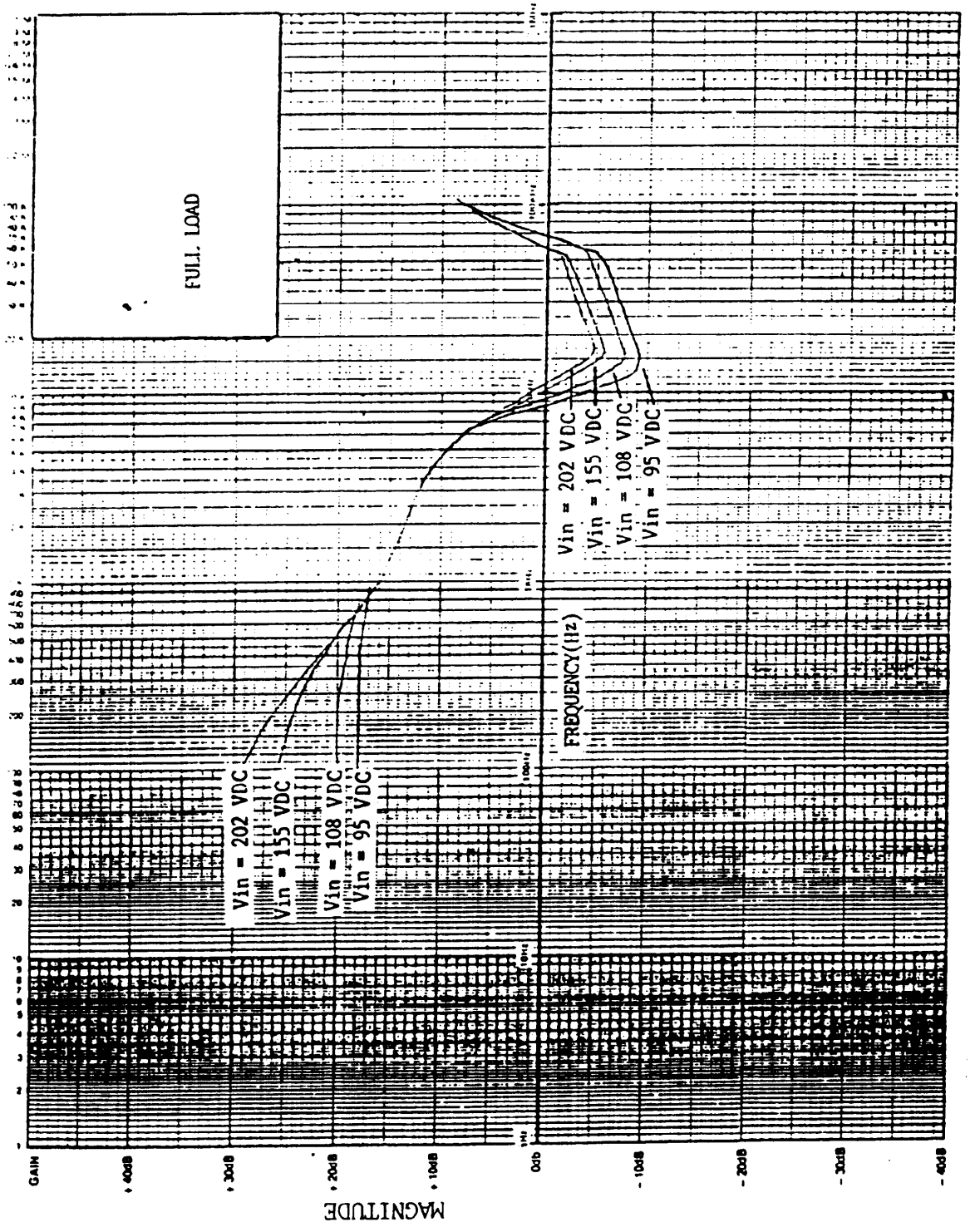


FIGURE 9. Minimum input impedance magnitude.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

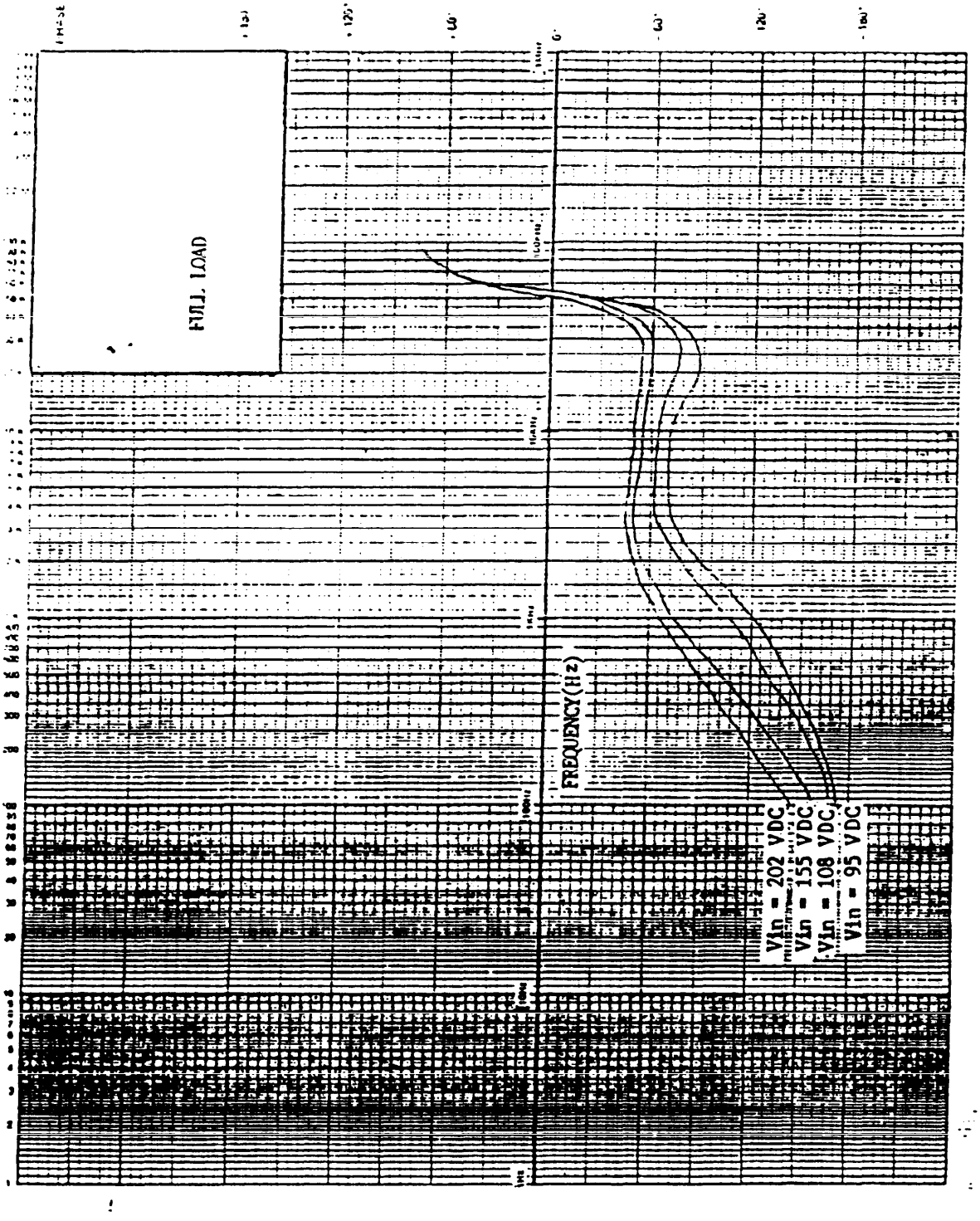


FIGURE 10. Minimum input impedance phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

(c) With input voltages of 108, 155, and 202 VDC. Also measurement at a minimum input voltage, just prior to the output voltage going out of regulation or turning off.

(d) At full load for all input voltages. For 108 VDC input voltage, also measure at 10 percent and 50 percent full load.

The Zi computer model shall be made to match the Zi measured data within ± 1.0 dB for gain and ± 6° for phase at frequencies up to 10 kHz. Above 10 kHz the required match shall be ± 2.0 dB for gain and ± 24° for phase. The measured data, the computer model, and curves obtained from the computer model shall be submitted to the SPSP-QAA with no restrictions on use by system manufacturers utilization for stability purposes.

3.17 Spike voltage. The power supply shall be designed to withstand, without damage, the 2500 volt spike (see MIL-P-24764) clipped to a 250 volt amplitude by external system or equipment devices. The duration of the spike shall be 162 ± 16 microseconds (us) at the 250 volt level.

3.18 Exceptions to MIL-P-24764. This power supply shall have the following exceptions to the requirements of MIL-P-24764:

(a) The power input circuit can withstand a spike voltage of only 250 volt peak (see 3.17).

(b) The input voltage turn-on and turn-off thresholds are different (see table I, test conditions GG1 and GG2).

(c) The remote voltage sensing shall have the capability of sensing for greater than 10 inches from the power supply and of compensating for greater than 3 percent voltage drop when the output is programmed for 5.05 volt (see 3.13).

(d) The steady state output voltage with open sensing leads shall be within plus 10 percent, minus 5 percent of the nominal output voltage instead of within plus or minus 5 percent.

(e) The overvoltage protection is not required to be a latching type.

(f) The EMI limits for requirements CE01, CE03, RE01 and RE02 of MIL-STD-461 are modified or extended (see 3.15).

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the currentversion before use.

(g) An external EMI filter is required to enable the power supply to meet requirement CE03 of MIL-STD-461.

(h) The remote ON/OFF function shall not control output-2 (in the dual output mode).

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-P-24764 shall apply.

4.2 Quality conformance tests. Quality conformance tests of all power supplies shall be in accordance with the procedures and equipment specified in MIL-P-24764.

4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Equivalent equipment may be used for quality conformance and qualification testing provided the correlation requirements of MIL-P-24764 are met.

4.4 Test procedures. The power supply shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the SPSP-QAA and the test methods conform to the requirements of MIL-P-24764.

4.4.2 Order of testing. Test conditions A through KK shall be performed in any order with the following exceptions. If test conditions A, BB, HH, and JJ are part of a sequence of tests, the order of testing shall be as follows: A, BB, HH, and JJ where applicable, followed by the remaining tests in the sequence. Test condition W shall be performed in accordance with 4.5.3.

4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-P-24764 with the modifications or additions specified in 4.5.1 through 4.5.7.2.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the currentversion before use.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-P-24764. Before starting these tests, test conditions A through V and X through KK in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25 ± 5°C mounting surface temperature. At turn on, following one hour stabilization at the low temperature, test conditions C1, C2, C5, C7, C9, C11, D2, D3, and P5 shall be performed. When electrical testing is called for at the end of each temperature stabilization period (at -10°C, 60°C and 25°C post operational temperature), test conditions C, D, G, H, M7, N1-N3, P1-P3, Q1-Q3, Q10-Q12, S1-S4, S17-S24, T, V1-V9, and GG1-GG3 shall be performed in the order specified in 4.4.2 using initial limits for the applicable operating temperature. When electrical testing is called for at 80°C, test conditions C3-C4, D1-D12, N1-N3, and Q1-Q3 shall be performed using the high temperature end-of-life limits. At all other times during the operating temperature tests, the power supply shall be operated in the test circuit shown on figure 11.

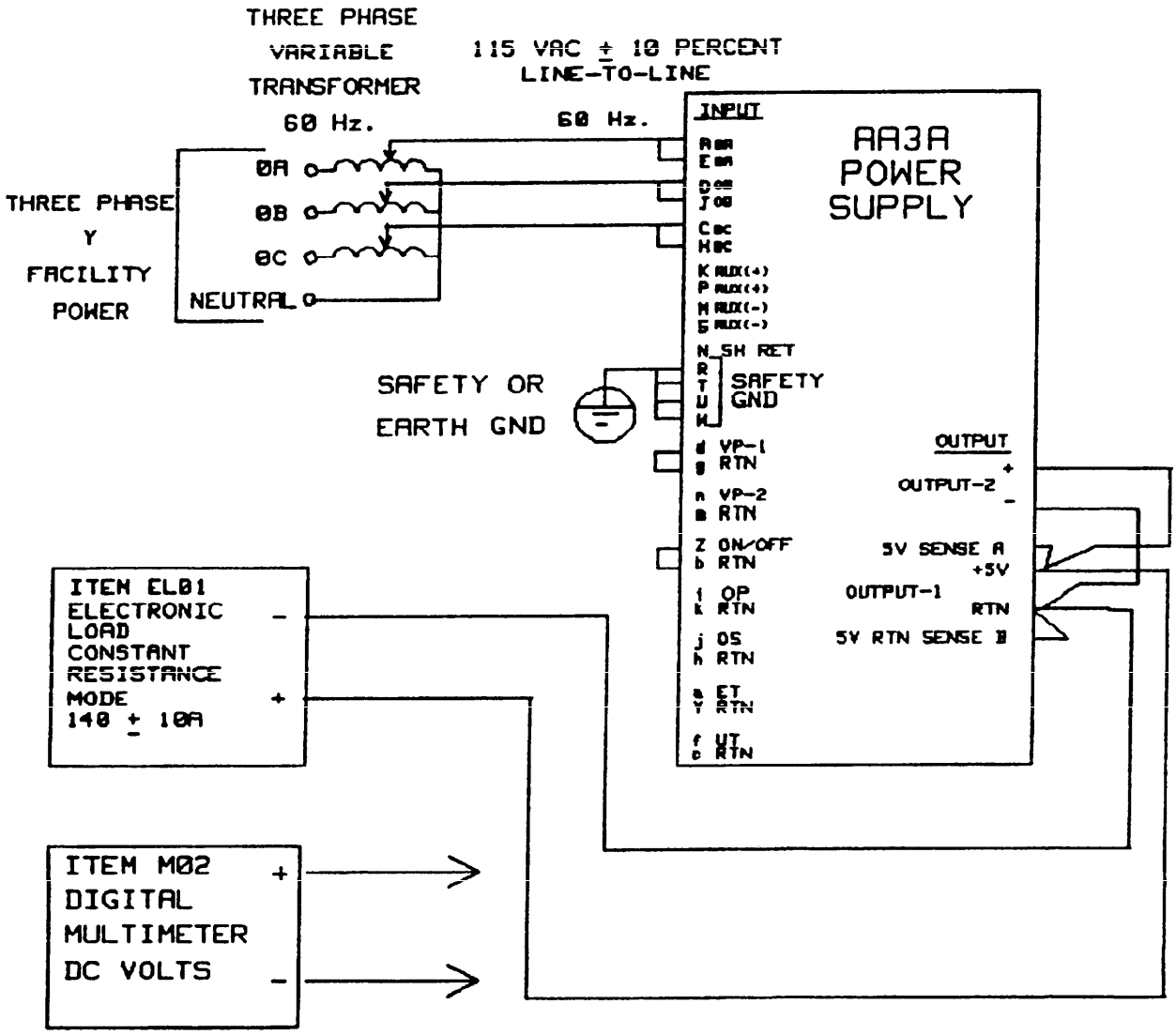
4.5.2 Component temperature tests. The component temperatures shall be measured in accordance with MIL-P-24764. The power supply shall be operated as shown on figure 11. These measurements shall be performed on each initial qualification submission that is the result of a new vendor or of a major redesign of a previously qualified vendor.

4.5.3 Over-temperature indicators. Operation of the excessive temperature indicator and unsafe temperature indicator shall be verified to be within the temperature limits specified in table I. Test conditions W1-W6 are non-operating tests designed to verify that the over temperature indicators operate. Test condition W shall only be performed after all other electrical and environmental tests are completed.

4.5.4 Vibration tests. The vibration tests shall be conducted in accordance with MIL-P-24764. During the vibration tests, the power supply shall be operated in the circuit shown on figure 12. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25 ± 5°C mounting surface temperature.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the currentversion before use.

voltage and current combinations at one time;



NOTES:

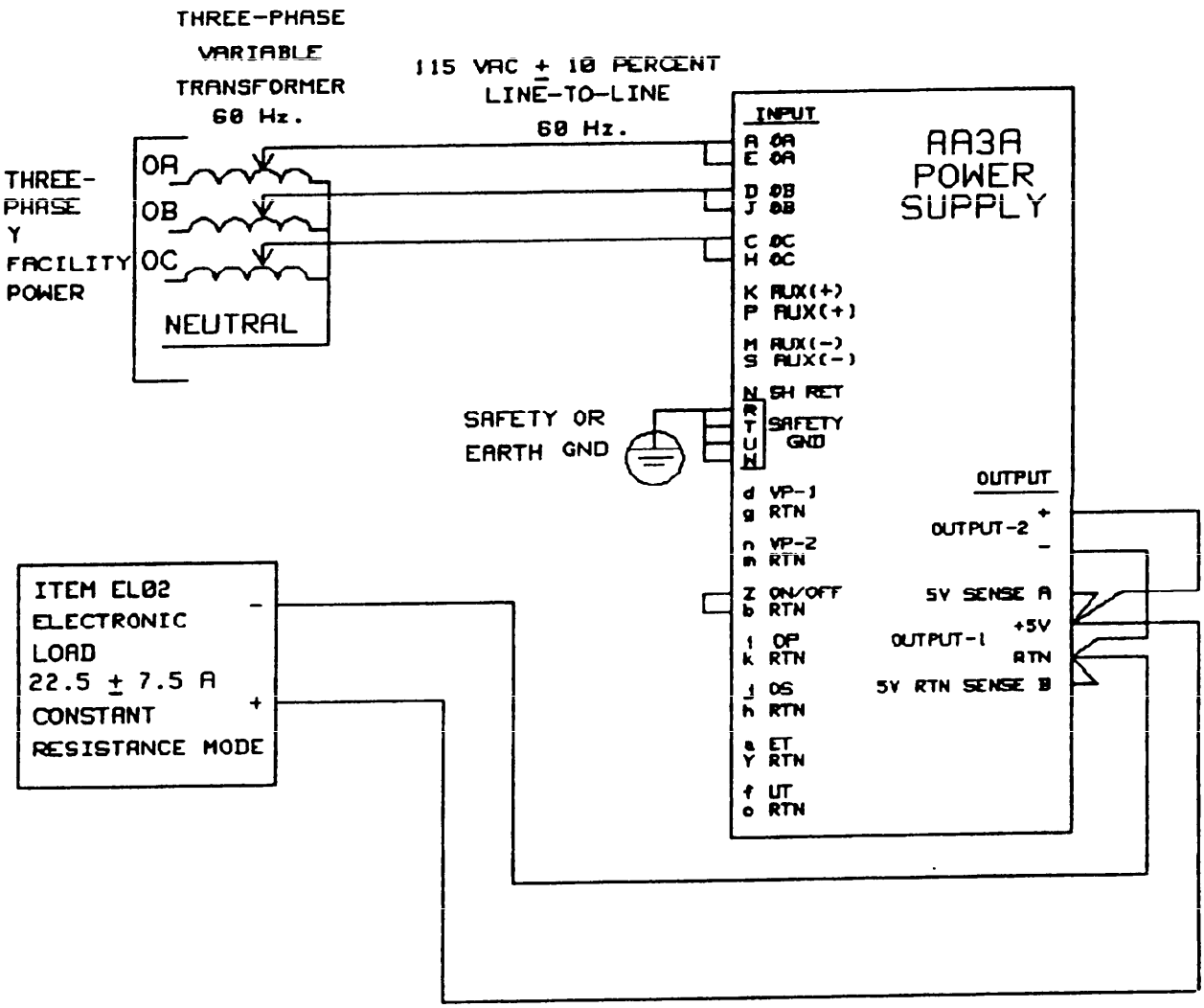
1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. For the long term stability test, the input voltage and output load current shall be within ± 1 percent when making output voltage measurements. Item M02 shall be used.

FIGURE 11. Operating temperature, component temperature, and long term stability test circuit.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.



NOTES:

1. See table II for output connector contact assignment.

2. Output loading is specified at 15 percent of rated load to minimize the requirement for externally supplied cooling. The test may be performed at any load level between 10 percent and 100 percent rated load; however, the power supply mounting surface temperature shall not exceed 60°C.

FIGURE 12. Vibration and shock test circuit.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

4.5.5 Shock tests. The mechanical shock tests shall be conducted in accordance with MIL-P-24764. During the shock tests, the power supply shall be operated in the circuit shown on figure 12. Following the shock test, the electrical tests in table I which are marked with an "X" in the test type column shall be performed using the initial limits at 25 ± 5°C mounting surface temperature.

4.5.6 Long term stability test. The long term stability test shall be conducted in accordance with MIL-P-24764. The power supply shall be operated as shown on figure 11. After operating for 20 ± 2 hours, the baseline output voltage reading shall be taken. At the conclusion of the test, prior to disconnecting the power supply or reducing the temperature, the output voltage readings shall again be recorded. The end-of-test voltage shall not change by more than 0.2 percent from the baseline voltage. Following the return of the power supply to 25°C, electrical tests A through V, X, Z, EE, FF, and GG1-GG7 shall be performed in the order specified in 4.4.2 using the 25 ± 5°C mounting surface temperature end-of-life requirements in table I.

4.5.7 EMI test. The EMI tests shall be conducted in accordance with MIL-P-24764, the procedures of MIL-STD-462, and as modified herein. For requirements CE01 and CE03 the power supply shall be tested using 115 VAC ± 2 percent line-to-line, 3 phase, 60 ± 1 Hz input voltage and tested using 155 VDC ± 2 percent input voltage. For requirement RE01 the supply shall be tested using the 115 VAC ± 2 percent line-to-line input voltage. At the option of the testing facility, for the remainder of the requirements, either the nominal AC or DC input voltage may be used. The loading shall be 90 to 100 percent of full load.

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T01:52Z
Check the source to verify that this is the currentversion before use.

4.5.7.1 RE01 test modification. The RE01 upper frequency shall be 10 MHz as shown on figure 8.

4.5.7.2 RE02 test modification. The test shall be performed for two conditions:

(a) The installed configuration shall have the power supply marking (see figure 1) facing the antenna, with the connector area and all leads completely shielded. The purpose of this configuration is to ensure that only the radiated emissions through the case and cover of the power supply are being measured.

(b) The connector opening configuration shall have the power supply connector opening facing the antenna and all leads up to the connector opening completely shielded. The purpose of this configuration is to ensure that only the radiated emissions from the power supply connector area are being measured.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-P-24764.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Power supplies specified herein are intended for use in military shipboard and subsurface ship systems and subsystems.

6.2 Acquisition requirements. Acquisition documents should specify the ordering data required in MIL-P-24764.

6.3 End-of-life. When any individual power supply parameter fails to conform to the end-of-life requirements in table I, the power supply is considered to have completed its useful life (see 3.9).

6.4 Power supply description.

Key code: AA3A

Case size: 2.48 inches (62.99 millimeters (mm)) by 6.80 inches (172.72 mm) by 10.8 inches (274.32 mm)

Mean time between failures (MTBF): 100,000 hours

Input voltage: Accepts any of the following voltages:

(a) 115 volts root mean square (VRMS) line-to-line, 3-phase, 47-63 Hz, delta.

(b) 155 VDC to any two AC inputs.

(c) 155 VDC to the auxiliary input.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

Outputs: Provides any one of the following output voltage and current combinations at one time:

(a)	Single output:	+ 5.05/+ 5.25 volts at 15 to 150 amperes.
(b)	Dual output:	+ 5.05/+ 5.25 volts at 8 to 80 amperes. + 5.05/+ 5.25 volts at 7 to 70 amperes.
(c)	Dual output:	+ 5.05/+ 5.25 volts at 8 to 80 amperes. - 5.05/- 5.25 volts at 7 to 70 amperes.

6.5 Power supply function.

6.5.1 Output programming. Output programming allows the power supply to operate as a single positive output of 150 amperes or as a dual output supply. When programmed for dual operation, output-1 (80 amperes output) is positive with respect to the output return and output-2 (70 amperes output) may be used as either positive or negative. When programmed for single output operation, output-1 (80 amperes) and output-2 (70 amperes) are used in parallel.

6.5.2 Voltage programming. When the supply is programmed for dual operation, each output is individually contact programmable for a nominal voltage of either 5.05 or 5.25 volts. When the power supply is operated as a single output supply, the output-1 voltage programming input (VP-1) will control the nominal output voltage. The output-2 voltage programming input (VP-2) will be disabled and will not affect the power supply output voltage when operated as a single output supply. The voltage programming-1 return is common with the output-1 sense return. The VP-2 return is common with the output-2 negative (-). The VP-2 return will be a negative voltage when output-2 is used as a negative output.

6.5.3 Remote ON/OFF control. The remote ON/OFF control applies to the 150 amperes output when the supply is programmed for single output operation. The remote ON/OFF only applies to output-1 (80 amperes) when the supply is programmed for dual output operation.

Source: https://assist.dla.mil -- Downloaded: 20260410-20T01:52Z
Check the source to verify that this is the current version before use.

6.5.4 Remote sense capability. The power supply remote sense capability applies to the fixed positive outputs only which are the 150 amperes output (single) and the 80 amperes output-1 (dual). The sense leads should not be left disconnected.

6.5.5 Stability with remote sense. Power supply stability may be adversely affected by certain load lead characteristics when the output voltage is remotely sensed. The inductance (L) of the load leads between the power supply output and the remote sense point together with the load capacitance (C) form an additional LC filter on the output of the power supply. When the corner frequency of this LC filter falls within the unity gain bandwidth of the power supply feedback loop, then the power supply phase margin is reduced. These conditions tend toward and can result in power supply instability. Caution should be exercised when utilizing remote sensing and having significant load lead inductance (L) and load capacitance (C). Power supply stability should be verified by analysis and also by dynamic load testing in the actual circuit configuration.

6.5.6 Output current rating. The single output is rated at 15 to 150 amperes. For dual operation, the output-1 is rated at 8 to 80 amperes and the output-2 is rated at 7 to 70 amperes. For operation between 0 and 15 amperes (single output) or 0 to 8 and 0 to 7 amperes (dual output), the 25°C initial load regulation and ripple tolerances are twice the tolerances for operation within the rated current limits.

6.6 Recovery from overvoltage. The output voltage is not required to recover automatically from an output overvoltage fault and may require the input voltage to be taken to zero and then reapplied to reset the output. Cycling of the remote ON/OFF control will not accomplish this reset function.

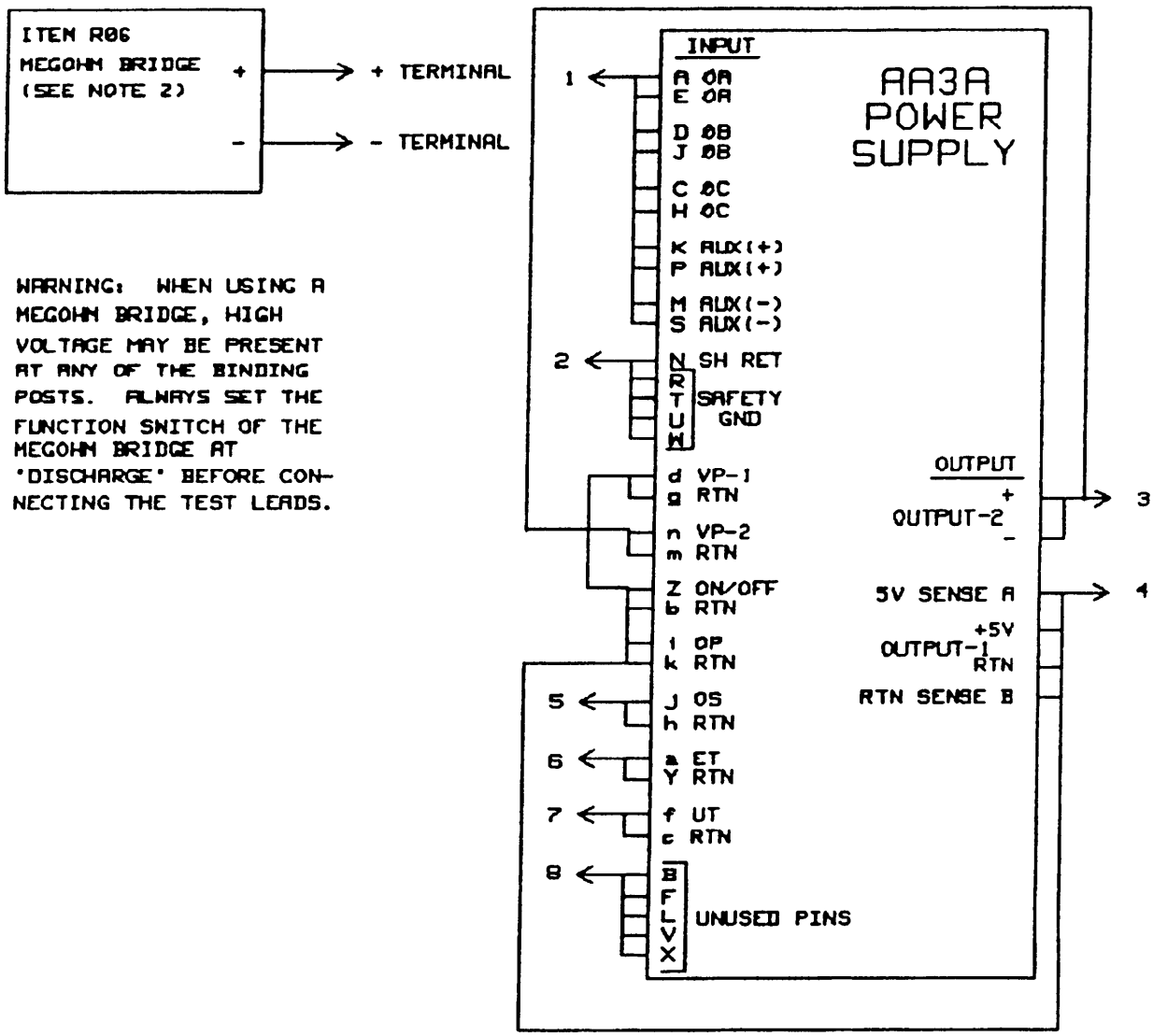
Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE IV. Isolation tests.

Test cond	Parameter	Item R06 Voltage ± 10%	Connect R06 to 1/ + terminal pointe:	- terminal pointe:	Initial limits 25 ± 5°C Mounting surface temperature Min	Max	Unit	Test fig
A1	Isolation-input to chassis and output	800 VDC	1	2,3,4	100	INF.	kOhm	13
A2	Isolation-output to chassis	50 VDC	3,4	2				
A3	Isolation-output to output	20 VDC	3	4				
A4	Isolation-excessive temp.	100 VDC	1-5, 7	6				
A5	Isolation-unsafe temp.	100 VDC	1-6	7				
A6	Isolation-output status	100 VDC	1-4, 6, 7	5				
A7	Isolation-input connector used to unused contacts	100 VDC	1-7	8	100		kOhm	
A8	Isolation-input to chassis and output	500 VDC	1	2,3,4	10		MOhm	
A9	Isolation-output to chassis	50 VDC	3,4	2				
A10	Isolation-excessive temp.	100 VDC	1-5, 7	6				
A11	Isolation-unsafe temp.	100 VDC	1-6	7				
A12	Isolation-output status	100 VDC	1-4, 6, 7	5				
A13	Isolation-input connector used to unused contacts	100 VDC	1-7	8	10	INF.	MOhm	13

1/ Two or more numbers in one block indicate that these points shall be connected together in addition to being connected to the + terminal or - terminal as applicable (for example, '3,4' under column "+ terminal points:' indicates point 3, connected to point 4, connected to the + terminal).

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.



NOTES:

1. See table II for output connector contact assignment.
2. See table IV for item R06 connections and voltage setting.

FIGURE 13. Isolation test circuit for table IV.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE V. Continuity test.

Test cond	Parameter	Connect to MB2 (Ohms mode)		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		+ to contact	- to contact	Min	Max		
B1	Chassis contacts to chassis continuity	Input R	Chassis	0.0	1.0	Ohm	-
B2	5V sense RTN to remote ON/OFF RTN continuity	Output B	Input b				
B3	5V sense RTN to voltage Program-1 RTN continuity	Output B	Input g				
B4	Out-2 (-) to voltage Program-2 RTN continuity	Output J	Input m				
B5	5V sense RTN to output program RTN continuity	Output B	Input k				
B6	Safety ground contact continuity	Input R	Input T				
B7	Safety ground contact continuity	Input R	Input U				
B8	Safety ground contact continuity	Input R	Input M				
B9	Safety GND to shield RTN contacts continuity	Input R	Input N				
B10	Phase A contacts continuity	Input A	Input E				
B11	Phase B contacts continuity	Input D	Input J				
B12	Phase C contacts continuity	Input C	Input H	0.0	1.0	Ohm	-

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE V. Continuity test - Continued.

Test cond	Parameter	Connect to M82 (Ohms mode)		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		+ To contact	- To contact	Min	Max		
813	Auxiliary DC(+) contacts continuity	Input K	Input P	0.0	1.0	Ohm	-
814	Auxiliary DC(-) contacts continuity	Input M	Input S				
815	Output-1 (+5V) contacts continuity	Output C	Output E				
816			Output H				
817			Output K				
818			Output M				
819			Output P				
820			Output S				
821			Output U				
822			Output W				
823			Output Y				
824	Output-1 (+5V) contacts continuity	Output C	Output a	0.0	1.0	Ohm	-

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE V. Continuity test - Continued.

Test cond	Parameter	Connect to M82 (Ohms mode)		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		+ To contact	- To contact	Min	Max		
825	Output-1 (+5V) contacts continuity	Output C	Output c	0.0	1.0	Ohm	-
826	Output-1 (5V) contacts continuity	Output C	Output e				
827	Output-1 RTN contacts continuity	Output D	Output F				
828			Output J				
829			Output L				
830			Output N				
831			Output R				
832			Output T				
833			Output V				
834			Output X				
835			Output Z				
836	Output-1 RTN contacts continuity	Output D	Output b	0.0	1.0	Ohm	-

Source: https://assist.dla.mil -- Downloaded: 202640-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE V. Continuity test - Continued.

Test cond	Parameter	Connect to M82 (Ohms mode)		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		+ To contact	- To contact	Min	Max		
837	Output-1 RTN contacts continuity	Output D	Output d	0.0	1.0	Ohm	-
838	Output-1 RTN contacts continuity	Output D	Output f				
839	Output-2 (+) contacts continuity	Output h	Output k				
840			Output n				
841			Output r				
842			Output t				
843			Output v				
844			Output x				
845			Output z				
846			Output BB				
847			Output DD				
848	Output-2 (+) contacts continuity	Output h	Output FF	0.0	1.0	Ohm	-

Source: https://assist.dla.mil -- Downloaded: 2026/0-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE V. Continuity test - Continued.

Test cond	Parameter	Connect to M82 (Ohms mode)		Initial limits 25 ± 9°C mounting surface temperature		Unit	Test fig
		+ To contact	- To contact	Min	Max		
849	Output-2 (-) contacts continuity	Output J	Output m	0.0	1.0	Ohm	-
850			Output p				
851			Output e				
852			Output u				
853			Output w				
854			Output y				
855			Output AA				
856			Output CC				
857			Output EE				
858	Output-2 (-) contacts continuity	Output J	Output HH	0.0	1.0	Ohm	-
859	Chassis contacts to chassis continuity	Input N	Chassis	0.0	0.1	Ohm	-
860		Input R					
861		Input T					
862		Input U					
863	Chassis contacts to chassis continuity	Input W	Chassis	0.0	0.1	Ohm	-

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE VI. Static load regulation test.

Test cond	Parameter	Item EL02-1 curr. +0,-2%	Item EL02-2 curr. +0,-2%	Connect 1/ 1 to points	Connect 1/ 2 to points	VP-1 input contact d to g	VP-2 input contact n to m	OP input contact i to k	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
C1	Load regulation 5.05 V, 15 A.	7.5A	7.5A	3,5	4,6	open	3/	open	5.00	5.10	VDC	14
C2	Load regulation 5.05 V, 15A A.	75A	75A			open	3/		5.00	5.10		
C3	Load regulation 5.25 V, 15 A	7.5A	7.5A			shorted	3/		5.20	5.30		
C4	Load regulation 5.25 V, 15A A.	75A	75A	3,5	4,6	shorted	3/	open	5.20	5.30		
C5	Load regulation 5.05 V, output-1	0A	7A	5	3,6	open	open	shorted	5.00	5.10		
C6	Load regulation 5.05 V, output-1	0A	70A			open	open	shorted	5.00	5.10		
C7	Load regulation 5.05 V, output-1	60A	7A						5.00	5.10		
C8	Load regulation 5.05 V, output-1	60A	70A	5					5.00	5.10		
C9	Load regulation -5.05 V, output-2	0A	7A	4					-5.10	-5.00		
C10	Load regulation -5.05 V, output-2	60A	7A	4	3,8	open	open	shorted	-5.10	-5.00	VDC	14

-See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE VI. Static load regulation test - Continued.

Test cond	Parameter	Item EL02-1 curr. +0,-2%	Item EL02-2 curr. +0,-2%	Connect 1/ 1 to points	Connect 1/ 2 to points	VP-1 input contact d to g	VP-2 input contact n to m	OP input contact i to k	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
C11	Load regulation -5.03V, output-2	0A	70A	4	3,6	open	open	shorted	-5.10	-5.00	VDC	14
C12	Load regulation -5.03V, output-2	60A	70A	4	3,6	open	open	shorted	-5.10	-5.00		
C13	Load regulation 5.25V, output-1	60A	70A	5	3,6	shorted	shorted	shorted	5.20	5.30		
C14	Load regulation -5.23V, output-2	60A	70A	4	3,6	shorted	shorted	shorted	-5.30	-5.20		
C15	Load regulation 5.05V, no load	0A	0A	3,5	4,6	open	3/	open	4.95	5.15		
C16	Load regulation 5.05V, output-1	0A	0A	5	3,6	open	open	shorted	4.95	5.15		
C17	Load regulation -5.03V, output-2	0A	0A	4	3,6	open	open	shorted	-5.15	-4.95		
C18	Load regulation, OS 5.05V no load	0A	0A	J	h 2/	open	open	shorted	0.0	0.4		
C19	Load regulation output-1, remote ON/OFF = OFF	--	70A	5	3,6	open	open	shorted	0.0	0.5		
C20	Load regulation output-2, remote ON/OFF = OFF	--	70A	4	3,6	open	open	shorted	-5.10	-5.00	VDC	14

-See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

1/ Two numbers in one block indicate that these two points shall be connected together in addition to being connected to point 1 or 2 as applicable (for example "3,5" under column "1 to points" indicates point 3 connected to point 5 connected to point 1).

2/ For test condition C18, keep 3 connected to 6, and connect 2 only to input contact h.

3/ Contacts can be either open or shorted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

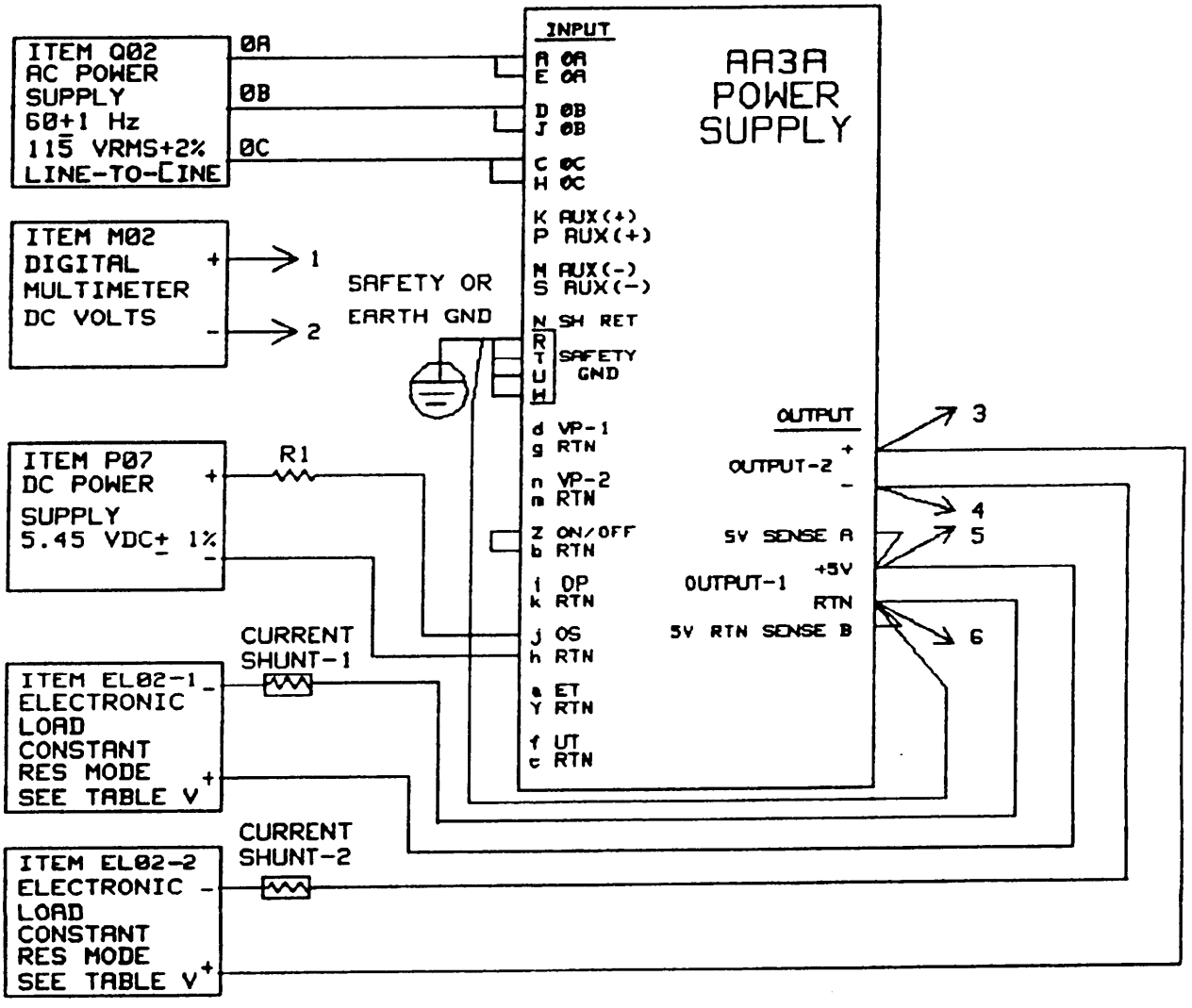


FIGURE 14. Static load regulation test circuit for table VI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. R1 = 1.69 kilohm (kOhm) ± 1 percent, 1/8 watt resistor.

6. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

7. For test conditions C19 and C20, disconnect the short between input contacts Z and b. Item EL02-1 shall be set to 80 amperes prior to disconnecting the short.

FIGURE 14. Static load regulation test circuit for table VI - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE VII. Line regulation test.

Test cond	Parameter	Item EL02-1 curr. +0,-2%	Item EL02-2 curr. +0,-2%	Connect 1/ 1 to pointe	Connect 1/ 2 to pointe	Input voltage ±1% 2/	OP input contact i tg k	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
D1	88 VRMS line reg.	7.5A	7.5A	3,5	4,6	86 VRMS	open	5.00	5.10	VDC	15
D2	86 VRMS line reg.	75A	75A			86 VRMS					
D3	144 VRMS line reg.	7.5A	7.5A			144 VRMS					
D4	144 VRMS line reg.	75A	75A	3,5	4,6	144 VRMS	open				
D5	88 VRMS line reg.	8A	7A	5	3,6	86 VRMS	shorted	5.00	5.10		
D6	88 VRMS line reg.	8A	7A	4		86 VRMS	shorted	-5.10	-5.00		
D7	88 VRMS line reg.	80A	70A	5		86 VRMS		5.00	5.10		
D8	88 VRMS line reg.	80A	70A	4		86 VRMS		-5.10	-5.00		
D9	144 VRMS line reg.	8A	7A	5		144 VRMS		5.00	5.10		
D10	144 VRMS line reg.	8A	7A	4	3,6	144 VRMS	shorted	-5.10	-5.00		
D11	144 VRMS line reg.	80A	70A	5	3,6	144 VRMS	shorted	5.00	5.10		
D12	144 VRMS line reg.	80A	70A	4	3,6	144 VRMS	shorted	-5.10	-5.00		
D13	108 VDC line reg.	7.5A	7.5A	3,5	4,6	108 VDC	open	5.00	5.10	VDC	15

-See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE VII. Line regulation test - Continued.

Test cond	Parameter	Item EL02-1 curr. +0,-2%	Item EL02-2 curr. +0,-2%	Connect 1/ 1 to points	Connect 1/ 2 to points	Input voltage ±1% 2/	OP input contact i to k	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
D14	100 VDC line reg.	7.5A	7.5A	3,5	4,6	108 VDC	open	5.00	5.10	VDC	15
D15	202 VDC line reg.	7.5A	7.5A			202 VDC					
D16	202 VDC line reg.	7.5A	7.5A			202 VDC					
D17	100 VDC aux. reg.	7.5A	7.5A			100 VDC					
D18	202 VDC aux. reg.	7.5A	7.5A			202 VDC					
D19	115 VRMS, 47 Hz.	7.5A	7.5A			115 VRMS					
D20	115 VRMS, 63 Hz.	7.5A	7.5A	3.5	4.6	115 VRMS	open				
D21	100 VDC line reg.	0A	7A	5	3,6	108 VDC	shorted	5.00	5.10		
D22		8A	7A	4				-5.10	-5.00		
D23		80A	70A	5				5.00	5.10		
D24	100 VDC line reg.	80A	70A	4		108 VDC		-5.10	-5.00		
D25	202 VDC line reg.	0A	7A	5		202 VDC		5.00	5.10		
D26		0A	7A	4				-5.10	-5.00		
D27		80A	70A	5				5.00	5.10		
D28	202 VDC line reg.	80A	70A	4	3,6	202 VDC	shorted	-5.10	-5.00	VDC	15

-See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 202640-20T01:52Z
Check the source to verify that this is the current version before use.

1/ Two numbers in one block indicate that these two points shall be connected together in addition to being connected to points 1 or 2, as applicable (for example, "3,5" under column "1 to points" indicates point 3 connected to point 5 connected to point 1).

2/ See figure 15, note 6 for connection of input power. The root mean square (RMS) voltage is measured line-to-line.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

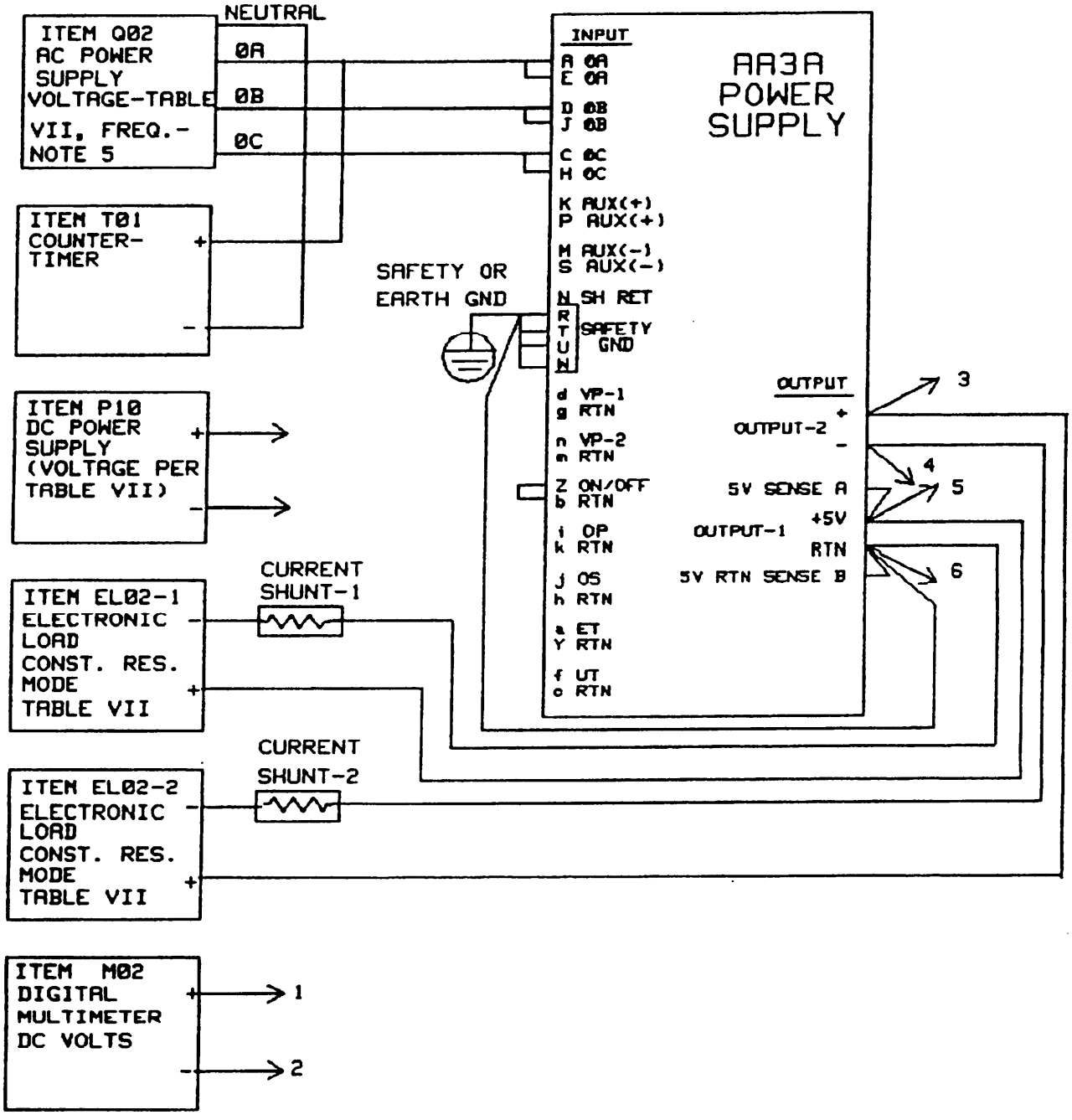


FIGURE 15. Line regulation test circuit for table VII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. The frequency of item Q02 shall be 60 ± 1 Hz for test conditions D1-D12, 47 Hz ± 0.3 Hz for test condition D19, and 63 ± 0.3 Hz for test condition D20.

6. Where VRMS input voltage is specified (test conditions D1-D12, D19, and D20), item Q02 connected as shown shall be used. For test conditions D13-D16 and D21-D28, disconnect item Q02 and connect item P10 + to input contacts A and E, and - to input contacts D and J. For test conditions D17 and D18, connect item P10 + to input contacts K and P, and - to input contacts M and S. Only one power source shall be connected at any one time.

FIGURE 15. Line regulation test circuit for table VII - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE VIII. Low input voltage and loss of phase test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
E	Low input voltage protection	5.00	5.10	VDC	16
F1	Loss of phase A protection	5.00	5.10	VDC	16
F2	Loss of phase B protection	5.00	5.10	VDC	16
F3	Loss of phase C protection	5.00	5.10	VDC	16

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

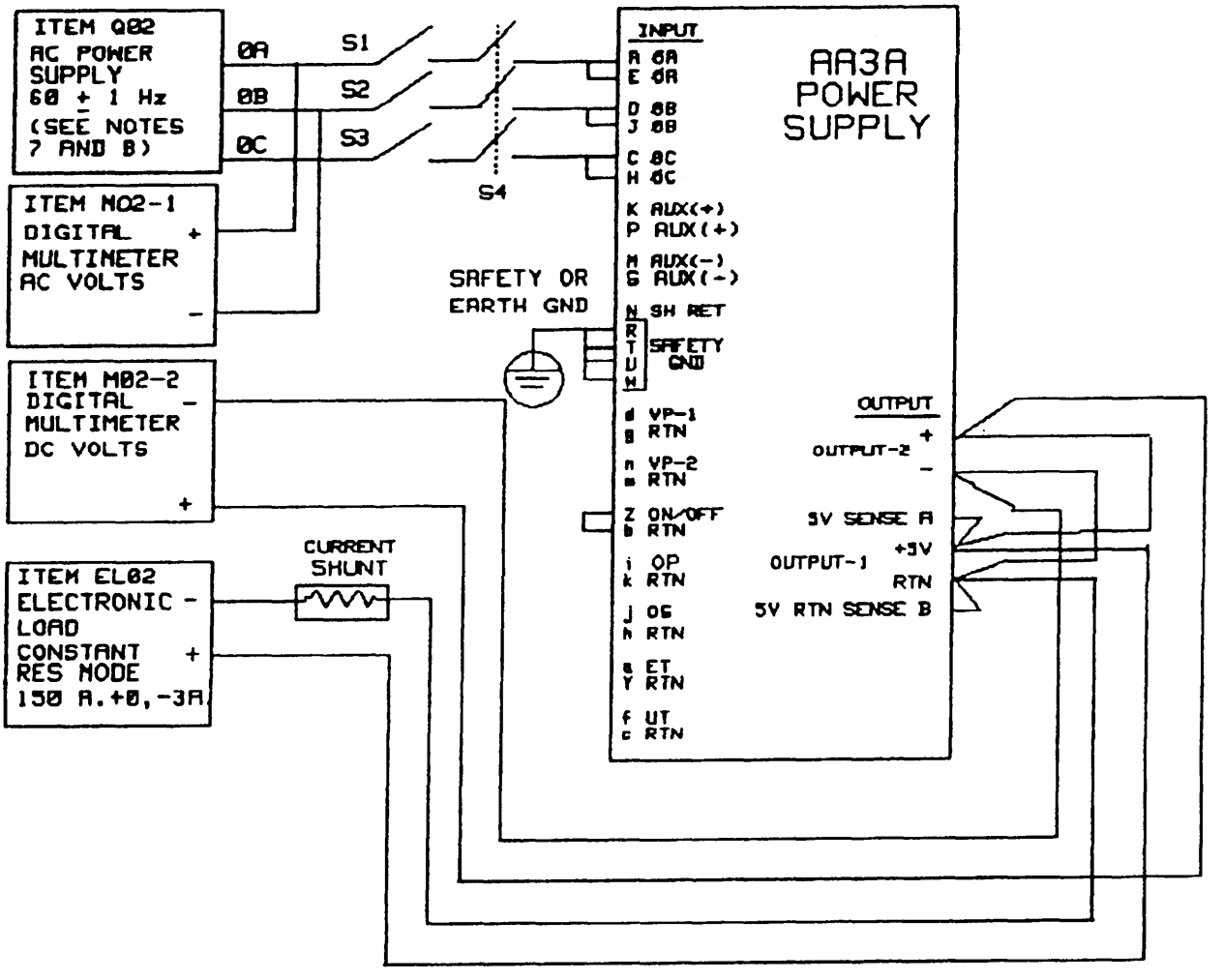


FIGURE 16. Low input voltage protection and loss of phase protection test circuit for table VIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. S1 = S2 = S3 = single pole single throw (spst), 160 VRMS, 15 amperes switch.

6. S4 = three pole single throw (3 pst), 160 VRMS, 15 amperes switch.

7. For test condition E, connect the circuits as shown with switches S1, S2, S3, and S4 closed. The item Q02 voltage shall be monitored with item M02-1. Open switch S4 and adjust the item Q02 voltage to 0 VRMS. Close switch S4 and slowly (less than 5 volts per second) increase the item Q02 voltage to 115 VRMS $\pm$ 2 percent line-to-line, then decrease it to 0 VRMS at the same rate. Open switch S4, and adjust the item Q02 voltage to 115 VRMS $\pm$ 2 percent line-to-line. Close switch S4. The item M02-2 voltage reading shall meet the test condition E limits.

8. Test conditions F1, F2 and F3 shall be performed as follows:

(a) For test condition F1, connect the circuit as shown with switches S1, S2, S3, and S4 closed. Open switch S1 for a minimum of 5 seconds, then close switch S1. The item M02-2 voltage reading shall meet the test condition F1 limits. The power supply is not required to meet the regulation limits during the loss of phase period.

(b) For test condition F2, repeat (a), except use switch S2 and test condition F2 limits.

(c) For test condition F3, repeat (a), except use switch S3 and test condition F3 limits.

FIGURE 16. Low input voltage protection and loss of phase protection test circuit for table VIII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE IX. Efficiency, power factor, and input current balance test.

Test cond	Parameter	VP1 input pin d to pin g	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			Min	Max		
G1	Efficiency, 5.05 V, 150 A, 1/ 2/	Open	70	100	%	17
G2	Efficiency, 5.05 V, 15 A, 1/ 2/	Open	60	100	%	17
G3	Efficiency, 5.25 V, 150 A, 2/	Closed	70	100	%	17
G4	Efficiency, 5.05 V, 80A; -5.05 V, 70 A 1/ 2/	Open	70	100	%	18
H	Power factor 3/	Open	0.8	1.0	-	17
J1	Input current balance 4/	Open	-5	5	%	17
J2	Input current balance 4/	Open	-5	5	%	17
J3	Input current balance 4	Open	-5	5	%	17

- See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

1/ At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input shown for test conditions G1, G2, and G4.

2/ Efficiency is calculated as follows for test conditions G1, G2, and G3.

$$\text{Efficiency} = \frac{(\text{Vo})(\text{Io})}{(\text{Item S03-1 watts}) + (\text{Item S03-2 watts})},$$

or for DC input $\text{Efficiency} = \frac{(\text{Vo})(\text{Io})}{(\text{Vin})(\text{Iin})}$,

where Vin is the input voltage and is measured with an item M02 multimeter and Iin is the input current and is measured with an item S03 wattmeter. The meters shall measure true RMS. Vo and Io are as defined in figure 17.

For test condition G4,

$$\text{Efficiency} = \frac{(\text{V1})(\text{I1}) + (\text{V2})(\text{I2})}{(\text{Item S03-1 watts}) + (\text{Item S03-2 watts})}.$$

or for DC input $\text{Efficiency} = \frac{(\text{V1})(\text{I1}) + (\text{V2})(\text{I2})}{(\text{Vin})(\text{Iin})}$.

Measure Vin with an item M02 digital multimeter and Iin with an item S03. V1, V2, I1 and I2 are as defined in figure 18.

3/ Power factor is calculated as follows:

$$\text{power factor} = \frac{(\text{Item S03-1 watts}) + (\text{Item S03-2 watts})}{\left|\overline{\text{Va}}^2 \times \text{Ia}^2 + \text{Vb}^2 \times \text{Ib}^2 + \text{Vc}^2 \times \overline{\text{Ic}}^2\right|^{1/2}}.$$

where Ia, Ib, and Ic are measured with a Pearson 411 current transformer and an item M02 multimeter, and Va, Vb, and Vc are line-to-line voltages measured with an item M02 multimeter as shown in figure 17. The meters shall measure true RMS.

4/ Current balance is calculated as follows:

$$J1 = \frac{(\text{Ia}-\text{Ib}) \times 100 \text{ percent}}{\text{Ia}},$$

$$J2 = \frac{(\text{Ia}-\text{Ic}) \times 100 \text{ percent}}{\text{Ia}}, \qquad J3 = \frac{(\text{Ib}-\text{Ic}) \times 100 \text{ percent}}{\text{Ib}},$$

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

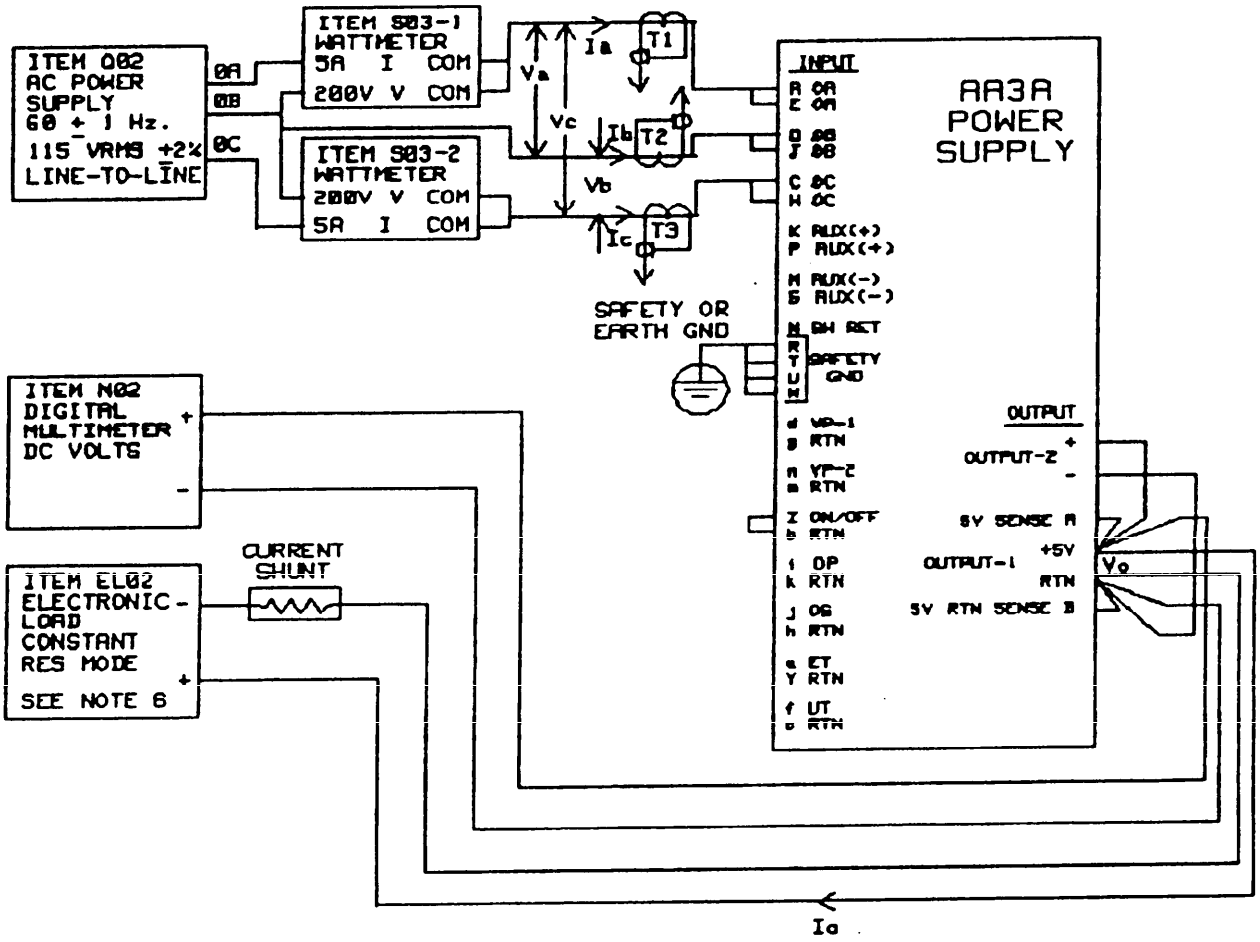


FIGURE 17. Efficiency and power factor test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. T1, T2, and T3 are Pearson Model 411 current transformers.

6. Item EL02 shall be set for 150 amperes +0 percent, -2 percent except for test condition G2, where it shall be set at 15 amperes ± 1 percent.

FIGURE 17. Efficiency and power factor test circuit for table IX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

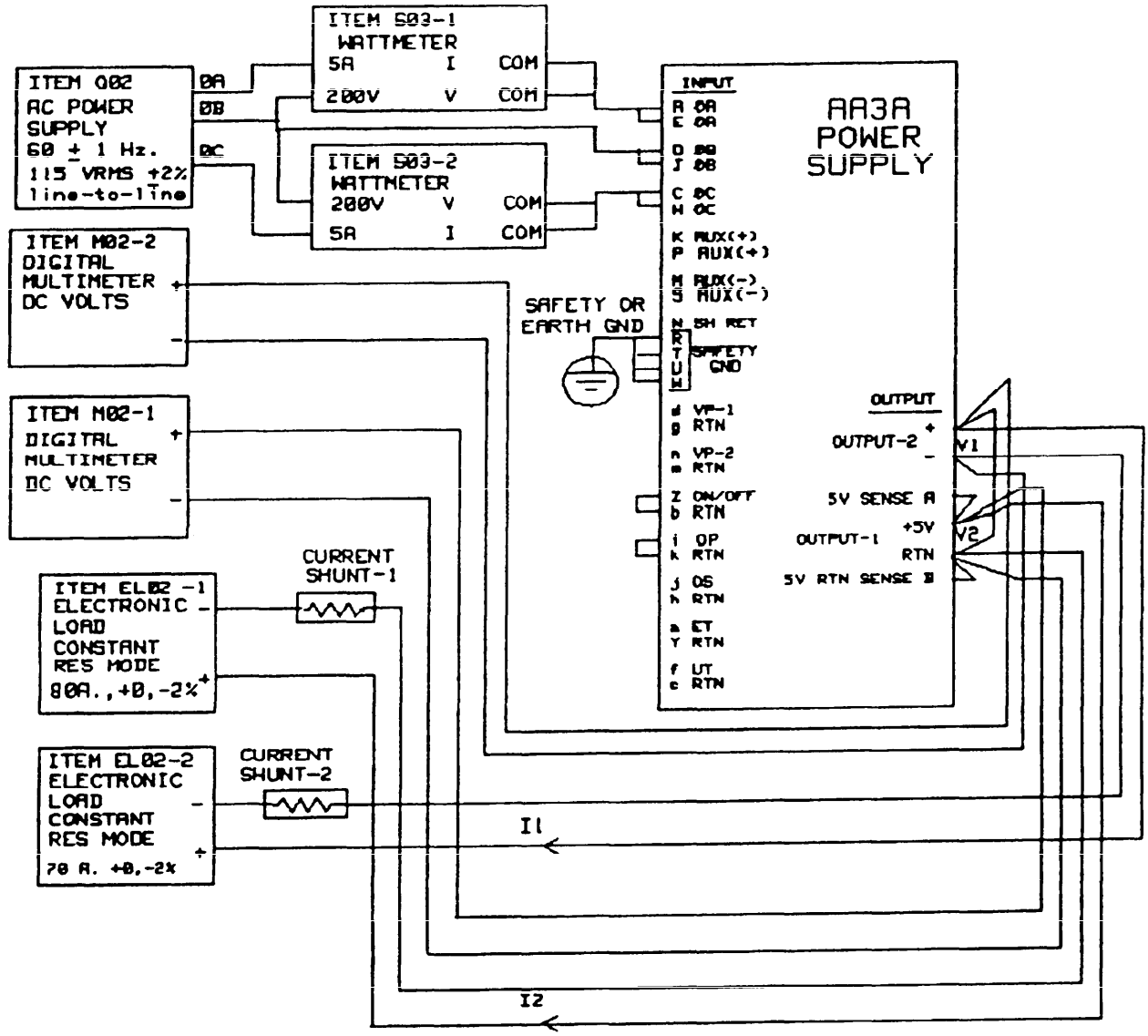


FIGURE 18. Efficiency test circuit for table IX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.
2. For cooling requirements, refer to 3.5.6.
3. Current shunts: 150A/50mV.
4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.
5. Item S03 meters are used for true power measurements.
6. Only one item M02 is required.

FIGURE 18. Efficiency test circuit for table IX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE X. Inrush current test.

Test cond	Parameter	Initial limits 25 + 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
K1	Inrush current-AC inputs	-	70	Apk	19, 20
K2	Inrush current-auxiliary input	-	70	Apk	19, 20

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

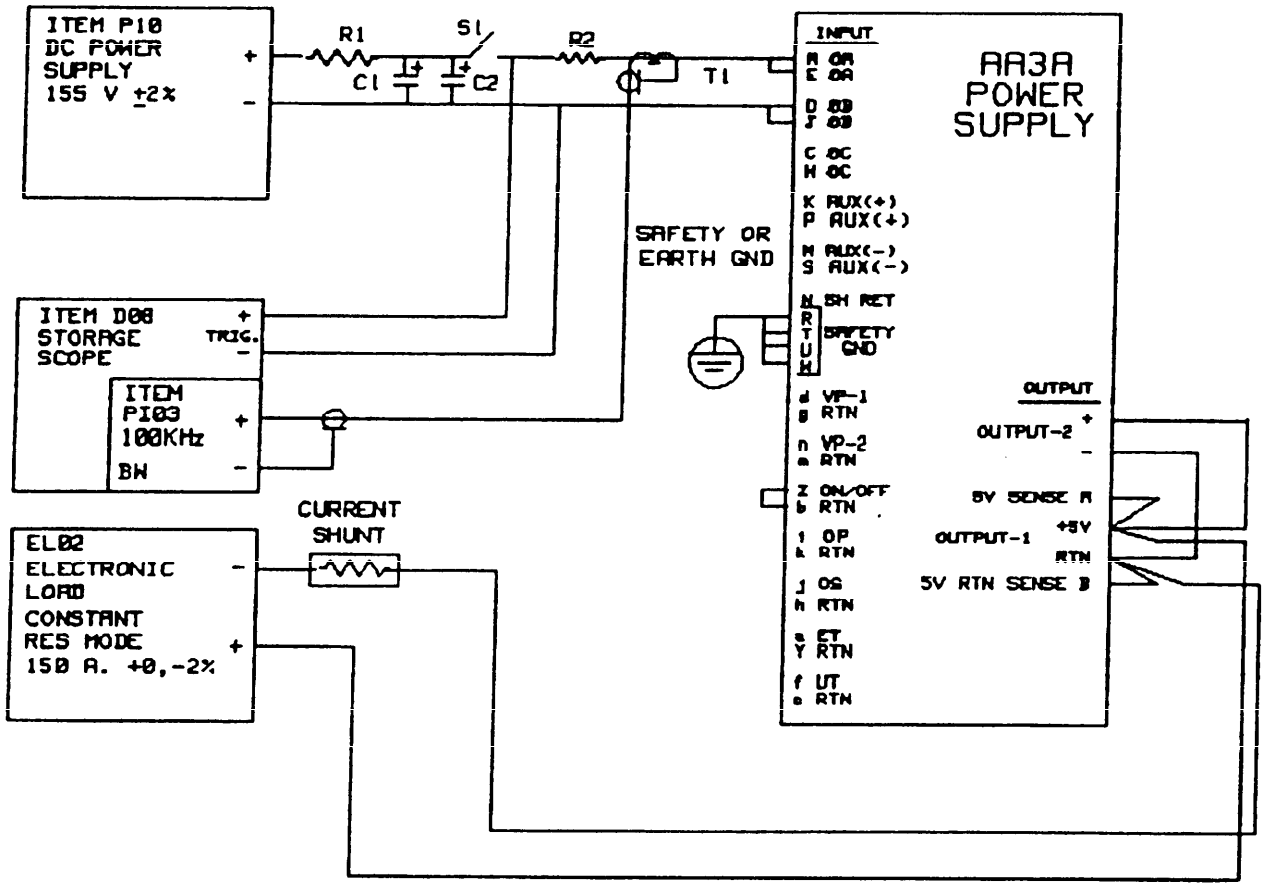


FIGURE 19. Inrush current test circuit for table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. S1 = spst, 225 VDC, 15 amperes switch.

6. T1 = Pearson Model 411 current transformer.

7. R1 = 1 Ohm $\pm$ 5 percent, 40 watt resistor.

8. R2 = 0.1 Ohm (represents maximum allowable series resistance).

9. C1 = C2 = 330 uF -15 percent, +50 percent, equivalent series resistance (ESR) of 0.4 Ohm max, 250 VDC capacitor (Sprague 80D331P250JE5D or equivalent).

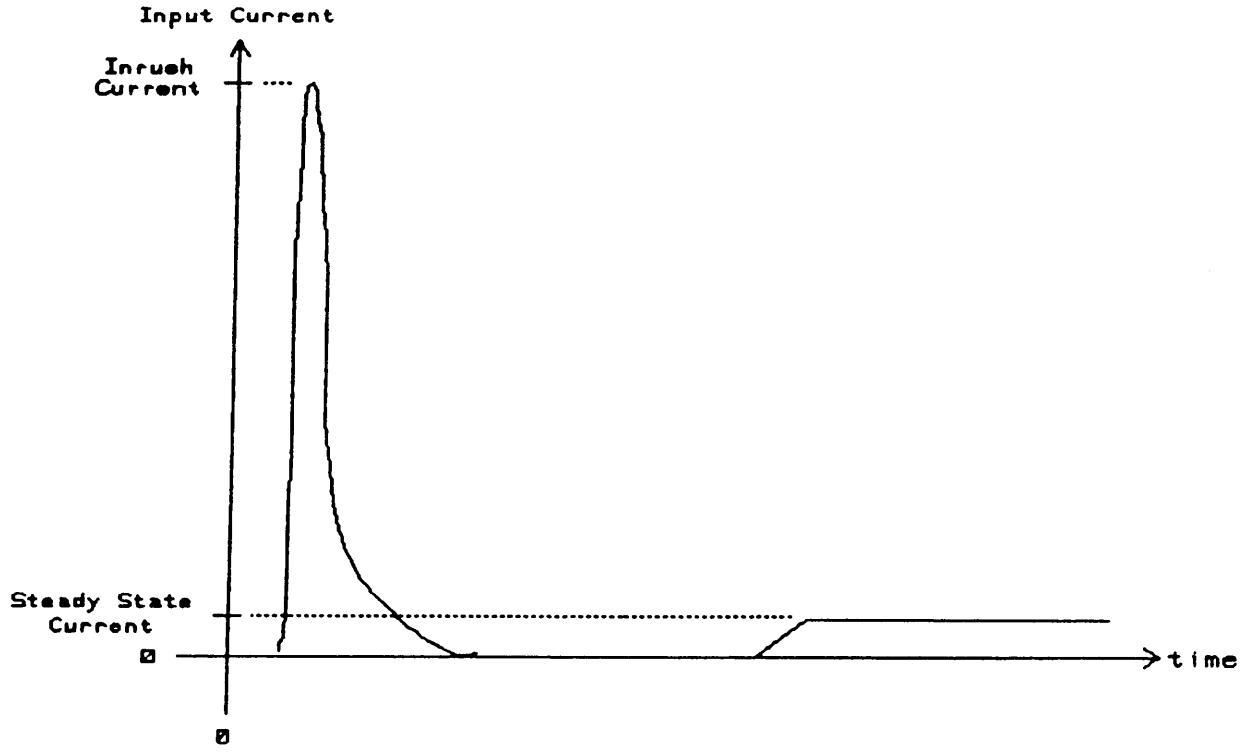
10. For test condition K1, connect item P10 to the AC inputs as shown. For test condition K2, connect item P10 to the auxiliary inputs: + to input contacts K and P, and - to input contacts M and S.

11. Apply power to the power supply under test by closing switch S1 with item P10 energized. Neither the power "OFF" period nor the power "ON" period shall be less than one second.

12. Keep the lead length between C1, C2, S1, and the power supply under test as short as possible.

FIGURE 19. Inrush current test circuit for table X - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.



NOTE:

1. The steady state current is not observable using the figure 19 test circuit, but is shown to indicate its relationship to the inrush current.

FIGURE 20. Inrush current test waveform for table X.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XI. Remote sense compensation test.

Test cond	Parameter	VP-1 Input contact d to g	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			Min	Max		
L1	Remote sense compensation, 5.05 V	open	5.00	5.10	VDC	21
L2	Remote sense compensation, 5.25 V	shorted	5.20	5.30	VDC	21

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

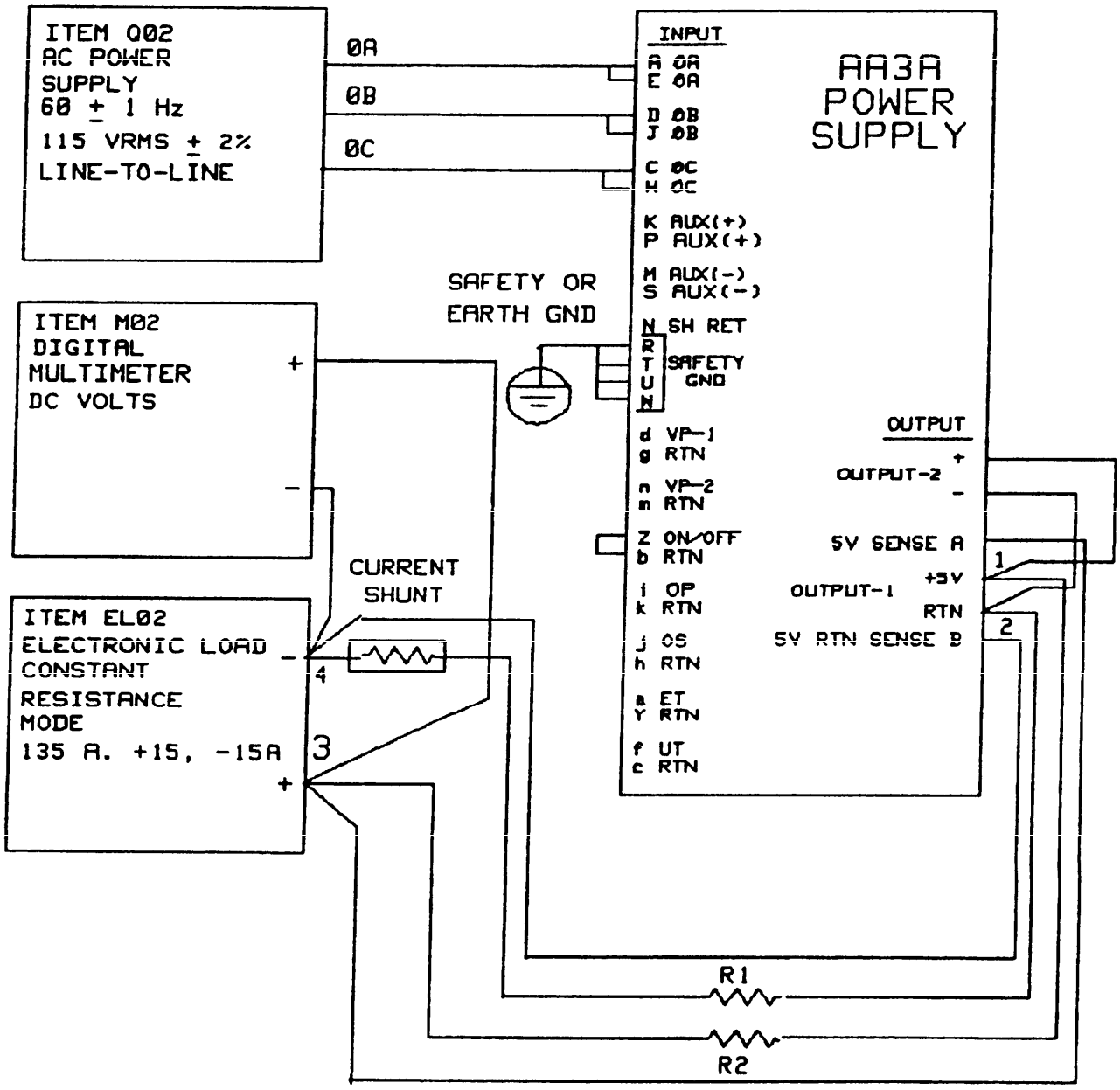


FIGURE 21. Remote sense compensation test circuit for table XI.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. For test condition L1; R1, R2, and the current shunt shall be the total resistance (including lead and contact resistance) to cause a 0.358 +.035, -.000 volt drop when the voltage at points 1 and 2 is compared to the voltage at points 3 and 4 (V1-2 minus V3-4 = 0.358 + .035, -.000 volts). The load current may be varied between the indicated values to assist in obtaining the correct voltage drop. For test condition L2, the drop across R1 plus the current shunt and R2 shall be 0.158 +.030, -.000 volts.

6. At the option of the testing facility, a DC input (item P10) of 155 VDC $\pm$ 2 percent connected between any two AC inputs may be used instead of the AC input specified.

FIGURE 21. Remote sense compensation test circuit for table XI - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T01:52Z
Check the source to verify that this is the currentversion before use.

TABLE XII. Open sense lead and reversal test.

Test cond	Parameter	Open switch	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			Min	Max		
M1	Open "+" Sense lead transient amplitude	S1	-400	100	mV	22,23
M2	Open "+" Sense lead transient time	S1	0	100	ms	
M3	Open "+" Sense lead voltage	S1	4.80	5.56	VDC	
M4	Open "-" Sense lead transient amplitude	S2	-400	100	mV	
M5	Open "-" Sense lead transient time	S2	0	100	ms	
M6	Open "-" Sense lead voltage	S2	4.80	5.56	VDC	
M7	Open "+" and "-" Sense lead voltage	S1& S2	4.80	5.56	VDC	
M8	Sense lead reversal protection	1/	5.00	5.10	VDC	22,23

1/ Both S1 and S2 are closed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

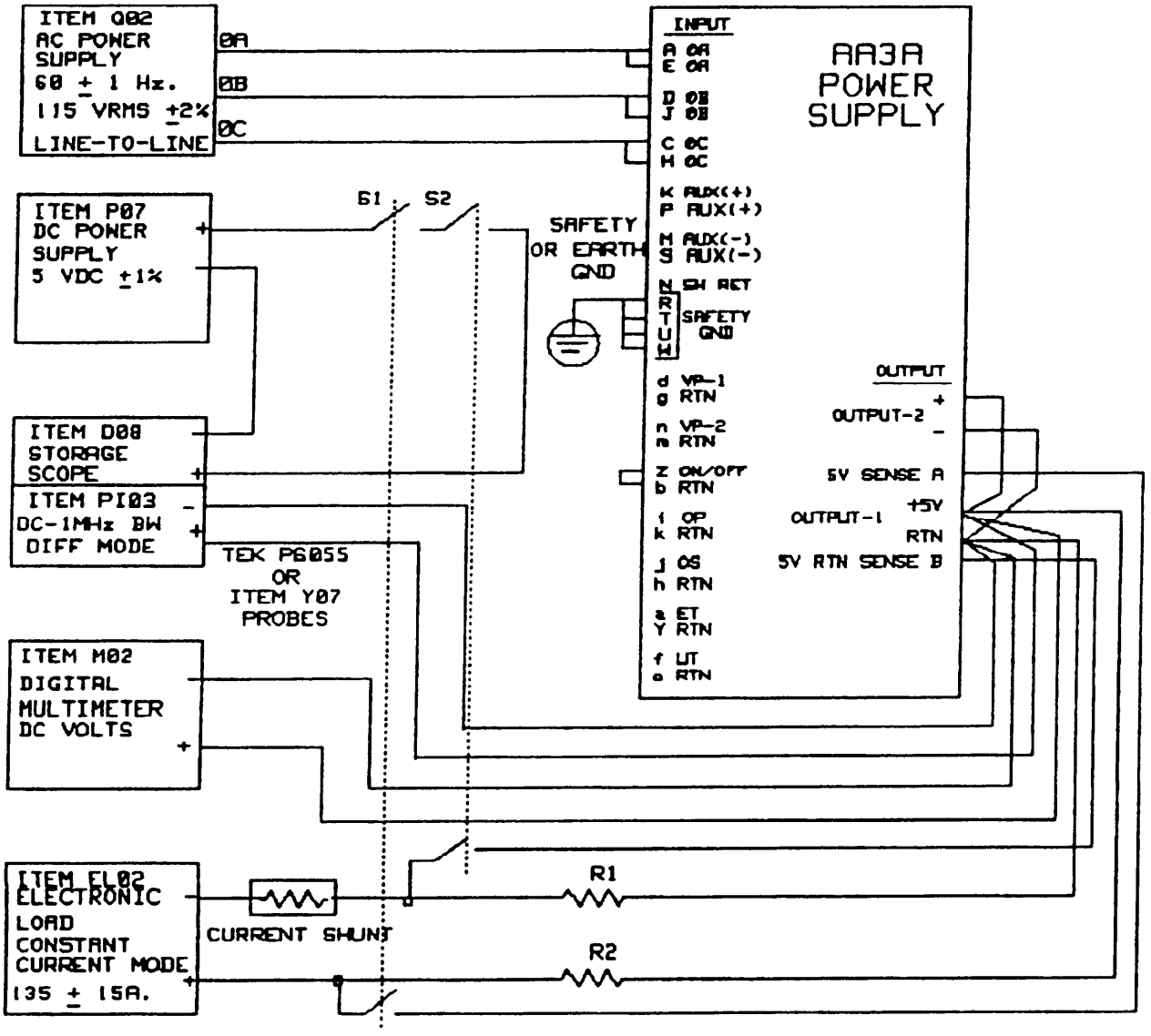


FIGURE 22. Open sense lead test circuit for table XII.

Source: https://assist.dla.mil -- Downloaded: 20260410-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. S1 = S2 = double pole single throw (dpst) switch.

6. At the option of the test facility, a DC input (item P10) of 155 volt ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

7. For test conditions M1 through M8; R1, R2 and the current shunt shall be the total resistance (including lead and contact resistance) to cause a load lead voltage drop of 0.358 + 0.035, -0.000 VDC. The load lead voltage drop is the difference between the voltage directly at the AA3A output voltage contacts and at the sense lead connection points. The load current may be varied between the indicated values to assist in obtaining the correct voltage drop.

8. Test conditions M1-M3 shall be performed as follows:

(a) Connect the circuit per figure 22, close switches S1 and S2, and apply power.

(b) Open switch S1.

(c) The maximum transient overshoot or undershoot shall not exceed the limits of M1. The power supply shall completely recover to and remain within the voltage limit of M3 and within the time limit of M2 (see figure 23). Test condition M3 voltage shall be measured with item MO2.

9. For test conditions M4-M6, repeat the procedure of note 6, except S2 shall be opened while S1 remains closed.

10. For test condition M7, open both S1 and S2 and measure the steady-state voltage with item MO2.

11. For test condition M8, reverse the sense leads, (that is, connect 5V sense to 5V RTN and RTN sense to 5V out) and leave connected for a minimum of 5 seconds. Then connect correctly and

FIGURE 22. Open sense lead test circuit for table XII - Continued.

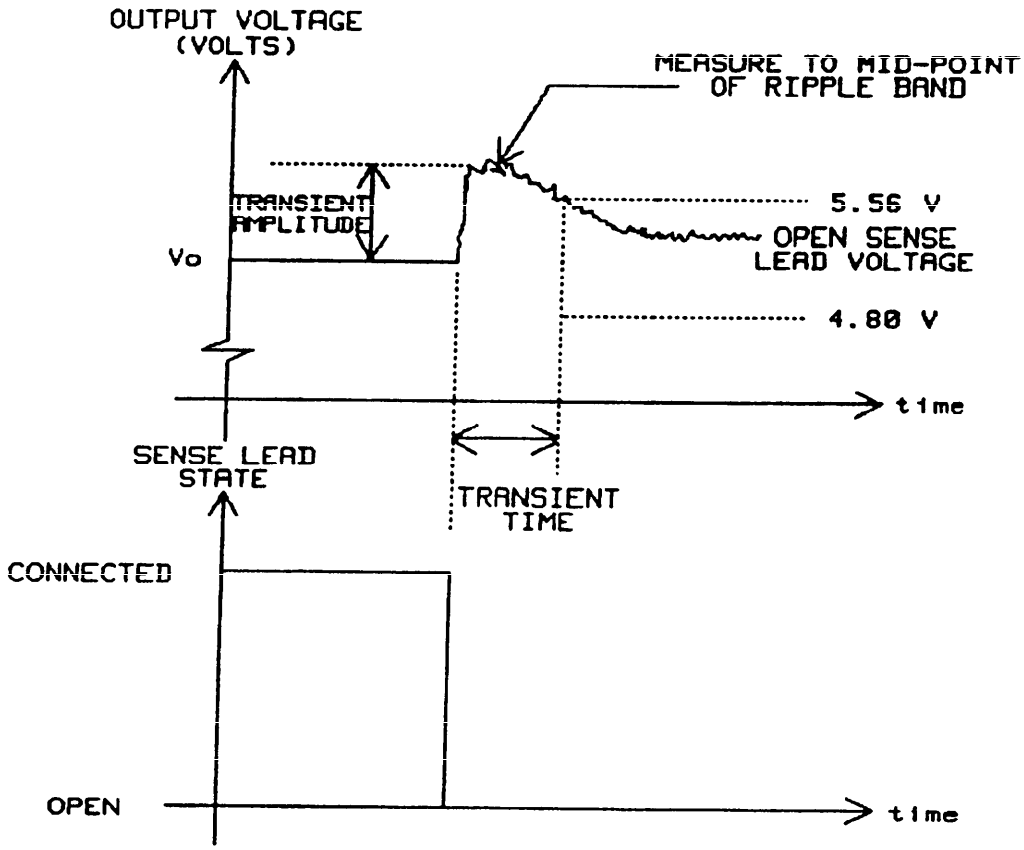
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

measure the output voltage, with item M02 connected across the item EL02 + and - terminals. Remove power (turn item Q02 off) each time before changing the sense lead connection.

FIGURE 22. Open sense lead test circuit for table XII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.



NOTES:

1. Transient amplitude is defined as the maximum change in voltage after opening the sense lead.

2. Transient time will be zero if the voltage stays within the 4.80 volt to 5.56 volt limits.

FIGURE 23. Open sense lead test definitions for table XII.

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T01:52Z
Check the source to verify that this is the currentversion before use.

TABLE XIII. Overload and short circuit protection test.

Test cond	Parameter	Measurement device	Connect 1 to	Connect 2 to	Initial limits 25 ± 3°C mounting surface temperature Min	Initial limits 25 ± 3°C mounting surface temperature Max	Unit	Test Fig
N1	Overload protection, 5 V, 150 A	shunt & V04	-	-	150	187.5	ADC	24
N2	Overload protection, 5 V, 80 A	shunt & V04	-	-	80	100	ADC	25
N3	Overload protection, -5 V, 70 A	shunt & V04	-	-	70	87.5	ADC	25
N4	Short circuit recovery, 5 V, 150 A	Item D08	-	-	0	100	ms	24
N5	Short circuit overshoot, 5 V, 150 A	Item D08	-	-	-253	253	mV	24
N6	Short circuit recovery, 5 V, 80 A	Item D08	5	6	0	100	ms	25
N7	Short circuit overshoot, 5 V, 80 A	Item D08	5	6	-253	253	mV	25
N8	Short circuit recovery, -5 V, 70 A	Item D08	3	4	0	100	ms	25
N9	Short circuit overshoot, -5 V, 70 A	Item D08	3	4	-253	253	mV	25
N10	Short circuit current, 150 A	shunt & V04	-	-	0	187.5	ADC	24
N11	Short circuit current, 80 A	shunt & V04	5	6	0	100	ADC	25
N12	Short circuit current, 70 A	shunt & V04	3	4	0	87.5	ADC	25

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

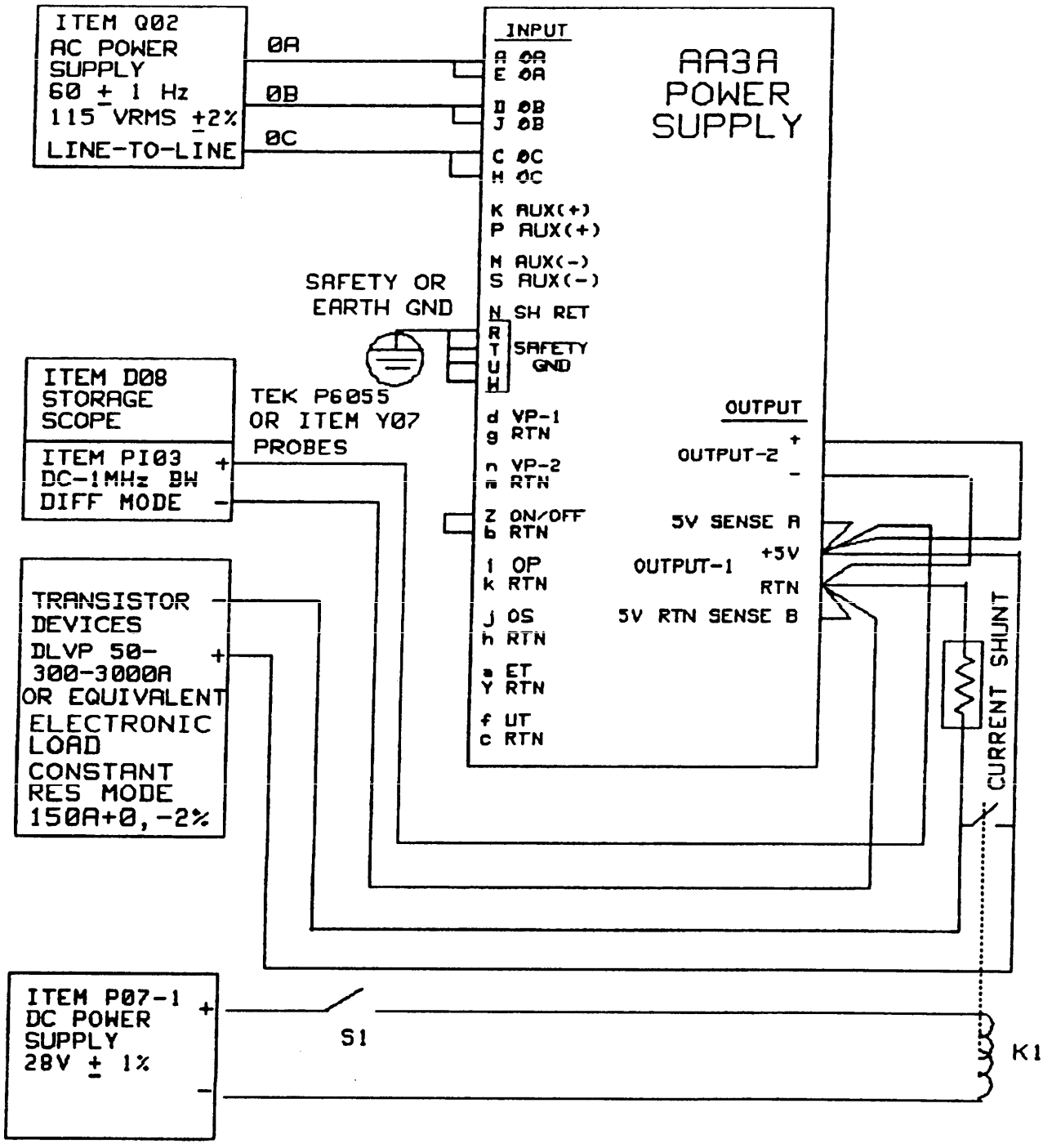


FIGURE 24. Overload and short circuit protection test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 200A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. S1 = spst switch.

6. K1 = Cutler Hammer MS24185D1 or equivalent contactor.

7. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

8. For test condition N1: Slowly increase the load current, as indicated by the shunt voltage, until the maximum current indication is obtained (this may be current saturation or prior to current shutdown). This maximum current shall fall within the specified limits. If the current exceeds 190 amperes, stop the test and do not perform short circuit tests on the failing output.

9. Keep the leads between the power supply under test and the electronic load as short as possible to minimize lead resistance.

10. Test conditions N4, N5, and N10 shall be performed as follows:

(a) Connect the circuit as shown and energize the power supply under test.

(b) Close the contactor K1 by closing S1. Short the 5V output for a minimum of 10 seconds. The short circuit current as monitored by the current shunt shall meet the limit of test condition N10.

(c) Remove the short and store the resulting transient on the storage scope (see figure 26). The power supply shall automatically recover within the regulation limits in the time limits of N4. The maximum overshoot after removal of the short shall meet the limits of N5.

FIGURE 24. Overload and short circuit protection test circuit for table XIII - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

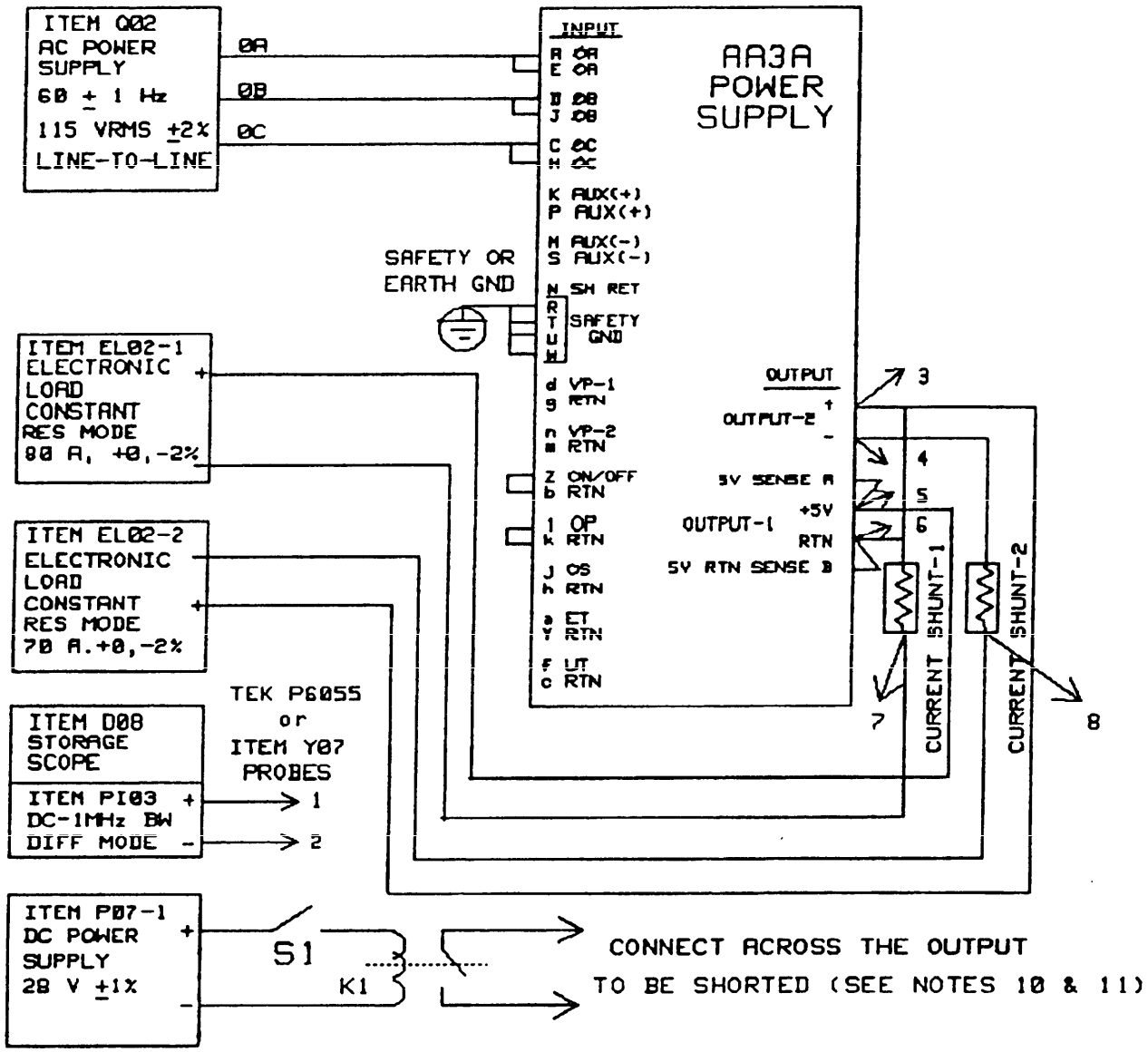


FIGURE 25. Overload and short circuit protection test circuit for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. S1 = spst switch.

6. K1 = Cutler Hammer MS24185D1 or equivalent contactor.

7. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC shown.

8. For test conditions N2-N3: Slowly increase the load current, as indicated by the shunt voltage, until a maximum current indication is obtained (this may be current saturation or prior to current shutdown). This maximum current shall fall within the specified limits. If the current exceeds 103 amperes for test condition N2 or 90 amperes for test condition N3, stop the test and do not perform short circuit tests on the failing outputs.

9. Keep the leads between the power supply under test and the electronic load as short as possible to minimize lead resistance.

10. Test conditions N6, N7, and N11 shall be performed as follows:

(a) Connect the circuit as shown with the contactor K1 connected between points 5 and 7 and energize the power supply under test.

(b) Close the contactor K1 by closing S1. Short the 5V output for a minimum of 10 seconds. The short circuit current as monitored by current shunt-1 shall meet the limit of test condition N11.

(c) Remove the short and store the resulting transient on the storage scope (see figure 26). The power supply shall automat-

FIGURE 25. Overload and short circuit protection test circuit for table XIII - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

ically recover within the regulation limits in the time limits of N6. The maximum overshoot after removal of the short shall meet the limits of N7.

11. For test conditions N8, N9, and N12, repeat the procedure of note 10 for the -5 volts, 70 amperes output with the contactor K1 connected between points 3 and 8.

FIGURE 25. Overload and short circuit protection test circuit for table XIII - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

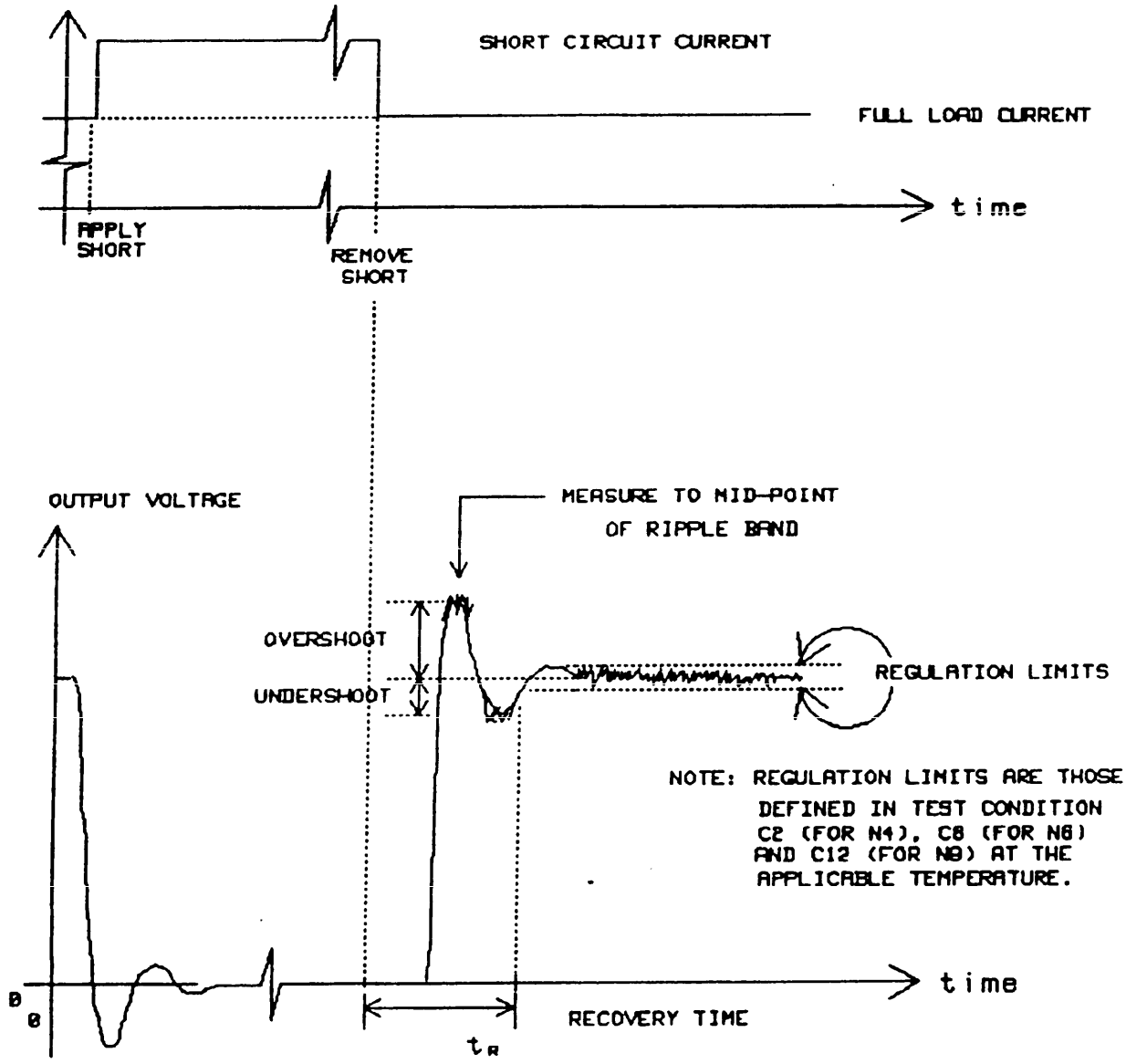


FIGURE 26. Overload and short circuit protection test definitions for table XIII.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XIV. Turn on/off transient and turn-on time test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
P1	Turn-on transient, 5 V, 150 A.	-250	250	mV	27,29
P2	Turn-on time, 5 V, 150 A.	15	250	ms	27,29
P3	Turn-off transient, 5 V, 150 A.	-250	250	mV	27,29
P4	Control turn-on transient	-250	250	mV	28,29
P5	Control turn-on time	15	250	ms	28,29
P6	Control turn-off transient	-250	250	mV	28,29

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

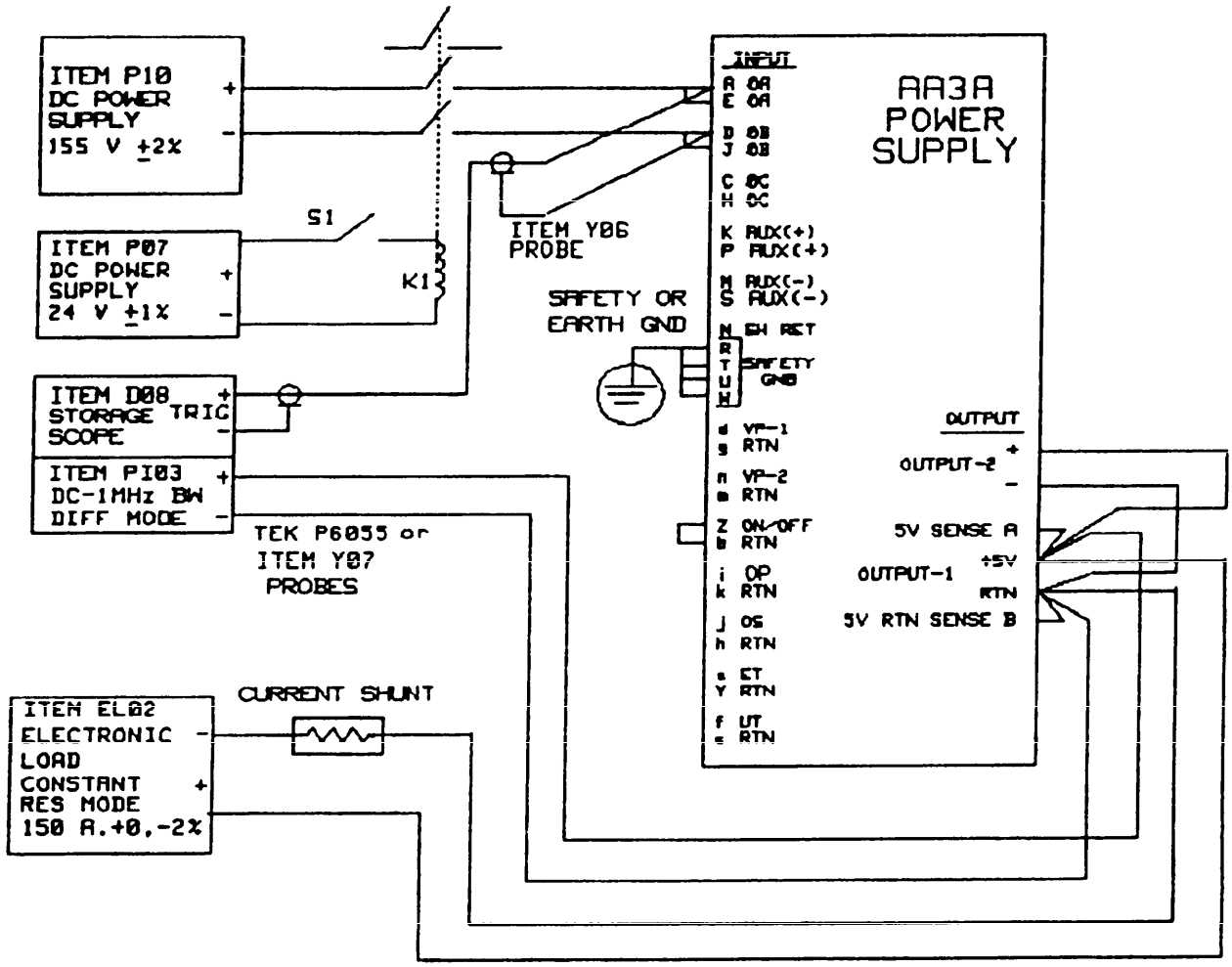


FIGURE 27. Turn on/off transient test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. S1 = spst switch.

6. K1 = 3pst mercury relay, Magnecraft WM35AAA-24D or equivalent.

7. At the option of the testing facility, an AC input (instead of the DC specified) of 3-phase, 115 VRMS ± 2 percent line-to-line, 60 ± 1 Hz can be applied at the phase A, B, and C inputs.

8. Refer to figure 29 for the definition of turn on/off transient and turn-on time. Each test shall be performed several times to ensure measurement of the highest transient, and longest times.

FIGURE 27. Turn on/off transient test circuit for table XIV - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

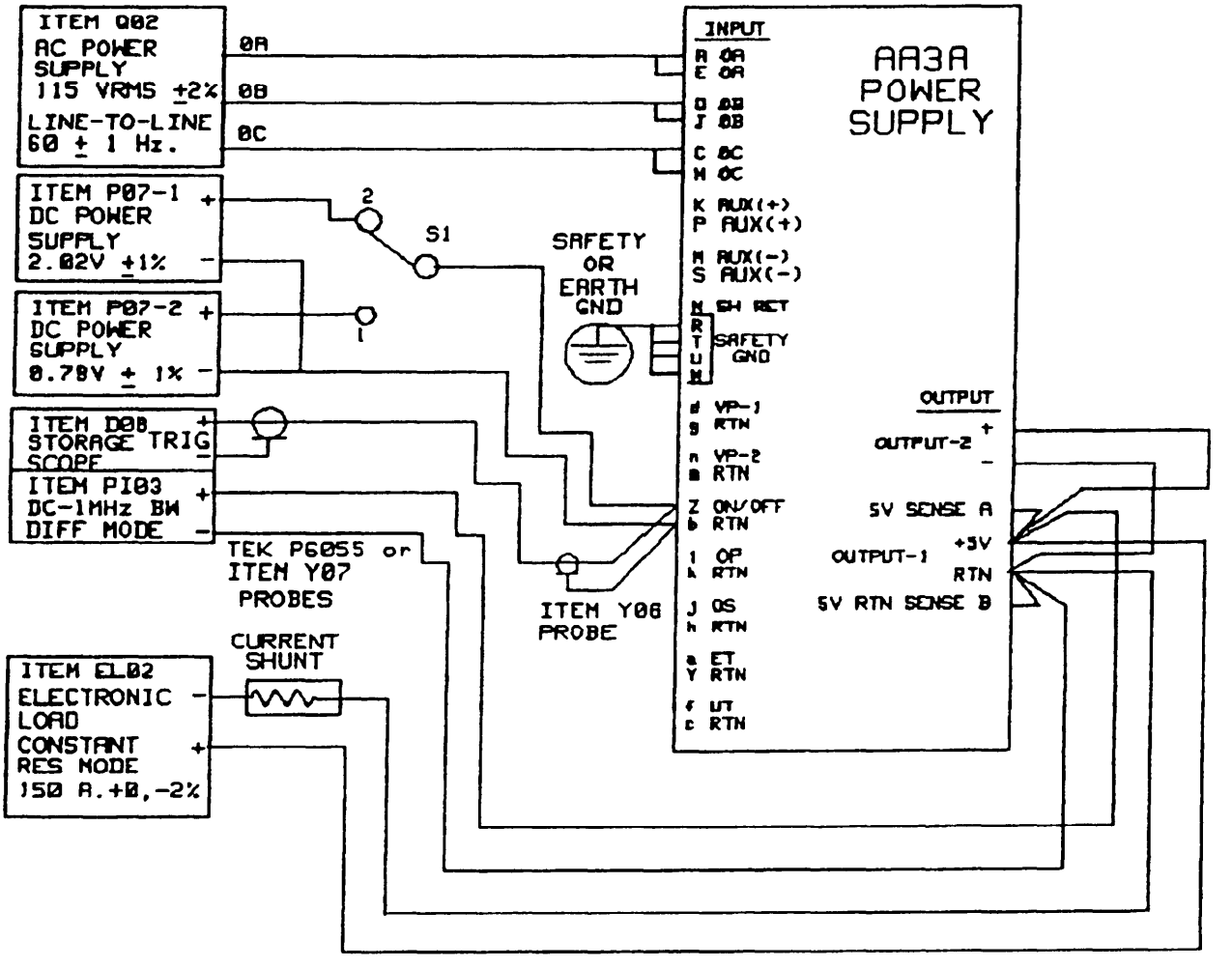


FIGURE 28. Control turn on/off transient test circuit for table XIV.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = single pole double throw (spdt) switch.

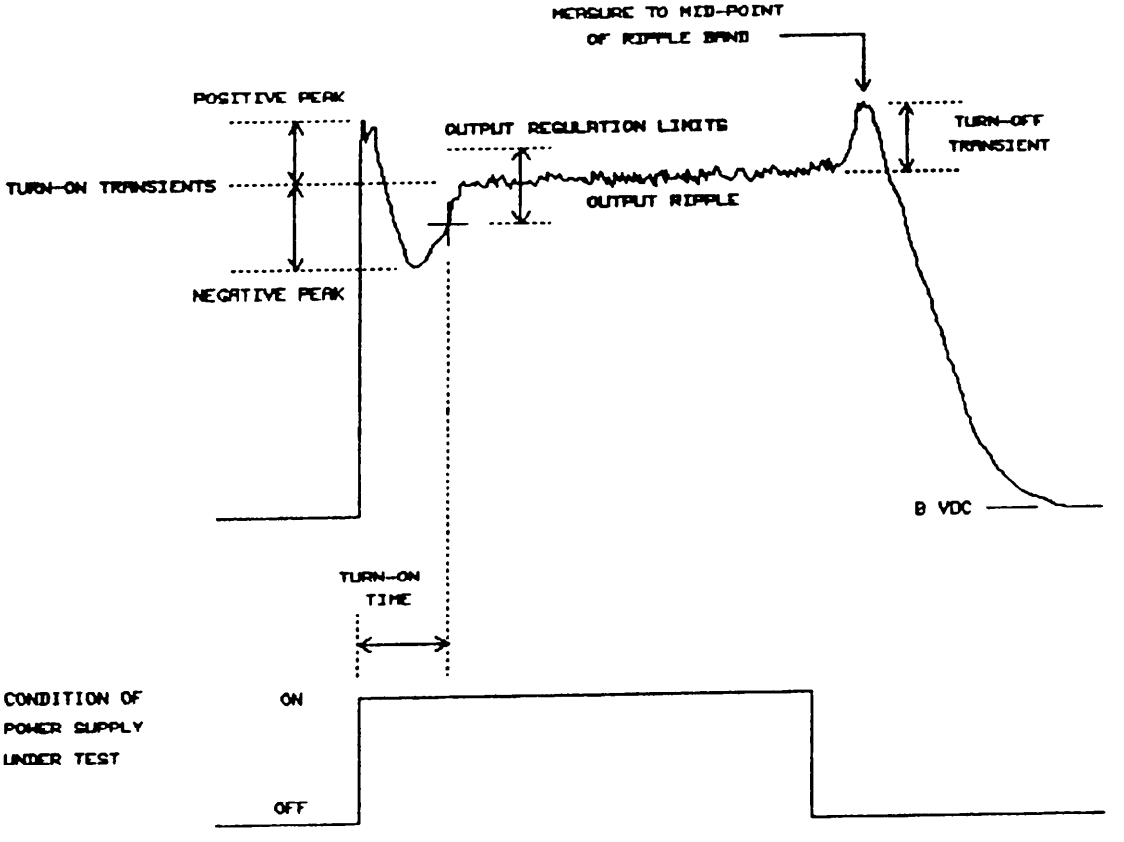
6. At the option of the testing facility, a DC input (item P10) of 155 volts ± 2 percent connected to any two AC inputs can be used instead of the AC input specified.

7. Refer to figure 29 for the definition of turn on/off transient and turn-on time. The test shall be performed several times to ensure measurement of the highest transient, and longest time.

8. The power supply is turned on when S1 is switched from position 2 to 1. The power supply is turned off when S1 is switched from position 1 to 2.

FIGURE 28. Control turn on/off transient test circuit for table XIV - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.



NOTES:

1. For turn-on transient, record the highest amplitude positive or negative peak.

2. The test condition P2 turn-on time is defined as the time from when input power is applied to the power supply until the output recovers to and remains within the regulation limits. The regulation limits shall be test condition C2 limits for +5.05 volt output (at the applicable temperature).

3. The test condition P5 control turn-on time is defined as the time from when voltage is applied to the control on/off input pins until the output recovers to and remains within the regulation limits (see note 2).

FIGURE 29. Turn-on and turn-off waveforms for table XIV.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XV. Ripple test.

Test cond	Parameter	Measurement device (Item DM)	AC input line-to-line voltage	Measurement frequency range (See 4.8.8.2) Low 1/	Measurement frequency range (See 4.8.8.2) High	Connect 1 to	Connect 2 to	Initial limits 25 ± 3°C mounting surface temperature Min	Initial limits 25 ± 3°C mounting surface temperature Max	Unit	Test fig
01	Ripple,total,150 A	Item PI03	115 VRMS ± 2%	10 Hz	1 MHz	-	-	0	65	mVp-p	30
02	Ripple,total,80 A	Item PI03			1 MHz	5	6		65		31
03	Ripple,total,70 A	Item PI03			1 MHz	3	4		65		31
04	Ripple,low frequency,150 A	Item PI03			10 kHz	-	-		50		30
05	Ripple,low frequency, 80 A	Item PI03			10 kHz	5	6		50		31
06	Ripple,low frequency,70 A	Item PI03		10 Hz	10 kHz	3	4		50		31
07	Ripple,high frequency,150 A	Item PI03		10 kHz	1 MHz	-	-		65		30
08	Ripple,high frequency, 80 A	Item PI03		10 kHz	1 MHz	5	6		65		31
09	Ripple,high frequency, 70 A	Item PI03		10 kHz	1 MHz	3	4		65		31
010	Ripple,total,150 A	Item PI01		10 Hz	20 MHz	-	-		300		30
011	Ripple,total,80 A	Item PI01				5	6		300		31
012	Ripple,total,70 A	Item PI01			20 MHz	3	4		300		31
013	Ripple,total,overload	Item PI02			5 MHz	-	-		200		30
014	Ripple,total, no load					-	-		130		30
015	Ripple,total,overload	Item PI02	115 VRMS ± 2%	10 Hz	5 MHz	5	6	0	200	mVp-p	31

-See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XV. Ripple test - Continued.

Test cond	Parameter	Measurement device (Item D08)	AC input line-to-line voltage	Measurement frequency range (3dB pt) Low 1/	Measurement frequency range (3dB pt) High	Connect 1 to	Connect 2 to	Initial limits 25 ± 3°C mounting surface temperature Min	Initial limits 25 ± 3°C mounting surface temperature Max	Unit	Test fig
Q16	Ripple, total, no load	Item PI02	115 VRMS ±2%	10 Hz	5 MHz	5	6	0	130	mVp-p	31
Q17	Ripple, total, overload					3	4	0	200		31
Q18	Ripple, total, no load		115 VRMS ±2%			2	4	0	120		21
Q19	Ripple, total, low input, 150 A		86 VRMS ±1%			-	-	0	130		30
Q20	Ripple, total, low input, 85 A					5	6	0	130		31
Q21	Ripple, total, low input, 70 A		86 VRMS ±1%			3	4	0	130		31
Q22	Ripple, total, high input, 150A		144 VRMS ±1%			-	-	0	130		30
Q23	Ripple, total, high input, 85 A					5	6	0	120		21
Q24	Ripple, total, high input 70 A	Item PI02	144 VRMS ±1%	10 Hz	5 MHz	3	4	0	130	mVp-p	31

1/ When using PI01 or PI02, use the AC coupling to obtain the 10 Hz lower 3dB point setting.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

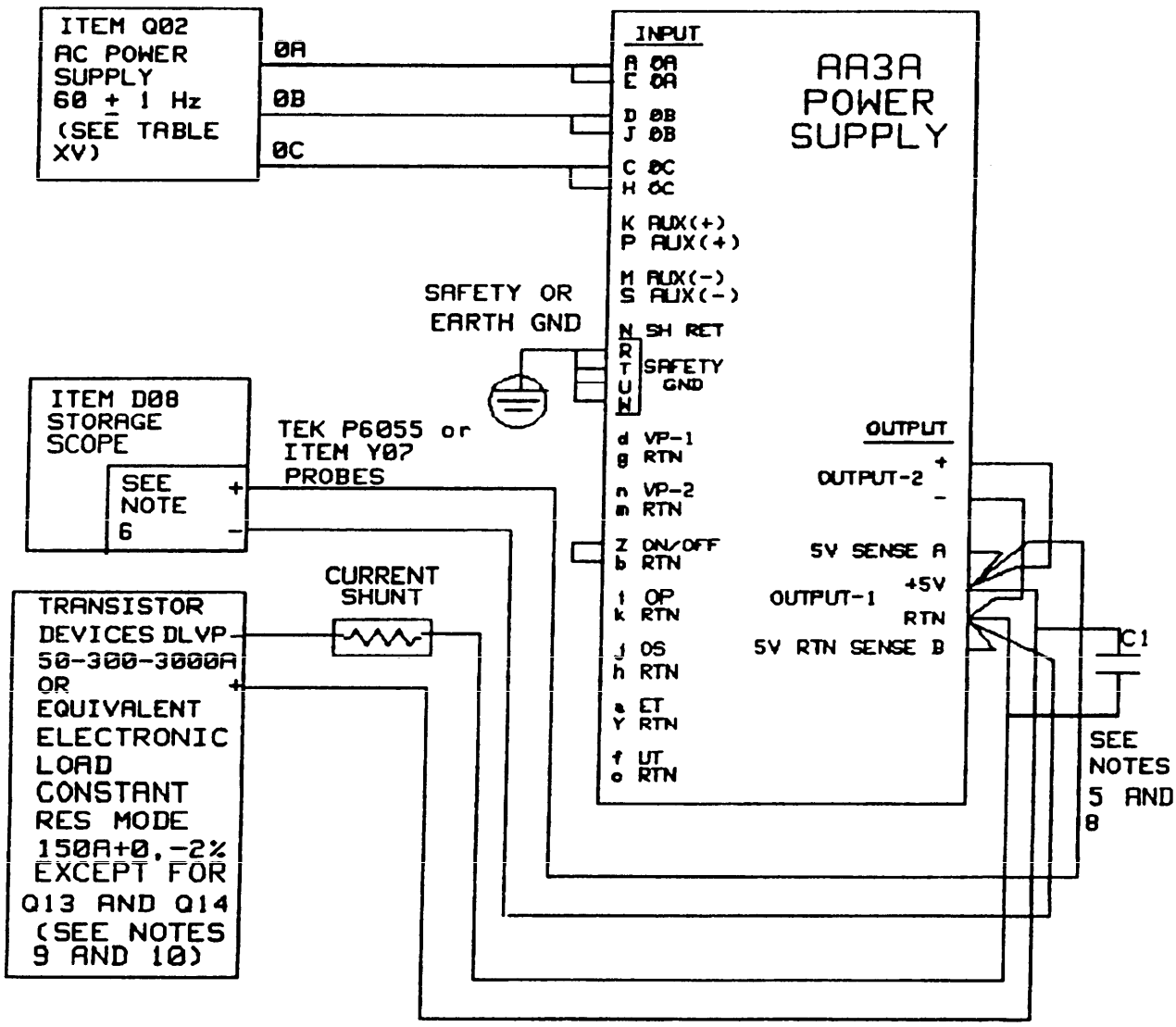


FIGURE 30. Ripple test circuit (single output) for table XV.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV (for test condition Q13, use 200A/50mV).

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. C1 = 1 uF, ± 10 percent, 50 VDC, ceramic, capacitor (use only for test condition Q10).

6. The smaller box within item D08 indicates a plug-in. See table XV for the plug-ins to be used. Use items PI02 and PI03 in the differential mode.

7. For all test conditions, record the total peak-to-peak ripple and noise observed on the oscilloscope with the frequency setting per table XV (including all spikes, hash, periodic, and aperiodic components). Use the single sweep mode. The sweep rate must be varied to obtain a maximum reading. The intensity must be turned up adequately to catch all spikes. The scope trigger should be varied to assure observation of both positive and negative peaks.

8. For test condition Q10, measurement shall be made with item D08 and item PI01 (bandwidth set to 20 megahertz (MHz)). The item Y07 shall be used so a differential measurement can be made. Capacitor C1 shall be connected within 2 inches (5.08 cm) of the scope probe tip.

9. For test condition Q13, the ripple measurement is made with the current protection circuit energized. The current on the output being measured shall be increased until one of the following things shall happen:

(a) to the point where the current saturates or begins to drop (in the case of current limiting or foldback),

(b) just prior to the point where the power supply shuts down (if the power shuts down on overcurrent),

FIGURE 30. Ripple test circuit (single output) for table XV - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) until the output voltage decreases to 95 percent of the value recorded at the 150 amperes level.

The ripple measurement shall be made at the condition which occurs first. In no case shall the current be increased beyond 187.5 amperes (125 percent).

10. For test condition Q14, adjust the electronic load to obtain 0 amperes load current.

FIGURE 30. Ripple test circuit (single output) for table XV - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

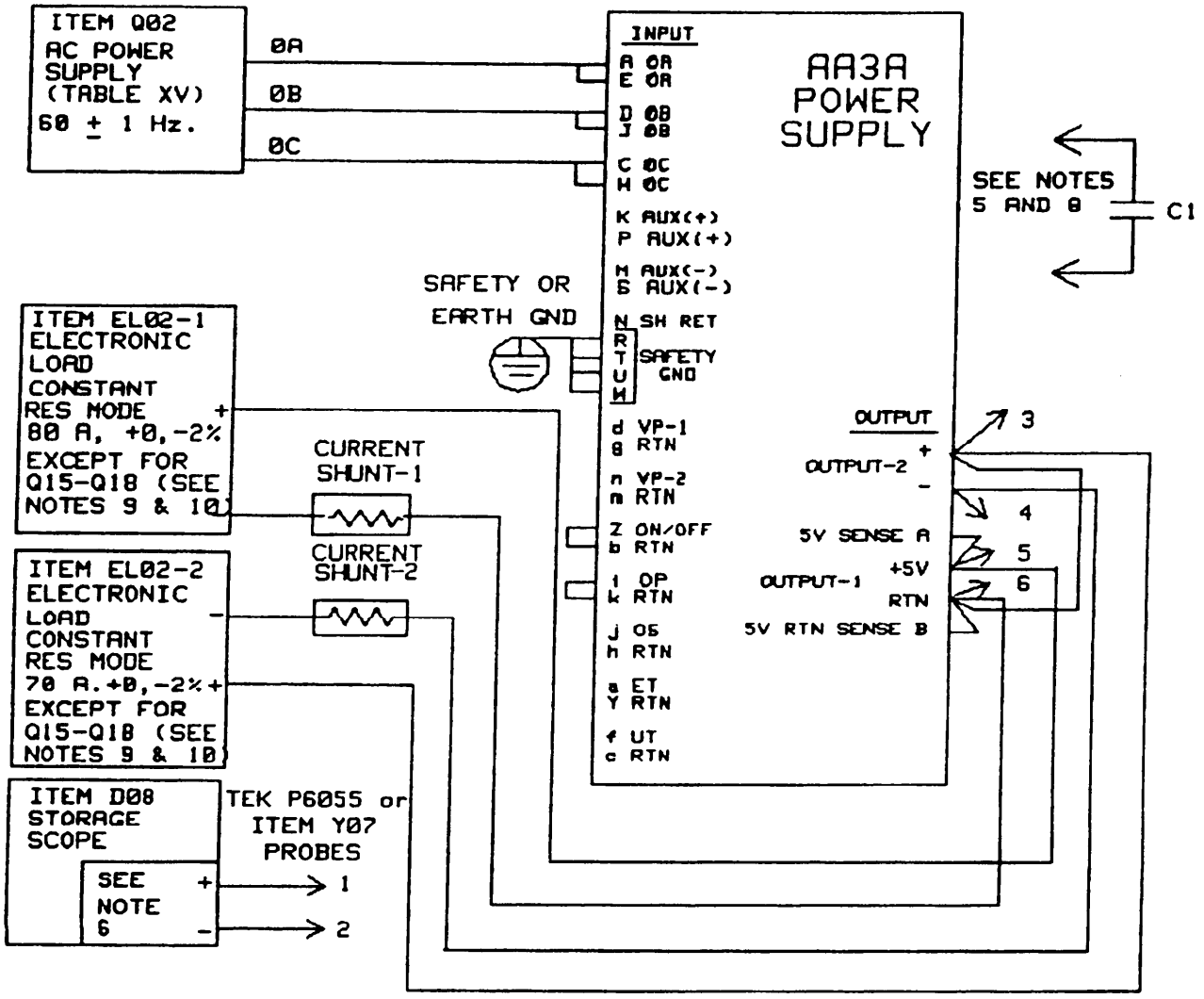


FIGURE 31. Ripple test circuit (dual output) for table XV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. C1 = 1 uF, ± 10 percent, 50 VDC, ceramic, capacitor (use only for test conditions Q11 and Q12).

6. The smaller box within item D08 indicates a plug-in. See table XV for the plug-ins to be used. Use items PI02 and PI03 in the differential mode.

7. For all test conditions, record the total peak-to-peak ripple and noise observed on the oscilloscope with the frequency setting per table XV (including all spikes, hash, periodic, and aperiodic components). Use the single sweep mode. The sweep rate must be varied to obtain a maximum reading. The intensity must be turned up adequately to catch all spikes. The scope trigger should be varied to assure observation of both positive and negative peaks.

8. For test conditions Q11 and Q12, measurement shall be made with item D08 and item PI01 (bandwidth set to 20 MHz). The item Y07 shall be used so a differential measurement can be made. Capacitor C1 shall be connected across the output being measured and within 2 inches (5.08 cm) of the scope probe tip.

9. For test conditions Q15 and Q17, the ripple measurement is made with the current protection circuit energized. The current on the output being measured shall be increased until one of the following things shall happen:

(a) to the point where the current saturates or begins to drop (in the case of current limiting or foldback),

(b) just prior to the point where the power supply shuts down (if the power supply shuts down on overcurrent),

FIGURE 31. Ripple test circuit (dual output) for table XV - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T01:52Z
Check the source to verify that this is the currentversion before use.

NOTES: - Continued.

(c) until the output voltage decreases to 95 percent of the value recorded at 80 amperes (test condition Q15) or 70 amperes (test condition Q17).

The ripple measurement shall be made at the condition which occurs first. In no case shall the current be increased beyond 100 amperes (test condition Q15) or 87.5 amperes (test condition Q17).

10. For test conditions Q16 and Q18, adjust items EL02-1 and EL02-2 to obtain 0 amperes load current.

FIGURE 31. Ripple test circuit (dual output) for table XV - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

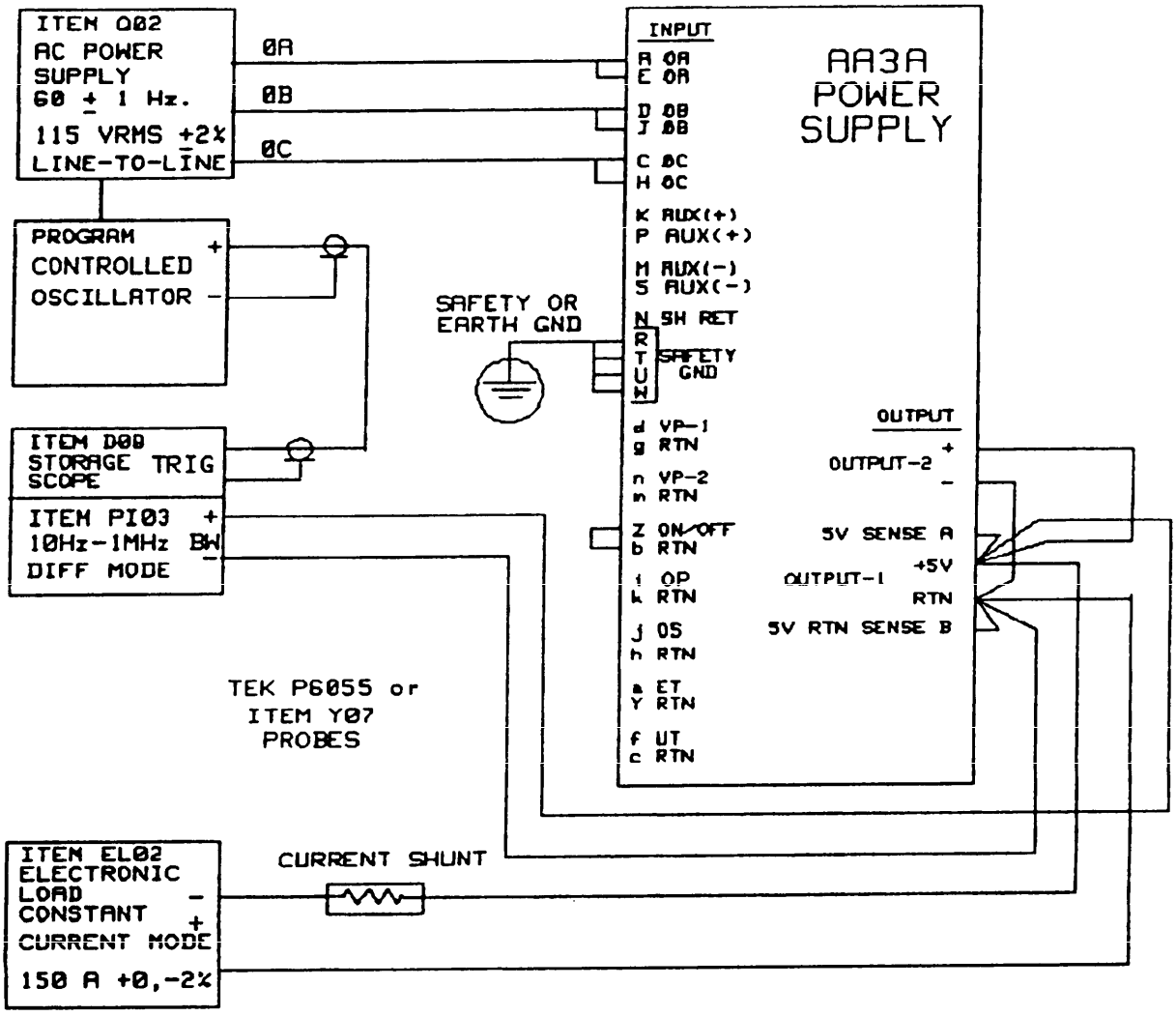
TABLE XVI. Dynamic line regulation test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
R1	Dynamic line regulation, +21% 1/	0	152	mV	32
R2	Dynamic line ringing (2nd half cycle) +21% 2/		76		
R3	Dynamic line regulation, -21% 1/		152		
R4	Dynamic line ringing (2nd half cycle) -21% 2/	0	76	mV	32

1/ +21 percent- from 115 VRMS to 139 VRMS; -21 percent- from 115 VRMS to 91 VRMS. See figure 33 for definitions.

2/ Any additional overshoot or ringing beyond the 2nd half cycle shall be within the total ripple envelope of test condition 01 peak-to-peak limits.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.



NOTES:

1. See table II for output connector contact assignment.

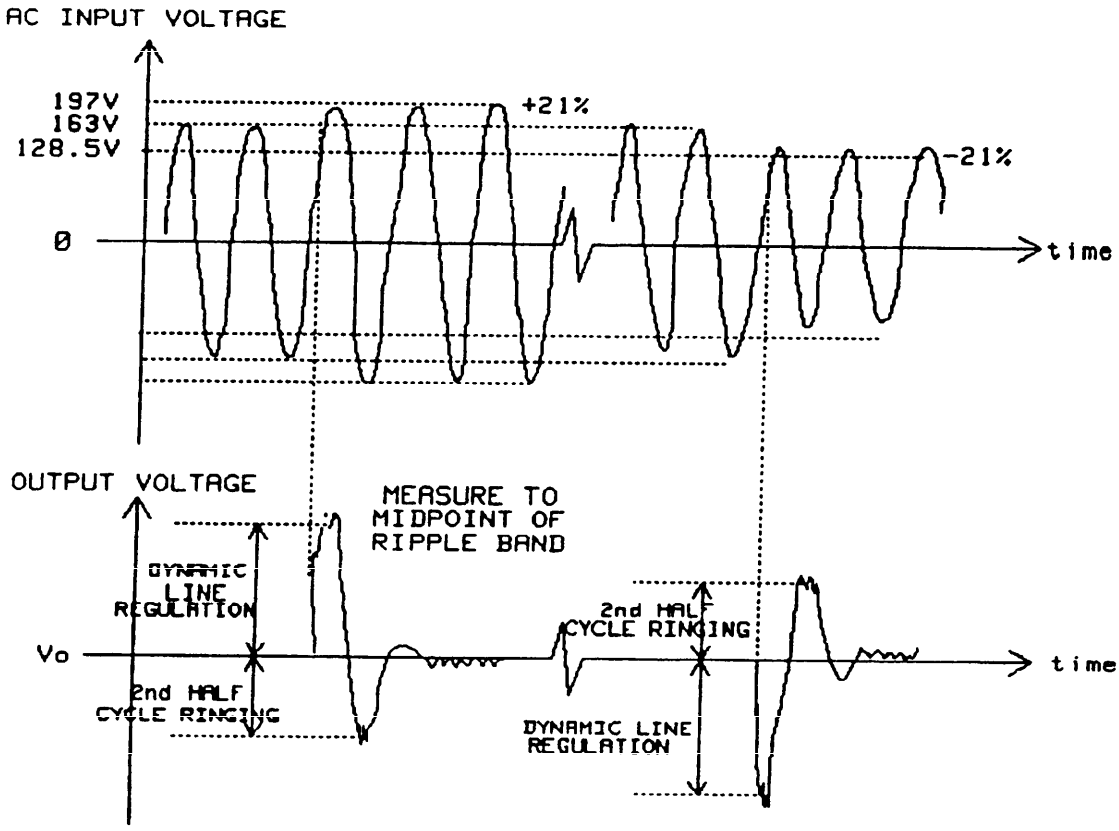
2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

FIGURE 32. Dynamic line regulation test circuit for table XVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.



NOTES:

1. The initial overshoot may be positive or negative.

2. Transition time for the ac amplitude shall be between 50 and 500 microseconds.

3. The ac input amplitude shall remain at each amplitude for a minimum of 2 seconds.

4. Nominal voltage is 115 VRMS (163 Vpk), +21 percent = 139 VRMS (197 Vpk), -21 percent = 91 VRMS (128.5 Vpk).

FIGURE 33. Dynamic line regulation test definitions for table XVI.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic load regulation tests.

Test cond	Parameter	Input pins j and k	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
S1	80% to 90%, 5.05V, sgl out, regulation	open	0	152	mV	34,37
S2	80% to 90%, 5.05V, ringing 1/			76		
S3	90% to 80%, 5.05V, sgl out, regulation			152		
S4	90% to 80%, 5.05V, ringing 1/			76		
S5	80% to 90%, 5.05V, sgl out, full cap load, regulation			152		
S6	80% to 90%, 5.05V, full cap load, ringing 1/			76		
S7	90% to 80%, 5.05V, sgl out, full cap load, regulation			152		
S8	90% to 80%, 5.05V, full cap load, ringing 1/			76		34,37
S9	80% to 90%, 5.25V, sgl out, max sense lead, regulation			162		35,37
S10	80% to 90%, 5.25V, max sense lead, ringing 1/			81		
S11	90% to 80%, 5.25V, sgl out, max sense lead, regulation			162		
S12	90% to 80%, 5.25V, max sense lead, ringing 1/	open		81		
S13	80% to 90%, 5.25V, dual out, max sense lead, regulation	short	0	162		
S14	80% to 90%, 5.25V, max sense lead, ringing, 1/	short	0	81		
S15	90% to 80%, 5.25V, dual out, max sense lead, regulation	short	0	162	mV	35,37

-See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XVII. Dynamic load regulation tests - Continued.

Test cond	Parameter	Input pins j and k	Initial limits 25 ± 5°C mounting surface temperature Min	Initial limits 25 ± 5°C mounting surface temperature Max	Unit	Test fig
S16	80% to 80%, 5.25V, max sense lead, ringing, 1/	short	0	81	mV	35,37
S17	80% to 90%, 5.05V, dual out, regulation			152		36,37
S18	80% to 90%, 5.05V, ringing 1/			76		
S19	90% to 80%, 5.05V, dual out, regulation			152		
S20	90% to 80%, 5.05V, ringing 1/			76		
S21	80% to 90%, -5.05V, dual out, regulation			152		
S22	80% to 90%, -5.05V, ringing 1/			76		
S23	90% to 80%, -5.05V, dual out, regulation 1/			152		
S24	90% to 80%, -5.05V, ringing 1/	short		76		36,37
S25	60% to 90%, 5.05V, sgl out, regulation	open		500		34,37
S26	60% to 90%, 5.05V, ringing 1/			250		
S27	90% to 60%, 5.05V, sgl out, regulation			500		
S28	90% to 60%, 5.05V, ringing 1/	open	0	250	mV	34,37

1/ Additional ringing beyond the 2nd half cycle shall fall within the regulation limits of test condition C1 or C3 (as applicable) and the peak-to-peak noise envelope of test condition C1.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

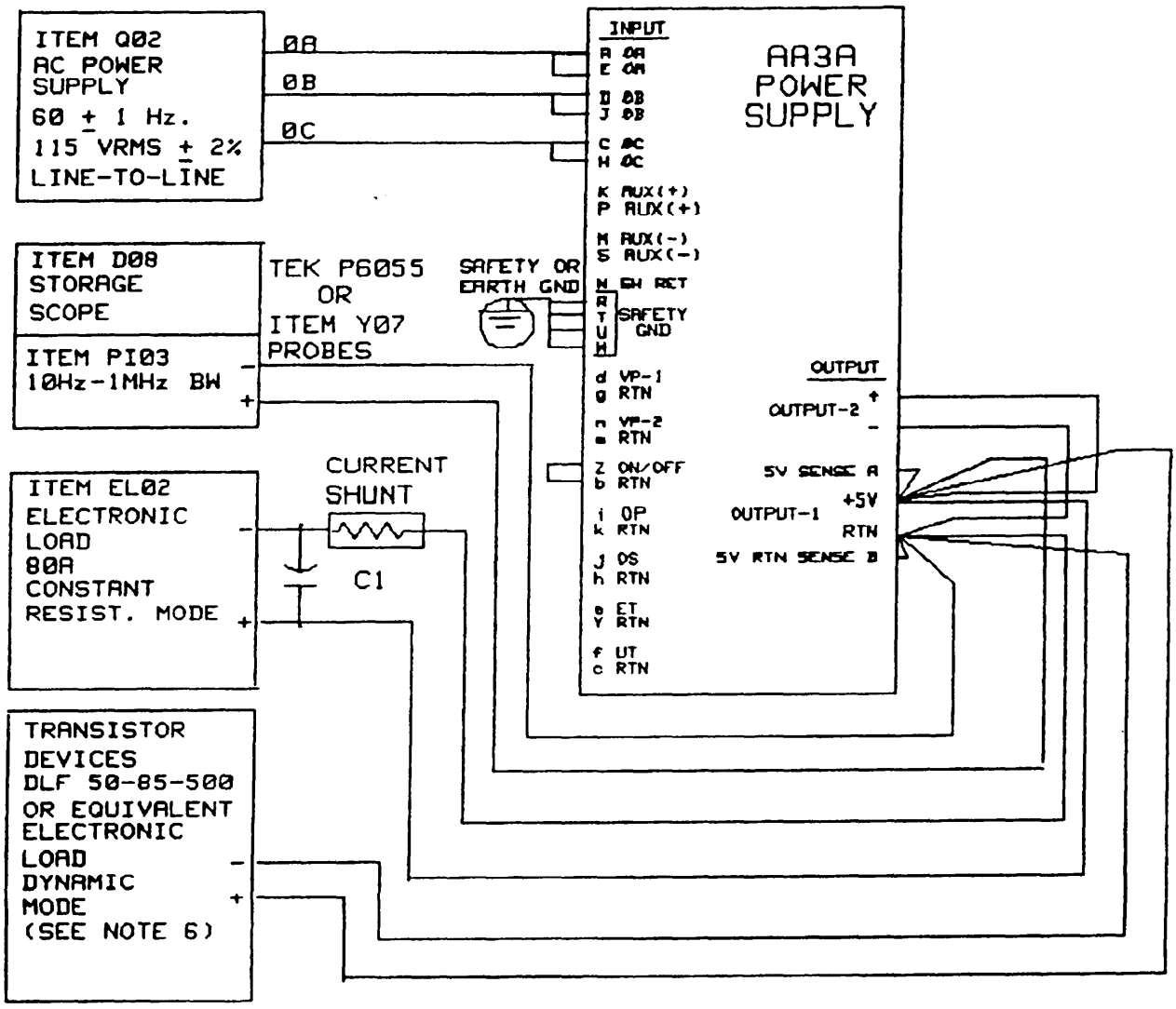


FIGURE 34. Dynamic load regulation test circuit for table XVII.

Source: https://assist.dla.mil -- Downloaded: 2026l0-20T01:52Z
Check the source to verify that this is the currentversion before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. C1 = 14250 uF, +20, -0 percent, 10 VDC, capacitor. Used in test setup for test conditions S5-S8 only.

6. The DLF 50-85-500 or equivalent electronic load shall be in the dynamic mode, switching between 40 amperes and 55 amperes for test conditions S1-S8. It shall be in the dynamic mode, switching between 10 amperes and 55 amperes for test conditions S25-S28.

7. At the option of the testing facility, a DC input (item P10) of 155 VDC $\pm$ 2 percent connected between any two AC inputs may be used instead of the AC input specified.

FIGURE 34. Dynamic load regulation test circuit for table XVII - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

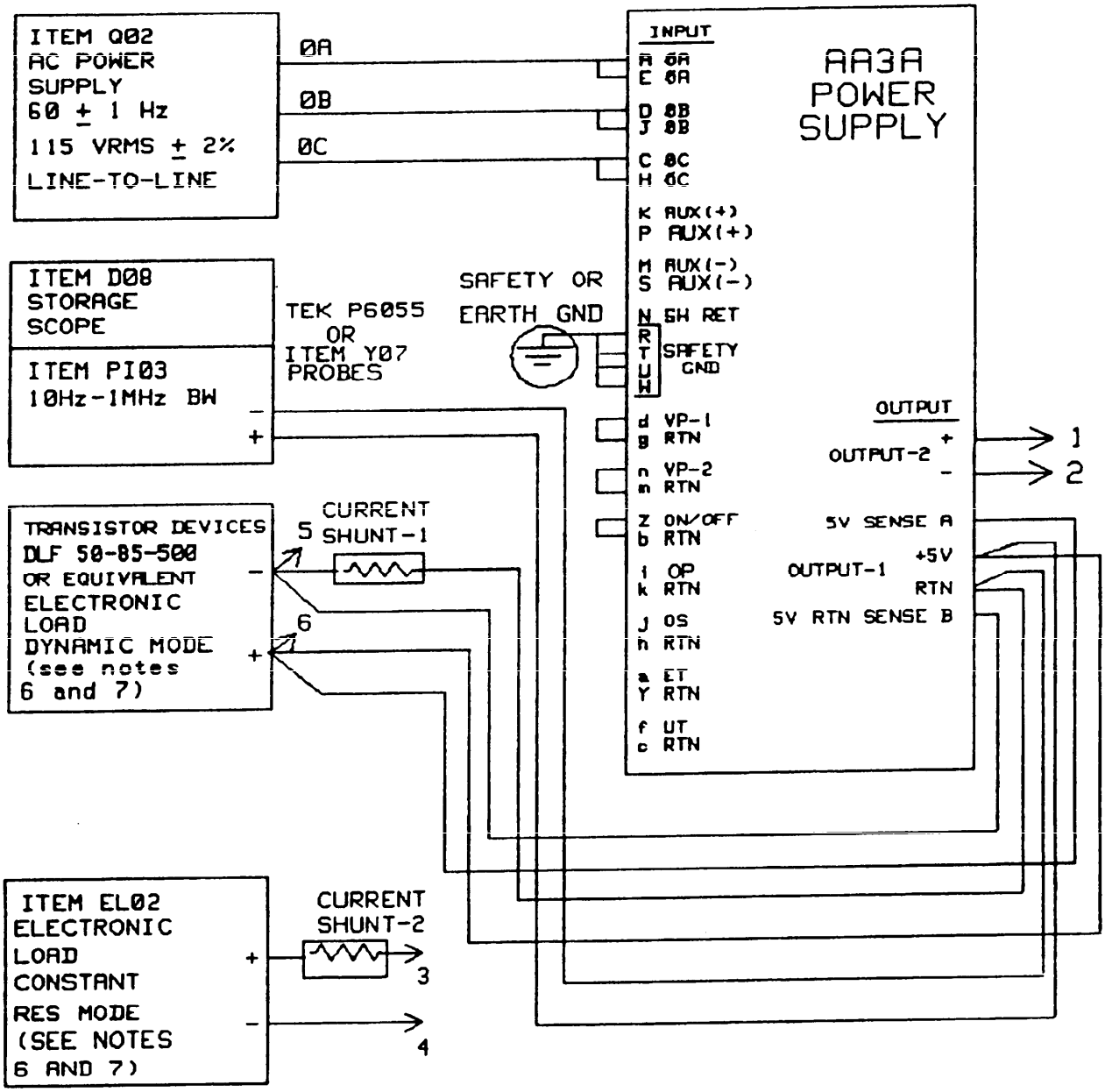


FIGURE 35. Dynamic load regulation (maximum sense leads) test circuit for table XVII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

6. For test conditions S9-S12, connect point 1 to +5V (output-1) and point 2 to RTN (output-1). Connect point 3 to point 5 and point 4 to point 6. Set item EL02 to 80 amperes. The DLF 50-85-500 or equivalent electronic load shall be in the dynamic mode switching between 40 and 55 amperes. Input contacts m and n are not required to be shorted.

7. For test conditions S13-S16, connect point 1 to point 3, and point 2 to point 4. Set item EL02 to 10 amperes. The DLF 50-85-500 or equivalent electronic load shall be in the dynamic mode, switching between 72 (90 percent full load) and 64 amperes (80 percent full load).

8. For test conditions S9-S16, the two sense leads shall be a twisted pair with a length of 56 ± 4 inches. The load lead voltage drop (difference between the voltage directly at the AA3A output voltage contacts and the sense lead connection point) shall be a maximum of 0.158 VDC. This voltage drop shall apply under steady state load current conditions with the current at 90 percent full load.

FIGURE 35. Dynamic load regulation (maximum sense leads) test circuit for table XVII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

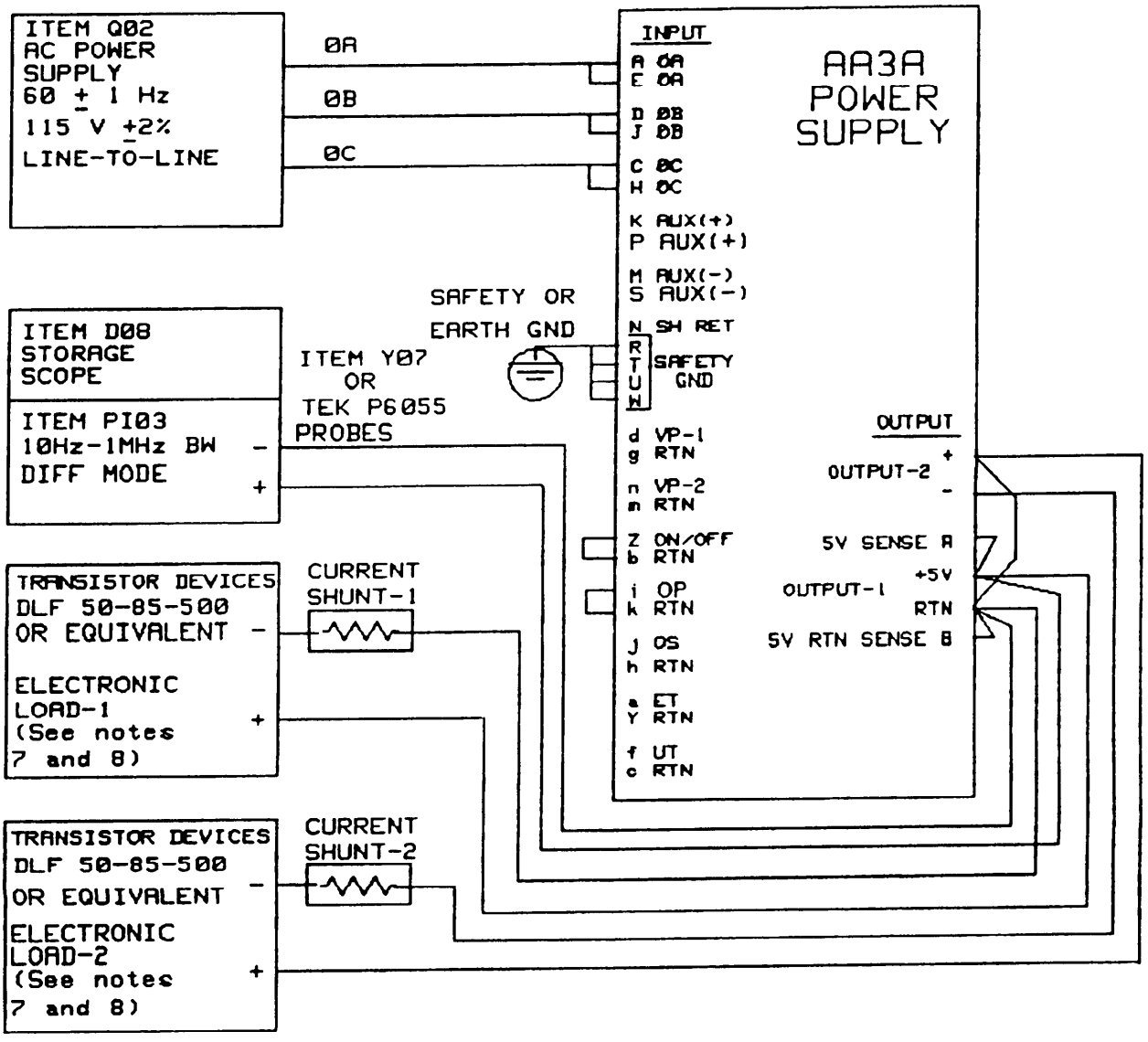


FIGURE 36. Dynamic load regulation test circuit for table XVII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

6. Keep all leads between the power supply under test and the electronic loads as short as possible.

7. For test conditions S17-S20, connect the scope probes as shown; electronic load-2 shall be in constant resistance mode at 10 amperes ± 10 percent. Electronic load-1 shall be in dynamic mode, switching between 72 (90 percent full load) and 64 amperes (80 percent full load).

8. For test conditions S21-S24, the scope probes shall be connected to OUTPUT-2; electronic load-1 shall be in constant resistance mode at 10 amperes ± 10 percent. Electronic load-2 shall be in dynamic mode, switching between 63 (90 percent full load) and 56 amperes (80 percent full load).

FIGURE 36. Dynamic load regulation test circuit for table XVII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

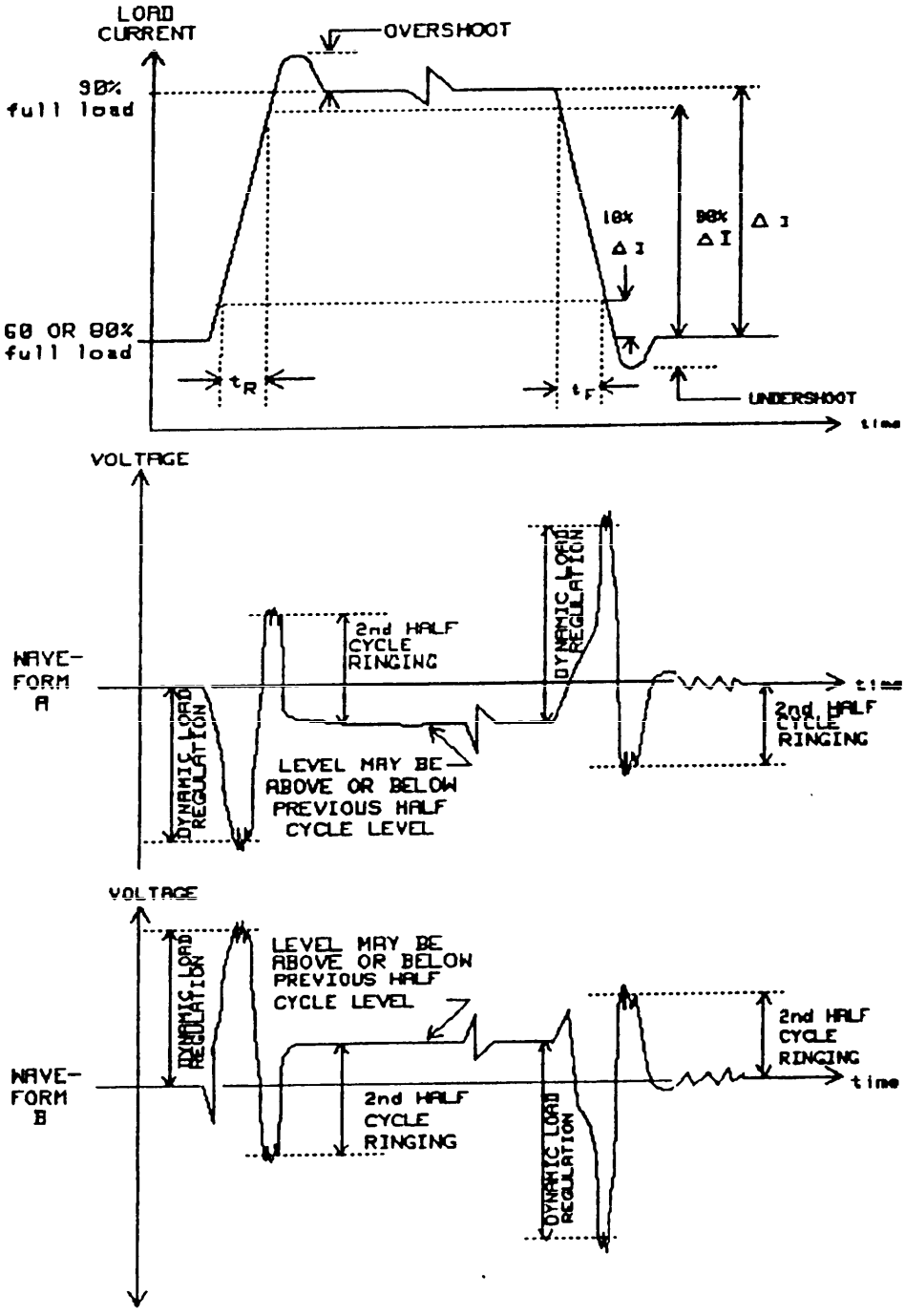


FIGURE 37. Dynamic load regulation test definitions for table XVII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. Rise time, tR, and fall time, tF, shall both be 60 us maximum.

2. The dynamic load may be a single pulse or repetitive plus or minus transitions with each state being maintained for no less than 1 millisecond (ms).

3. The load current overshoot or undershoot shall be minimized to the greatest extent possible.

4. Measurement of the dynamic load regulation and 2nd half cycle ringing shall be to the mid-point of any ripple occurring on the output voltage waveform.

5. Waveform A is an illustration of a possible output voltage response to the change in load current. For test conditions S9 through S16, the waveform may appear as illustrated by waveform B instead of that shown by waveform A.

FIGURE 37. Dynamic load regulation test definitions for table XVII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XVIII. Turn-on time into capacitive load test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
T	Turn-on time into capacitive load	15	250	ms	38

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

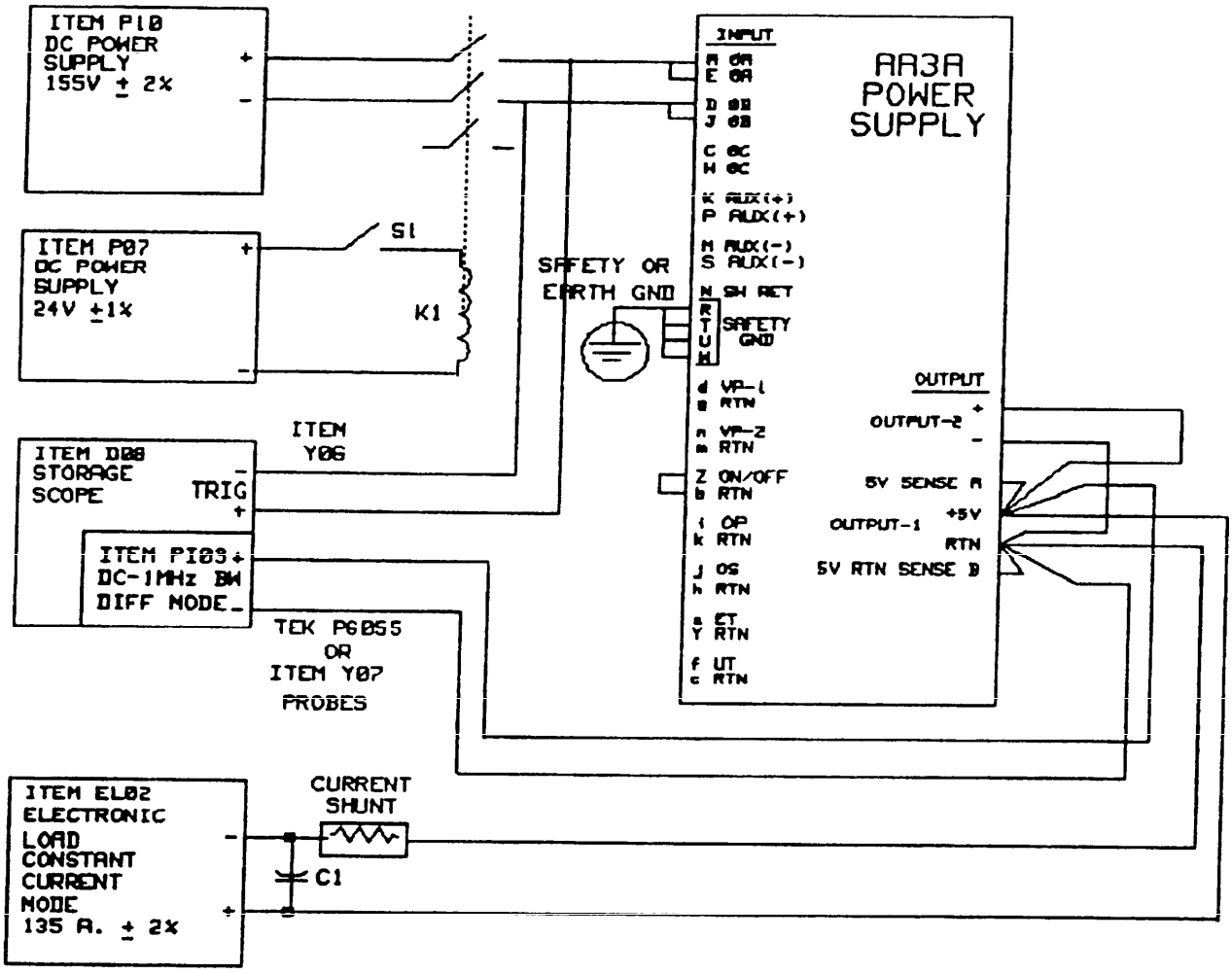


FIGURE 38. Turn-on time into capacitive load test circuit for table XVIII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. C1 = 14250 uF, +20, -0 percent, 10 VDC, capacitor.

6. S1 = spst switch.

7. K1 = 3pst mercury relay, Magnecraft WM35AAA-24D or equivalent.

8. At the option of the testing facility, an AC input (instead of the DC input specified) of 3-phase, 115 VRMS $\pm$ 2 percent, line-to-line, 60 $\pm$ 1 Hz voltage can be applied at the phase A, B, and C inputs.

9. Prior to connecting capacitor C1 across the output, adjust item EL02 load for 135 amperes with the power supply energized. Remove the input power by opening S1 and connect the capacitor C1 across the output. Close S1 and verify that the turn-on time meets the limits of table XVIII.

10. Turn-on time is the time from when power is applied to the input until the output reaches and remains within the regulation limits. The regulation limits shall be test condition C2 limits.

11. Item EL02 shall act as a constant current load while the 5 volt output is rising. Therefore, above a power supply output terminal voltage of 2.0 volts, the load current shall be rising to its set value at a rate two times, or greater, than the rate of voltage.

FIGURE 38. Turn-on time into capacitive load test circuit for table XVIII - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the currentversion before use.

TABLE XIX. Undervoltage and overvoltage output status test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
U1	Undervoltage-output status LED off	4.39	4.80	VDC	39
U2	Overvoltage-output status LED off 1/	5.57	6.45	VDC	
U3	Status LED check	Status LED on		-	
V1	Output status signal, logic low (5 mA)	0.0	0.4	VDC	
V2	Output status signal, logic high	29.0	30.5		
V3	Output status signal, sgl out undervoltage	4.39	4.80		39
V4	Output status signal, out-1 undervoltage	4.39	4.80		40
V5	Output status signal, out-2 undervoltage	4.39	4.80		40
V6	Output status signal, sing. out overvoltage 1/	5.57	6.45		39
V7	Output status signal, out-1 overvoltage 1/	5.57	6.45		40
V8	Output status signal, out-2 overvoltage 1/	5.57	6.45	VDC	40
V9	Output status signal leakage current	0.0	4.4	uADC	40
V10	Output status signal settling time (high to low)	0.0	2.0	ms	41
V11	Output status signal settling time (low to high)	0.0	2.0	ms	41
V12	Output status signal, logic low (0.5 mA)	0.0	0.4	VDC	41

- See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

1/ Test conditions U2 and V6-V8 verify that the output status LED and output status signal detect an overvoltage condition. Additionally, the power supply vendor will verify by 100 percent (X) testing, that the output voltage is limited as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

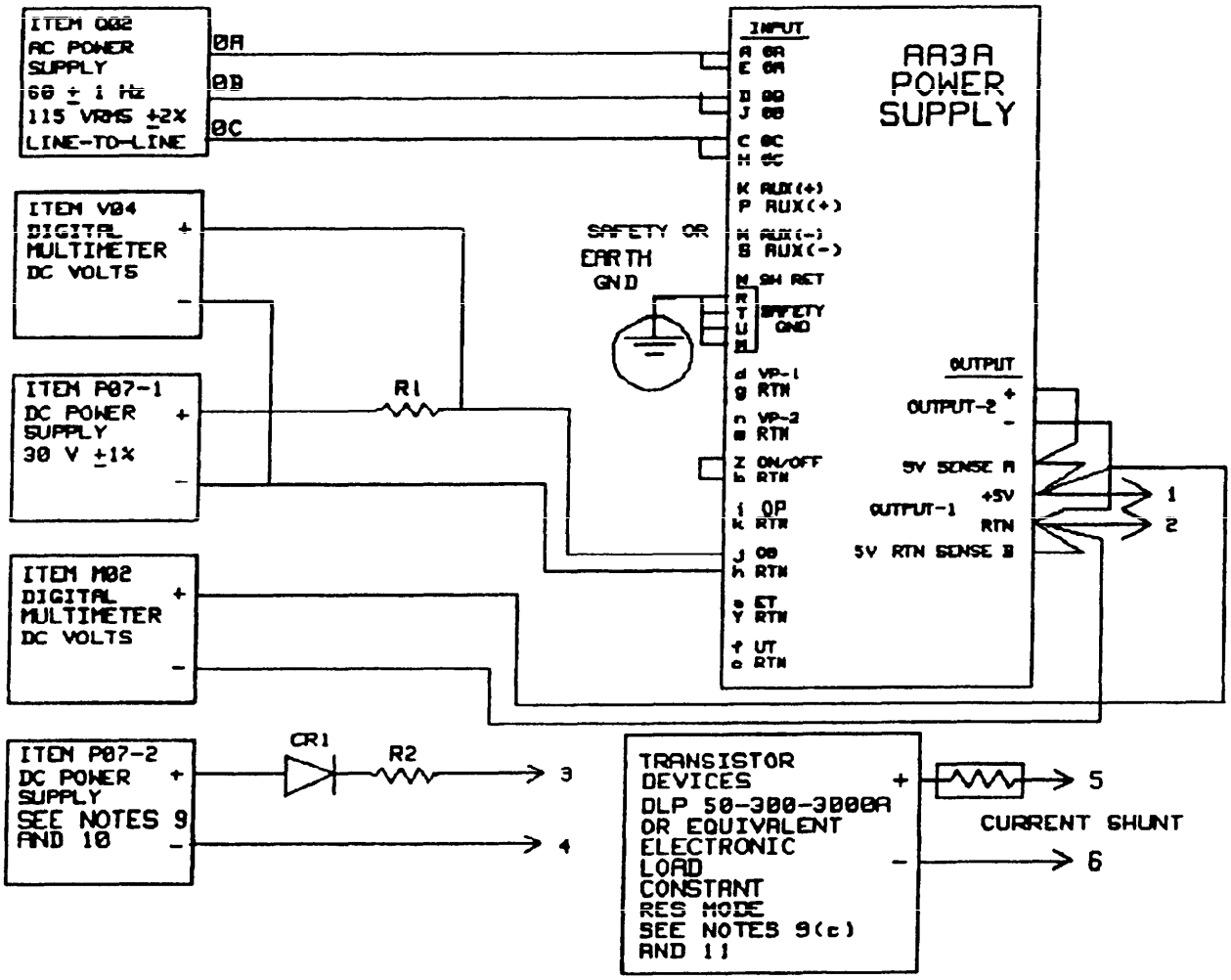


FIGURE 39. Undervoltage and overvoltage (single output) test circuit for table XIX.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 200A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. R1 = 5.9 kOhm ± 1 percent, 1/8 watt resistor.

6. R2 = 5 Ohm ± 1 percent, 8 watt resistor.

7. CR1 = 1N5616 or equivalent.

8. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

9. The output voltage can be reduced for undervoltage testing in one of three ways (the method to use will depend on the design of the AA3A power supply being tested):

(a) Method 1. Connect item P07-2 plus terminal to input contact d and minus terminal to input contact g (CR1 and R2 shall not be used). Beginning with item P07-2 voltage at 0 volt and item P10 ON, gradually increase the item P07-2 voltage until the output voltage is 5.05 ± 0.10 volts. Further increase of the item P07-2 voltage will cause the output voltage to drop as required. Items EL02-1 and EL02-2 are not required.

(b) Method 2. Prior to connecting an external power supply to the output, reduce the input voltage (as applied with item P10) until the AA3A output voltage drops to less than 4.39 volt but not low enough for the internal housekeeping supply to stop working. Verify that the output status LED is OFF and that the output status signal is high before proceeding with the test. Connect point 1 to point 3 and point 2 to point 4. Adjust item P07-2 until the item M02 voltage is 5.05 ± 0.05 volts prior to performing each test. This external supply is used to decrease the output voltage for testing. Items EL02-1 and EL02-2 are not required.

FIGURE 39. Undervoltage and overvoltage (single output) test circuit for table XIX - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(c) Method 3. Connect point 1 to point 5 and point 2 to point 6. For each test, begin slowly increasing the load current from 150 amperes and the output voltage shall gradually drop as required for testing. Do not exceed 190 amperes load current. Items P07-2, CR1 and R2 are not required.

10. The output voltage is increased for overvoltage testing by connecting point 1 to point 3 and point 2 to point 4. Adjust item P07-2 to 5.05 $\pm$ 0.05 volts prior to performing each test. This external supply is used to increase the output voltage for testing. Items EL02-1 and EL02-2 are not required.

11. Perform test conditions U3 and V1 before performing the undervoltage and overvoltage tests. For U3 and V1, connect point 1 to point 5 and point 2 to point 6. Adjust the electronic load to a current of 150 +0, -3 amperes. Using item M02, ensure that the output is within regulation limits for the applicable temperature (test condition C2). For test condition U3, verify that the output status LED is on. For test conditon V1, the item V04 reading shall be within the table XIX limits.

12. For test condition U1, slowly decrease the output voltage from nominal 5.05 volts until the output status LED goes off. The item M02 voltage at which that occurs shall meet the table XIX limits.

13. For test condition U2, increase the output voltage from nominal 5.05 volts until the output status LED turns off. The item M02 voltage at which that occurs shall be within table XIX limits. Do not raise the voltage at points 1 and 2 (as measured with item M02) above 6.6 volts.

14. Perform the test conditions V2 and V3 tests as follows:

(a) For test condition V2, decrease the output voltage from nominal 5.05 volts until the output status signal, as monitored by item V04, goes high (29.0 to 30.5 VDC). This item V04 voltage shall meet the table XIX limits.

(b) For test condition V3, decrease the output voltage from nominal 5.05 volts until the item V04 voltage goes high (within the test condition V2 limits). The item M02 voltage when this occurs shall meet the table XIX limits.

FIGURE 39. Undervoltage and overvoltage (single output) test circuit for table XIX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

15. For test condition V6, increase the output voltage from nominal 5.05 volts until the output status signal goes high (item V04 voltage within the test condition V2 limits). The item M02 voltage at which that occurs shall meet the table XIX limits. Do not raise the voltage at points 1 and 2 (as measured with item M02) above 6.6 volts.

16. The output may not automatically recover when the overvoltage is removed. However, it shall recover when the input voltage is taken to zero and then reapplied.

FIGURE 39. Undervoltage and overvoltage (single output) test circuit for table XIX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

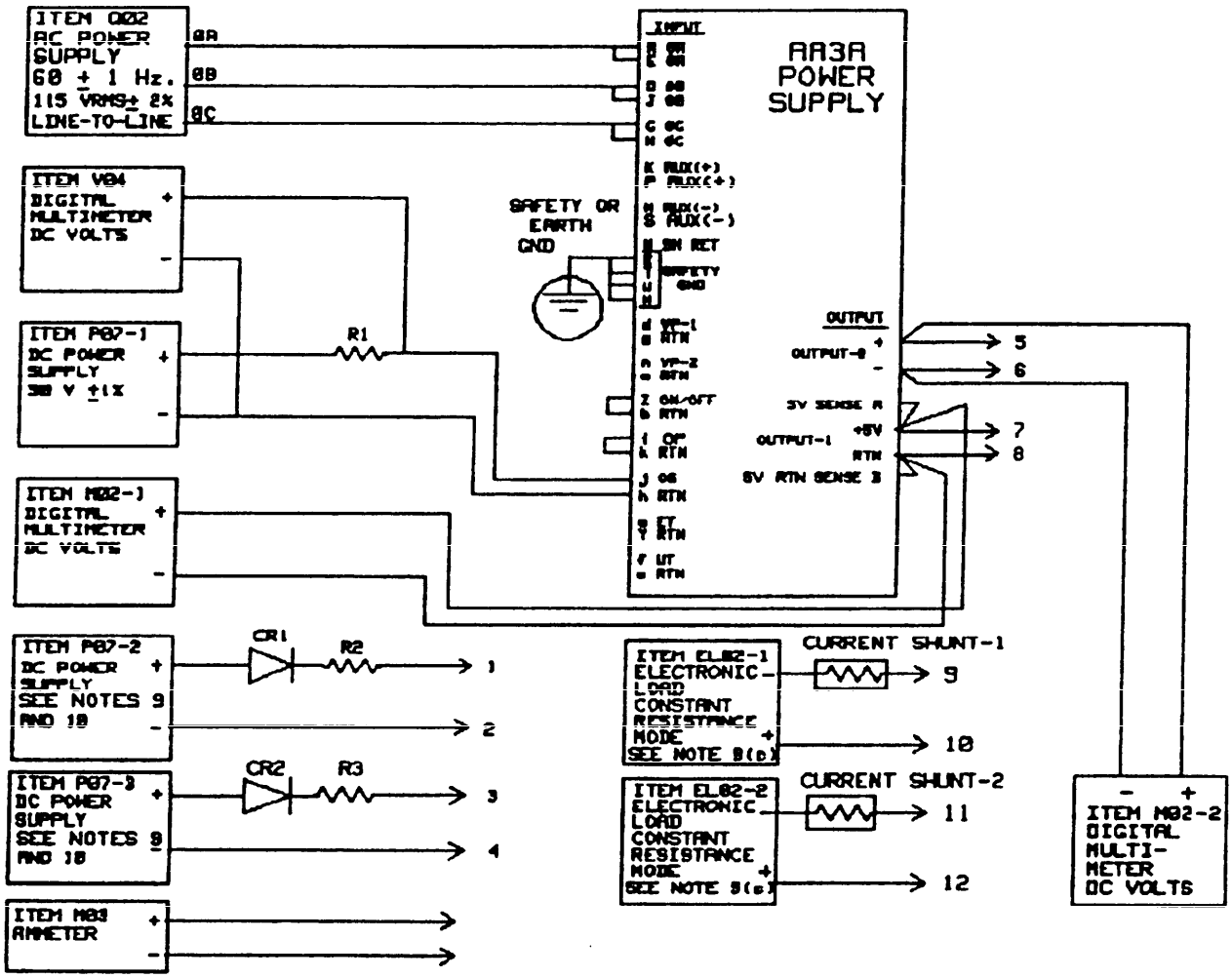


FIGURE 40. Undervoltage and overvoltage (dual output) test circuit for table XIX.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. R1 = 5.9 kOhm ± 1 percent, 1/8 watt resistor.

6. R2 = R3 = 5 Ohm ± 1 percent, 8 watt resistor.

7. CR1 = CR2 = 1N5616 or equivalent.

8. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

9. The output voltage can be reduced for undervoltage testing in one of three ways (the method to use will depend on the design of the AA3A power supply being tested):

(a) Method 1. Connect item P07-2 plus terminal to input contact d and minus terminal to input contact g. Connect item P07-3 plus terminal to input contact n and minus terminal to input contact m. CR1, CR2, R2 and R3 shall not be used. Begin with both item P07-2 and P07-3 voltages at 0 volt and item P10 ON. Gradually increase the item P07-2 (for output-1) and item P07-3 (for output-2) voltages until the voltage on both outputs is 5.05 ± 0.10 volts. Further increase of these P07-2 and P07-3 voltages will cause the output voltage to drop as required. Items EL02-1 and EL02-2 are not required.

(b) Method 2. Prior to connecting an external power supply to the output, reduce the input voltage (as applied with item P10) until both output voltages drop to less than 4.39 volt but not low enough for the internal housekeeping supply to stop working. Verify that the output status LED is off and that the output status signal is high before proceeding with the test. Connect point 1 to point 5, point 2 to point 6, point 3 to point 7, and point 4 to point 8. Adjust items P07-2 and P07-3 until items M02-2 and M02-1, respectively, are 5.05 ± 0.05 volts

FIGURE 40. Undervoltage and overvoltage (dual output) test circuit for table XIX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

prior to performing each test. Items P07-2 and P07-3 voltages are decreased to decrease the output-2 and output-1 voltages, respectively, for testing. Items EL02-1 and EL02-2 are not used.

(c) Method 3. Connect point 7 to point 9, point 8 to point 10, point 5 to point 11, and point 6 to point 12. Set item EL02-1 to 80 amperes and item EL02-2 to 70 amperes prior to each test. The load current on the output being tested is gradually increased from the starting current, causing the output voltage to drop as required for testing. Do not exceed 103 amperes on item EL02-1 (output-1) or 90 amperes on item EL02-2 (output-2).

10. The output voltage is increased for overvoltage testing by connecting point 1 to point 5, point 2 to point 6, point 3 to point 7, and point 4 to point 8. Adjust items P07-2 and P07-3 to 5.05 $\pm$ 0.05 volts prior to performing each test. Items P07-2 and P07-3 voltages are increased to increase the output-2 and output-1 voltages, respectively, for testing.

11. Perform test conditions V4 and V5 undervoltage tests as follows:

(a) Test condition V4. Keep output-2 voltage (item M02-2) at 5.05 $\pm$ 0.10 volts. Decrease the output-1 voltage from nominal 5.05 volts until the item V04 voltage goes high (within the test condition V2 limits). The item M02-1 voltage when this occurs shall meet the table XIX limits.

(b) Test condition V5. Repeat (a) above except keep the output-1 voltage (item M02-1) at 5.05 $\pm$ 0.10 volts and decrease the output-2 voltage (item M02-2).

12. Perform test conditions V7 and V8 as follows:

(a) Test condition V7. Keep output-2 voltage (item M02-2) at 5.05 $\pm$ 0.10 volts. Increase the output-1 voltage from nominal 5.05 volts until the item V04 voltage goes high (within the test condition V2 limits). The item M02-1 voltage when this occurs shall meet the table XIX limits.

(b) Test condition V8. Repeat (a) above except keep the output-1 voltage (item M02-1) at 5.05 $\pm$ 0.10 volts and increase the output-2 voltage (item M02-2).

FIGURE 40. Undervoltage and overvoltage (dual output) test circuit for table XIX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

13. Test condition V9. Connect the item M03 ammeter between resistor R1 and item P07-1. Keep the output-1 voltage (item M02-1) at 5.05 ± 0.10 volts. Decrease the output-2 voltage (item M02-2) from nominal 5.05 volts until the item V04 voltage goes high (within the test condition V2 limits). The item M03 current reading when this happens shall meet the table XIX limits. Disconnect item V04 while making the current reading.

14. The output may not automatically recover when the over-voltage is removed. However, it shall recover when the input voltage is taken to zero and reapplied.

FIGURE 40. Undervoltage and overvoltage (dual output) test circuit for table XIX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

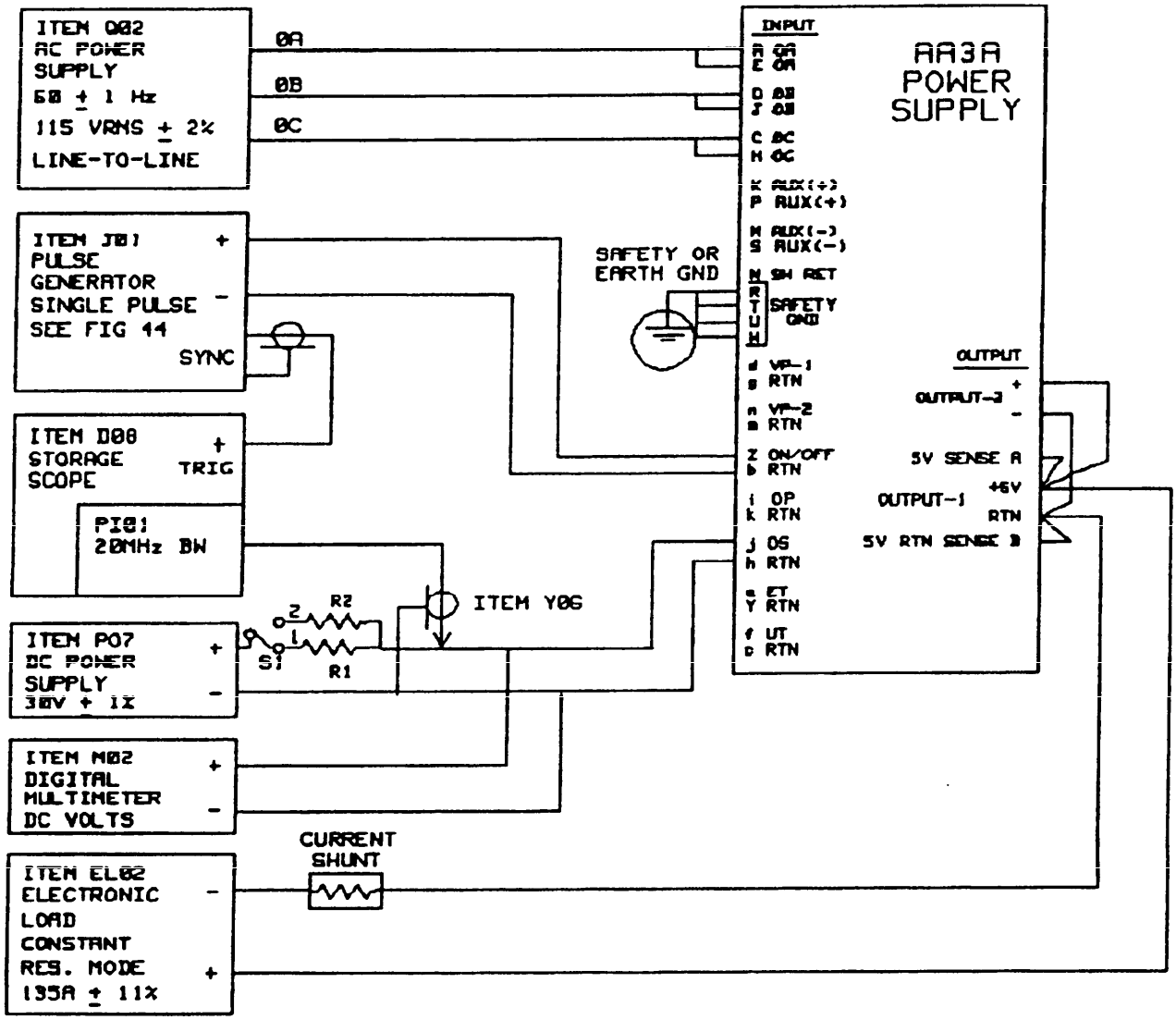


FIGURE 41. Output status signal settling time and logic low test circuit for table XIX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. R1 = 5.9 kOhm ± 1 percent, 1/4 watt resistor for test conditions V10 and V11 (set S1 to 1).

6. R2 = 60.4 kOhm ± 1 percent, 1/10 watt resistor for test condition V12 (set S1 to 2).

7. S1 = spdt switch.

8. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

9. For test conditions V10 and V11, set up the power supply input voltage and load current with item J01 disconnected and input contacts Z and b shorted (power supply on). Disconnect the short after setup. Set switch S1 to position 1.

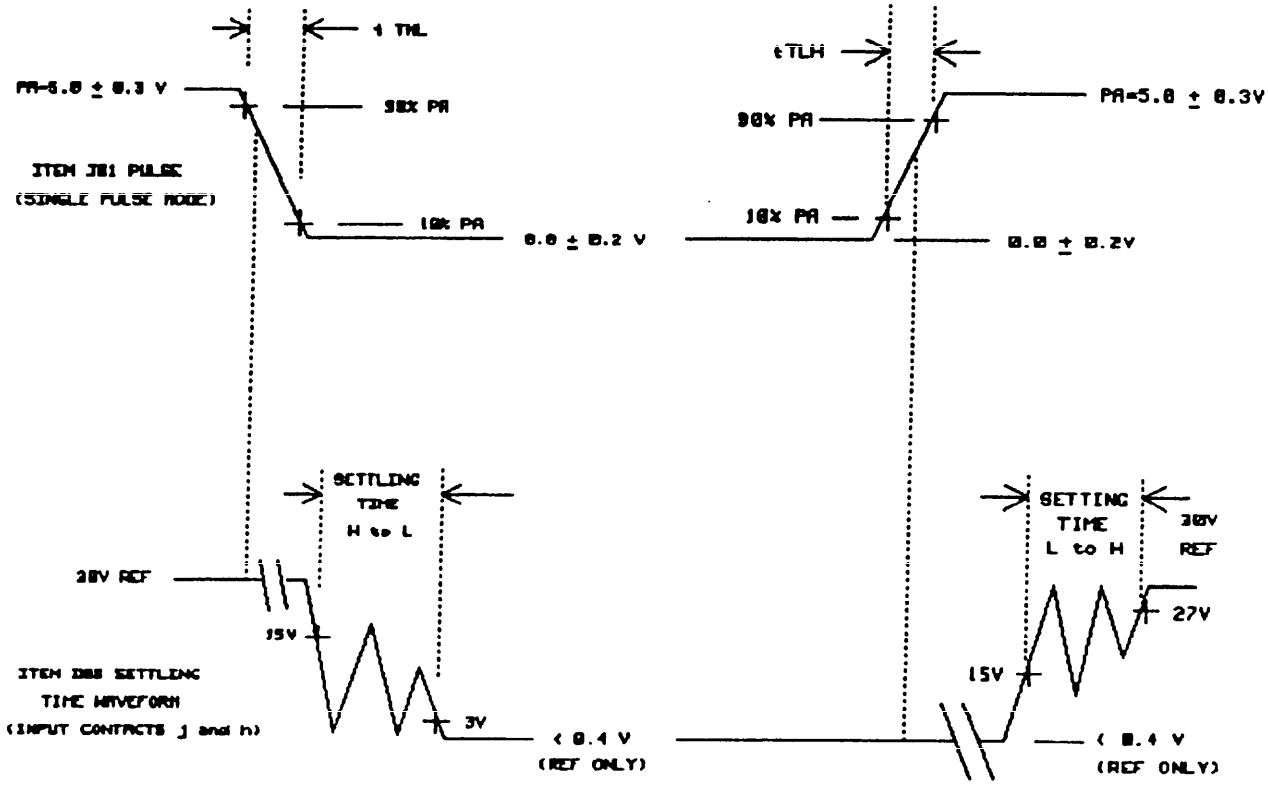
(a) For test condition V10, set up item J01 for a single high to low level pulse in accordance with figure 42. Connect item J01 as shown and trigger item J01 for a single pulse. Using the storage mode of item D08, measure the settling time (see figure 42).

(b) For test condition V11, repeat (a) above except for a single low to high level pulse from item J01.

10. For test condition V12, set switch S1 to position 2, disconnect item D08 and item J01 and short input contact Z to b. Connect OS RTN (input contact h) to SAFETY GND (input contact R). Using item M02, measure the voltage across input contacts j and h.

FIGURE 41. Output status signal settling time and logic low test circuit for table XIX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.



NOTES:

1. Item J01 pulse parameters:

   tTHL = tTLH = 10 ± 5 us

FIGURE 42. Output status settling time test circuit waveforms for table XIX.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XX. Excessive and unsafe temperature indication test.

Test cond	Parameter	Initial limits elevated ambient temperature		Unit	Test fig
		Min	Max		
H1	Excessive temperature indication (5 mA current) 2/	0.0	0.4	VDC	43
H2	Excessive temperature indication (500 mA current) 2/	0.0	0.4	VDC	
H3	Unsafe temperature indication (5 mA current) 2/	0.0	0.4	VDC	
H4	Unsafe temperature indication (500 mA current) 2/	0.0	0.4	VDC	
H5	Unsafe-excessive differential (5 mA current) 1/	1.0	25	°C	
H6	Indicator reset (5 mA current)	60	—	°C	
H7	Unsafe-excessive differential (500 mA current) 1/	1.0	25	°C	
H8	Indicator reset (500 mA current)	60	—	°C	43

-See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T01:52Z
Check the source to verify that this is the current version before use.

1/ Test conditions W5 and W7 are calculations. Subtract the trip point temperature measured in test condition W1 (W2) from the trip point temperature measured in test condition W3 (W4).

2/ Test conditons W2 and W4 verify that the closed contact resistance is a maximum of 0.8 ohms with a contact current of 500 mA. Test conditions W1 and W3 verify that the closed contact resistance is a maximum of 80 ohms with a contact current of 5 mA.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

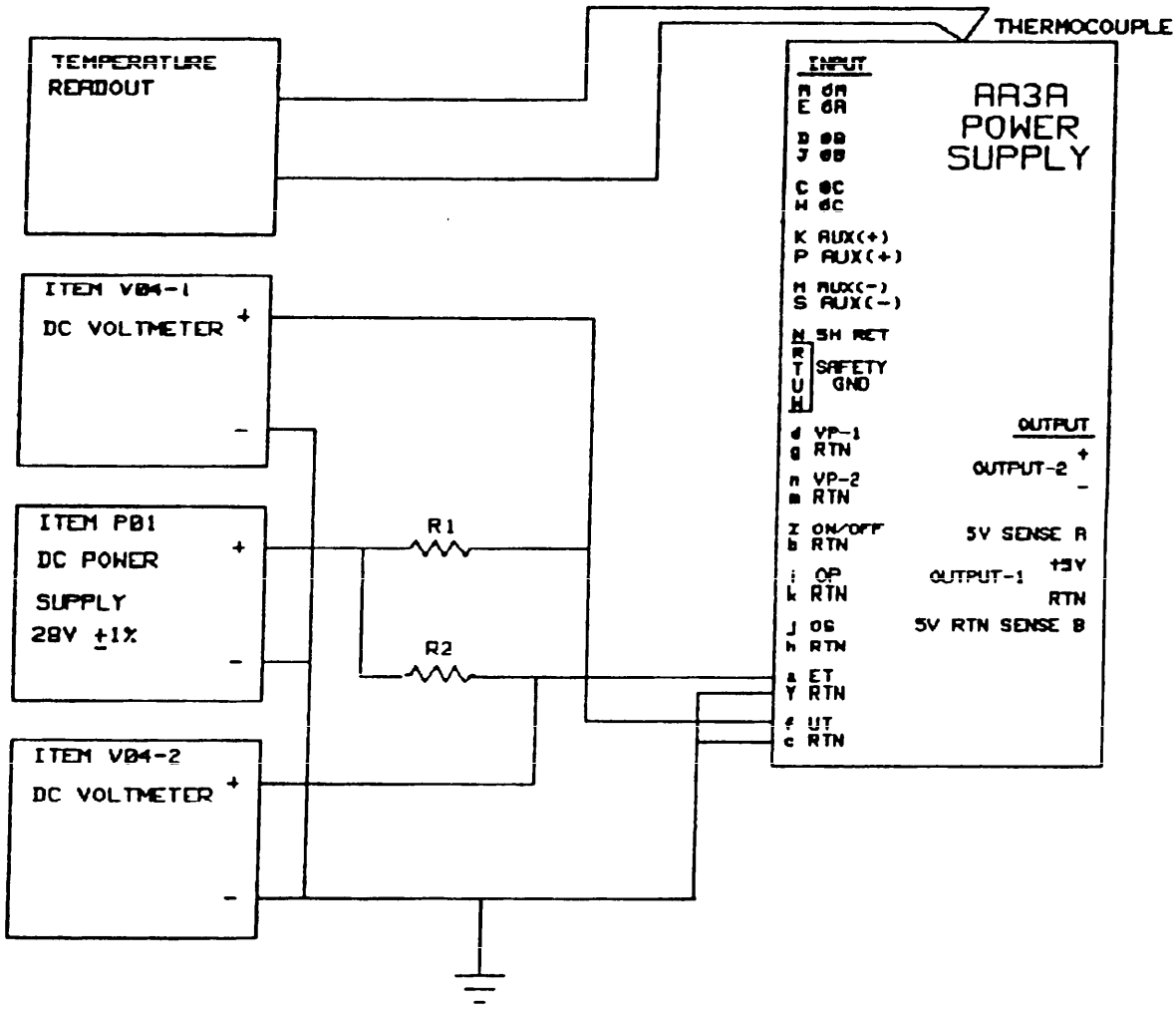


FIGURE 43. Excessive and unsafe temperature indication test circuit for table XX.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. Perform tests W1-W8 only after all other electrical and environmental tests have been performed.

2. R1 = R2 = 56 Ohm ± 1 percent, 15 watt resistor for test conditions W2, W4 and W8. R1 = R2 = 5.62 kOhm ± 1 percent, 1/4 watt resistor for test conditions W1, W3 and W6.

3. The power supply shall be unenergized during this test.

4. The temperature shall be monitored with a thermocouple connected to the power supply frame. Do not attach the thermocouple to the mounting surface.

5. Test conditions W1, W3, W5, and W6 shall be performed as follows and shall be performed prior to performing test conditions W2, W4, W7, and W8:

(a) Place the power supply in an oven and raise the temperature to 65 ± 2°C at a rate not to exceed 1°C per minute. Let the power supply soak for a minimum of 1 hour at this temperature. Verify that items V04-1 and V04-2 both read 28.0 ± 0.5 volts (thermal switches open).

(b) Begin increasing the oven temperature at a rate not to exceed 1°C per 15 minutes, until V04-2 voltage switches to a low level (excessive temperature thermal switch closed). This low level voltage shall meet test condition W1 limits. Record the power supply frame temperature at which the thermal switch closed for use in test condition W5 calculation. Do not exceed 140°C.

(c) Continue increasing the oven temperature, as noted in (b) above, until item V04-1 voltage switches to a low level (unsafe temperature thermal switch closed). This low level shall meet test condition W3 limits. Record the power supply frame temperature at which the thermal switch closed for use in test condition W5 calculation. Do not exceed 140°C.

FIGURE 43. Excessive and unsafe temperature indication test circuit for table XX - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(d) Begin decreasing the oven temperature at a rate not to exceed 1°C per 15 minutes, until V04-1 and V04-2 both read 28.0 ± 0.5 volts (thermal switches open). The power supply frame temperature at which both thermal switches were open shall meet the limits of test condition W6.

6. Repeat note 5 procedure for test conditions W2, W4, W7, and W8.

FIGURE 43. Excessive and unsafe temperature indication test circuit for table XX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XXI. Output voltage hold-up time.

Test cond	Parameter	Initial limits 25 ± 5°C Mounting surface temperature		Unit	Test fig
		Min	Max		
X	Output voltage hold-up time, 100 VDC input	50	-	us	44

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

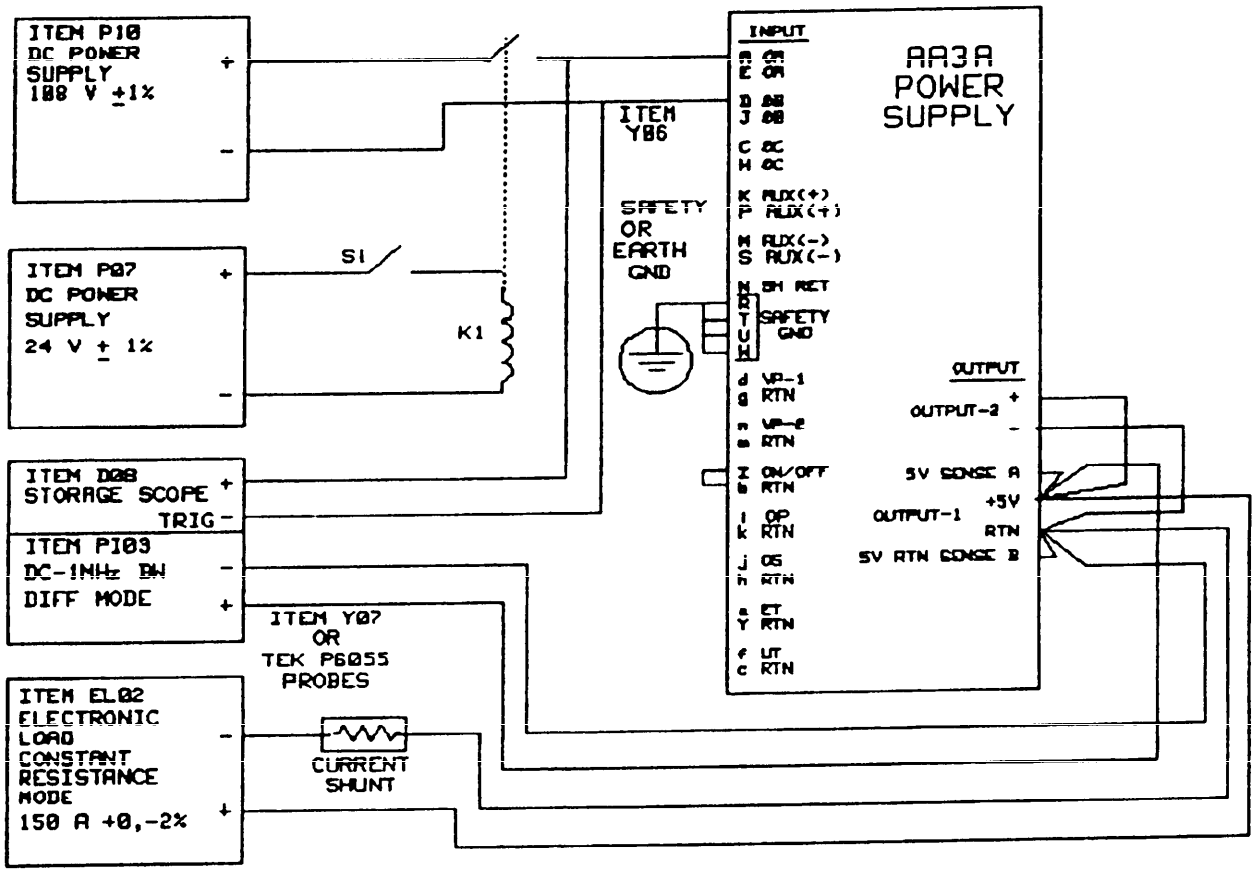


FIGURE 44. Output voltage hold-up time test circuit for table XXI.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

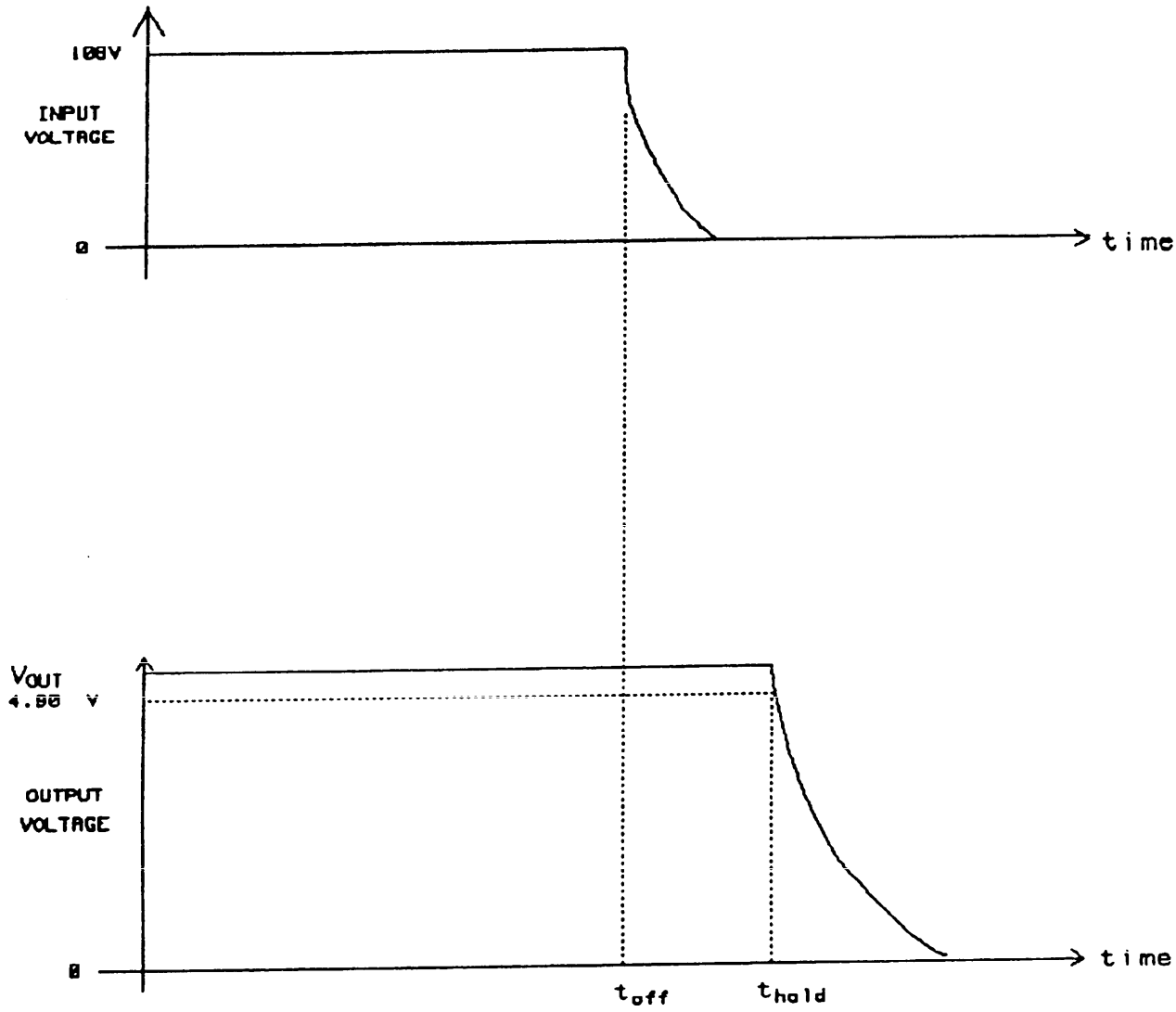
5. S1 = spst switch.

6. K1 = spst mercury relay, Magnecraft WM35A-24D or equivalent.

7. Connect the circuit as shown with switch S1 closed (input power applied). Open switch S1 (disconnecting input power) and measure the hold-up time as defined on figure 45.

FIGURE 44. Output voltage hold-up time test circuit for table XXI - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.



NOTE:

1. Hold-up time is defined as t hold minus t off.

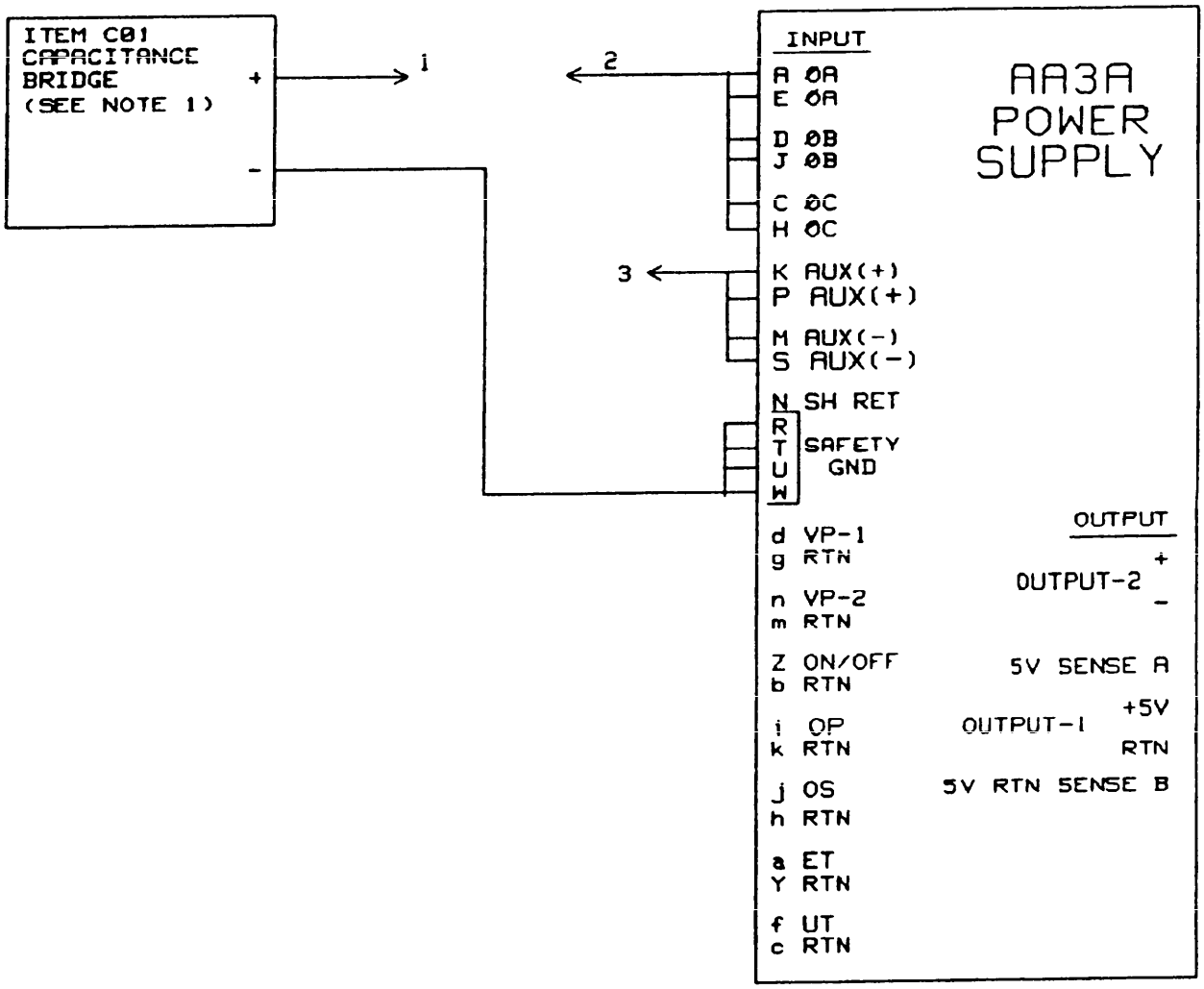
FIGURE 45. Output voltage hold-up time definition for table XXI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XXII. Input to chassis capacitance test.

Test cond	Parameter	Initial limits 25 ± 5°C Mounting surface temperature		Unit	Test fig
		Min	Max		
Y1	AC input to chassis capacitance	0.0	0.03	uF	46
Y2	Auxiliary input to chassis capacitance	0.0	0.05	uF	46

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.



NOTES:

1. Item C01 set-up: frequency = 1kHz ± 5 percent, voltage = 0.3 volts peak maximum.

2. For test condition Y1, connect point 1 to point 2. The item C01 capacitance reading shall meet the test condition Y1 limits.

3. For test condition Y2, connect point 1 to point 3. The item C01 capacitance reading shall meet the test condition Y2 limits.

FIGURE 46. Input to chassis capacitance test circuit for table XXII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XXIII. Output impedance test. 1/

Test cond	Parameter	Input contacts i and k	Connect 2/		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			1 to points	2 to points	Min	Max		
Z1	Output impedance, sgl output	Open	3,5	4,6	0.0	3.58	mV	47
Z2	Output impedance, output-1	Shorted	5	6	0.0	4.36	mV	47
Z3	Output impedance, output-2	Shorted	3	4	0.0	4.62	mV	47

1/ This test verifies a maximum output impedance of 30 mOhm in accordance with the relationships on figure 47, note 8(e). The mV limits in this table apply only while the power supply load resistances are within the figure 47 specified values at all frequencies.

2/ Two numbers in one block indicate that these two points shall be connected together in addition to being connected to points 1 or 2, as applicable (for example, "3,5" under column "1 to points" indicates point 3 connected to point 5 connected to point 1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

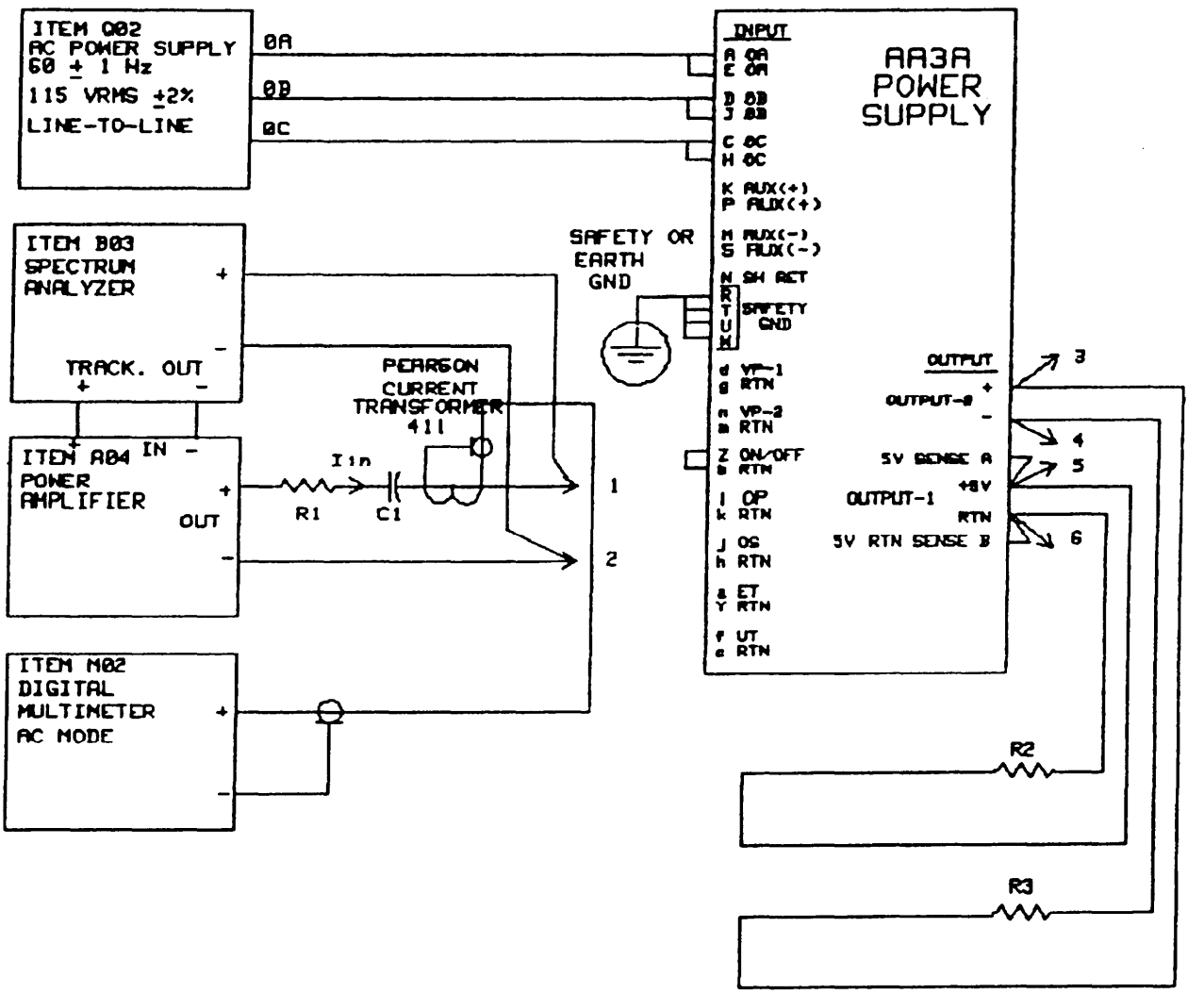


FIGURE 47. Output impedance test circuit for table XXIII.

Source: https://assist.dla.mil -- Downloaded: 202640-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. R1 = 10 Ohm ± 5 percent, non-inductive, 0.5 watt resistor.

4. R2 = 0.08 Ohm ± 5 percent, 325 watt resistor, non-inductive.

5. R3 = 0.10 Ohm ± 5 percent, 300 watt resistor, non-inductive.

6. C1 = 6000 uF + 75 percent, -10 percent, 25 volts capacitor.

7. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

8. The output impedance test shall be performed as follows:

(a) Make the required connections for test condition Z1, with the item A01 amplitude adjusted to 0V, and apply input power to the power supply. The DC load current on the power supply will be 114 amperes ± 10 percent for test condition Z1, 63 amperes ± 10 percent for test condition Z2, and 50.5 amperes ± 10 percent for test condition Z3.

(b) AC current adjustment: Set the spectrum analyzer to the manual mode and at a fixed frequency between 100Hz and 1kHz. Use the model 411 current transformer and item M02 to monitor the AC current (conversion factor = 0.1 volts per ampere). Use the item B03 tracking oscillator level control and the item A01 amplitude control to increase the AC current into the power supply output to 200 milliampere (mA) ± 3 percent.

(c) Set the item B03 spectrum analyzer to the sweep mode and find maximum Vo (as measured by item B03) over the frequency range of 10Hz to 25kHz. Sweep the frequency over a minimum of two bands:

10 Hz to 5 kHz, resolution bandwidth = 10 Hz.
5 kHz to 25 kHz, resolution bandwidth = 30 Hz.

FIGURE 47. Output impedance test circuit for table XXIII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

The AC current shall be maintained at a constant amplitude of 200 mA over the sweep range. The frequency of maximum Vo is the frequency of maximum output impedance. Read the maximum value displayed on item B03 (read the average value at the peak if there is noise in the signal). This shall be within the limits in table XXIII. Record the frequency of maximum Vo.

(d) Repeat the above procedure for test conditions Z2 and Z3.

(e) To calculate the maximum output impedance from the maximum Vo;

$$Zo = \frac{\text{item B03 voltage}}{Iin - \frac{\text{item B03 voltage}}{R2||R3}} = \frac{\text{item B03 voltage}}{0.2A. - \frac{\text{item B03 voltage}}{R2||R3}}$$

for test condition Z1,

$$Zo = \frac{\text{item B03 voltage}}{Iin - \frac{\text{item B03 voltage}}{R2}} = \frac{\text{item B03 voltage}}{0.2A. - \frac{\text{item B03 voltage}}{R2}}$$

for test condition Z2, and

$$Zo = \frac{\text{item B03 voltage}}{Iin - \frac{\text{item B03 voltage}}{R3}} = \frac{\text{item B03 voltage}}{0.2A. - \frac{\text{item B03 voltage}}{R3}}$$

for test condition Z3.

The resistance in the formulas (R2 in parallel with R3, R2, or R3) includes all impedances (may be frequency dependent) from directly at the power supply positive output terminal, through the resistor and to the negative terminal.

FIGURE 47. Output impedance test circuit for table XXIII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XXIV. Open load lead and return protection test.

Test cond	Parameter	Initial limits 25 + 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
AA1	Open 5.05V load lead protection	5.00	5.10	VDC	48
AA2	Open 5.05V RTN load lead protection	5.00	5.10	VDC	48

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

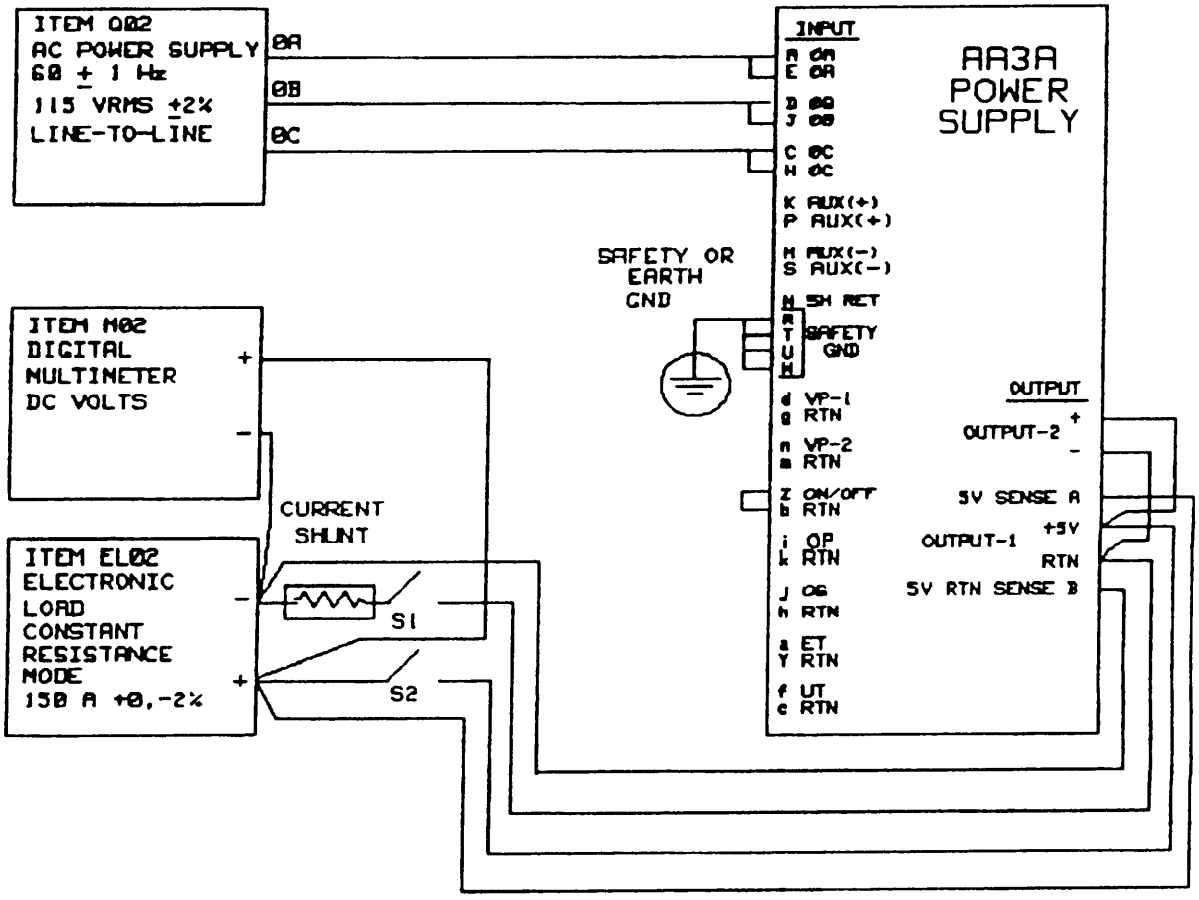


FIGURE 48. Open load lead and return protection test circuit for table XXIV.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = S2 = spst switch.

6. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

7. The two sense leads shall be a twisted pair with a maximum length of 60 inches.

8. The maximum load lead voltage drop to the sense lead connection point (with S1 and S2 closed) shall be 0.358 VDC.

9. The tests shall be performed as follows:

(a) Connect the circuit as shown with both S1 and S2 closed.

(b) For test condition AA1, open the +5V load lead (represented by opening S2) for a minimum of 5 seconds. Reconnect the load lead and measure the output voltage with item M02. The item M02 reading meeting the table XXIV limits verifies the power supply has not been damaged by opening the load lead.

(c) For test condition AA2, repeat (b) above but by opening the RTN load lead (represented by opening S1) for a minimum of 5 seconds.

10. Opening of the load lead may cause the power supply to go into a shutdown mode and require a manual reset. If this is the case, the input voltage shall be taken to zero and then reapplied prior to performing the test condition AA1 or AA2 measurements.

FIGURE 48. Open load lead and return protection test circuit for table XXIV - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XXV. Spike voltage protection test.

Test cond	Parameter	Initial limits 25 + 5°C mounting surface temperature		Unit	Test Fig
		Min	Max		
BB1	Spike voltage protection, AC input, initial transient (during first 20 us).	-253	253	mV	49,50
BB2	Spike voltage protection, AC input, secondary transient.	-126	126	mV	
BB3	Spike voltage protection, auxiliary input, initial transient. (during first 20 us).	-253	253	mV	
BB4	Spike voltage protection, auxiliary input, secondary transient.	-126	126	mV	49,50

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

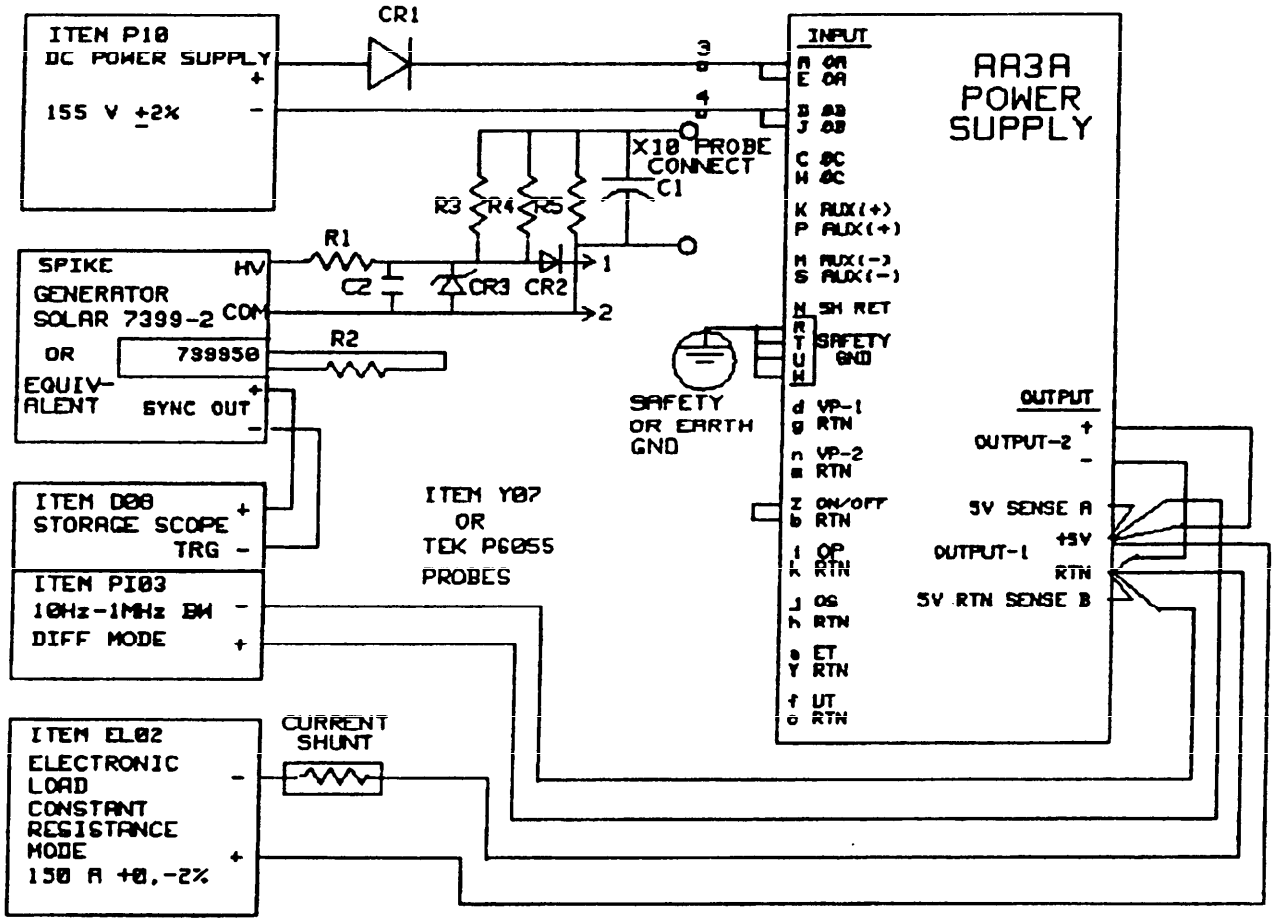


FIGURE 49. Spike voltage protection (AC input) test circuit for table XXV.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. CR1 = CR2 = International rectifier, 12FL100S05 or equivalent.

6. CR3 = Transorb, general semiconductor PIP120 or equivalent (see note 12(b)).

7. R1 = 7 Ohm ± 1 percent, 10 watt resistor.

8. R2 = 0.5 Ohm ± 5 percent, 250 watt resistor (see note 12(b)).

9. R3 = R4 = 180 kOhm ± 5 percent, 1 watt carbon composition resistor. R5 = 10 kOhm ± 5 percent, 1/2 watt carbon composition resistor. C1 = 27 picofarad ± 10 percent, 200 volts capacitor.

Select the resistors so:

$$\frac{R3//R4}{R5} = \frac{90\ k\text{Ohm}}{10\ k\text{Ohm}} = 9.0 \pm 1 \text{ percent}$$

A X100 probe may be used directly in lieu of these components.

10. C2 = Selected (start at approximately 1 uF), 400 volts, (see note 12 (b)).

11. WARNING: High voltage is present during this test. Exercise caution and observe the procedures given in the spike generator instruction book. The spike generator shall be in the single pulse mode. Allow a minimum of 30 seconds between spikes while adjusting the waveform.

FIGURE 49. Spike voltage protection (AC input) test circuit for table XXV - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

12. The input voltage waveform shown on figure 50 shall be obtained at points 1 and 2 (as measured with an oscilloscope connected to the X10 PROBE CONNECT points) prior to connection to the power supply under test.

(a) Adjust the spike generator to charge its internal capacitors to approximately 1500 volts (using the spike generator meter) prior to discharging to produce the spike.

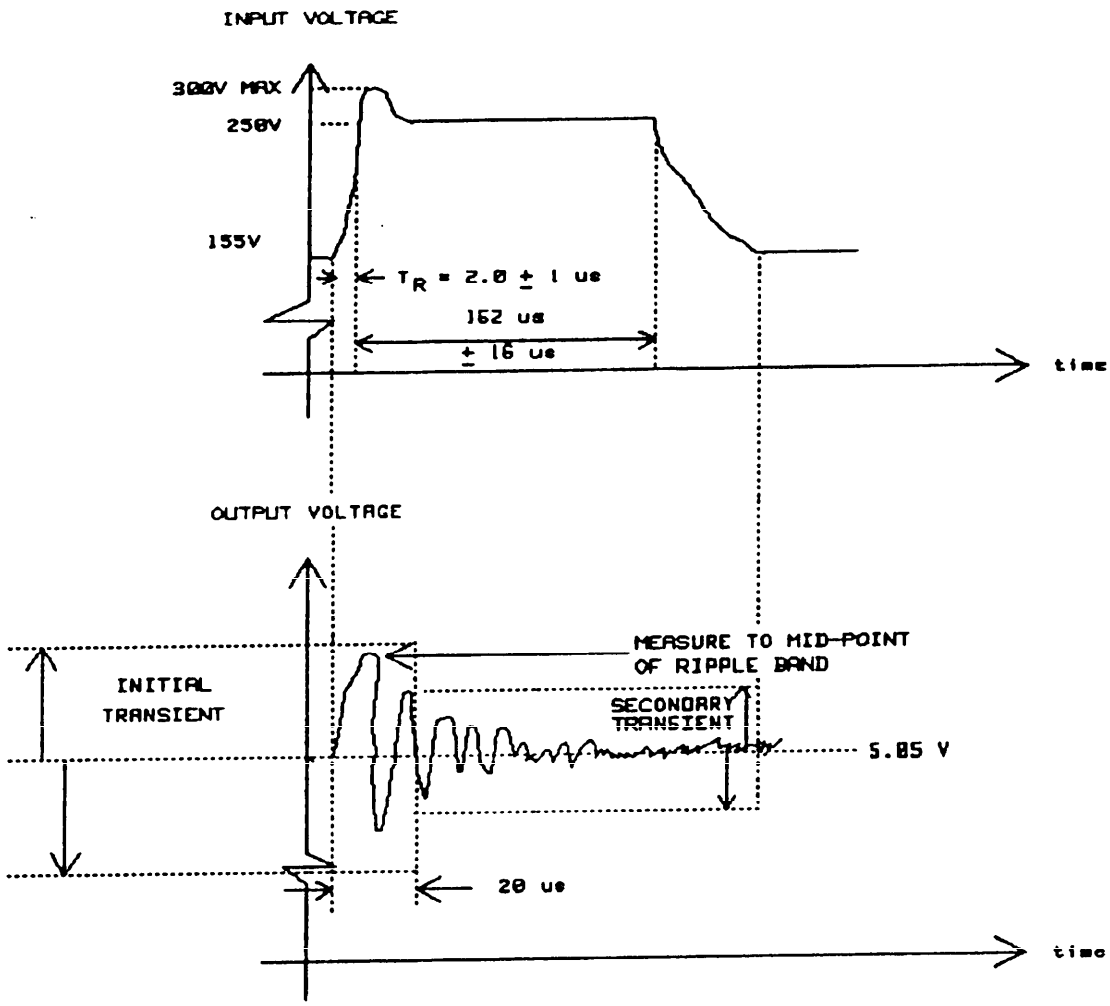
(b) Trigger the spike generator and observe that the input voltage waveform as shown on figure 50 (except the waveform will start at 0 volts and decay to 0 volts) is obtained. Select C2 to give the specified rise time. Increasing or decreasing resistor R2 will increase or decrease the pulse width. Selection of CR3 (or paralleling of more than one transorb) will set the 250 volts peak level.

13. Perform test conditions $\overline{BB1}$ and $\overline{BB2}$ (after the input waveform has been adjusted according to note 12) as follows: Connect point 1 to point 3 and point 2 to point 4. Apply a single voltage spike to the power supply under test and observe that the output waveform meets the test limits. Do not reapply the input spike voltage to the power supply at intervals of less than one minute (the spike voltage and current limiting resistor R1 have been scaled down, but simulate the 2500 volt spike voltage of DOD-STD-1399, Section 300 clipped to 250 volts).

14. Perform test conditions BB3 and BB4 (after the input waveform has been adjusted according to note 12) as follows: Connect the circuit as shown except point 3 shall be connected to input contacts K and P and point 4 shall be connected to input contacts M and S. Repeat the procedure of note 13.

FIGURE 49. Spike voltage protection (AC input) test circuit for table XXV - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.



NOTES:

1. The initial transient shall be defined as the highest output transient (plus or minus) occurring during the first 20 microseconds after application of the input spike voltage.

2. Secondary transient shall be defined as the highest output transient (plus or minus) occurring between 20 us after the spike application and the end of the input spike.

FIGURE 50. Spike voltage protection definition for table XXV.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XXVI. Reverse voltage protection and on/off parallel outputs test.

Test cond	Parameter	Connect 1/		Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		1 to points	2 to points	Min	Max		
CC	Reverse voltage prot., aux input	-	-	5.00	5.10	VDC	31
001	Reverse voltage prot., on output, 150 A	3,5	4,6	-1.0	0.8		32
002	Reverse voltage bias, turn-on	3,5	4,6	5.00	5.10		
003	Reverse voltage prot., on output, 80 A	5	6	-1.0	0.8		
004	Reverse voltage bias, turn-on	5	6	5.00	5.10		
005	Reverse voltage prot., on output, 70 A	3	4	-1.0	0.8		
006	Reverse voltage bias, turn-on	3	4	5.00	5.10		32
007	ON/OFF parallel outputs, single output	3,5	4,6	5.00	5.10		32
008	ON/OFF parallel outputs, OS high	-	-	2.4	5.5	VDC	
009	ON/OFF parallel outputs, OS LED off	-	-	output status LED is off		-	
0010	ON/OFF parallel outputs, dual, output-1, off	5	6	5.00	5.10	VDC	
0011	Positive voltage bias, turn-on, single output	-	-	4.95	5.15	VDC	
0012	Positive voltage bias, turn-on, output-1	-	-	4.95	5.15	VDC	
0013	Positive voltage bias, turn-on, output-2	-	-	4.95	5.15	VDC	32

1/ Two numbers in one block indicate that these two points shall be connected together in addition to being connected to points 1 or 2, as applicable (for example, "3,5" under column "1 to points" indicates point 3 connected to point 5 connected to point 1).

Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

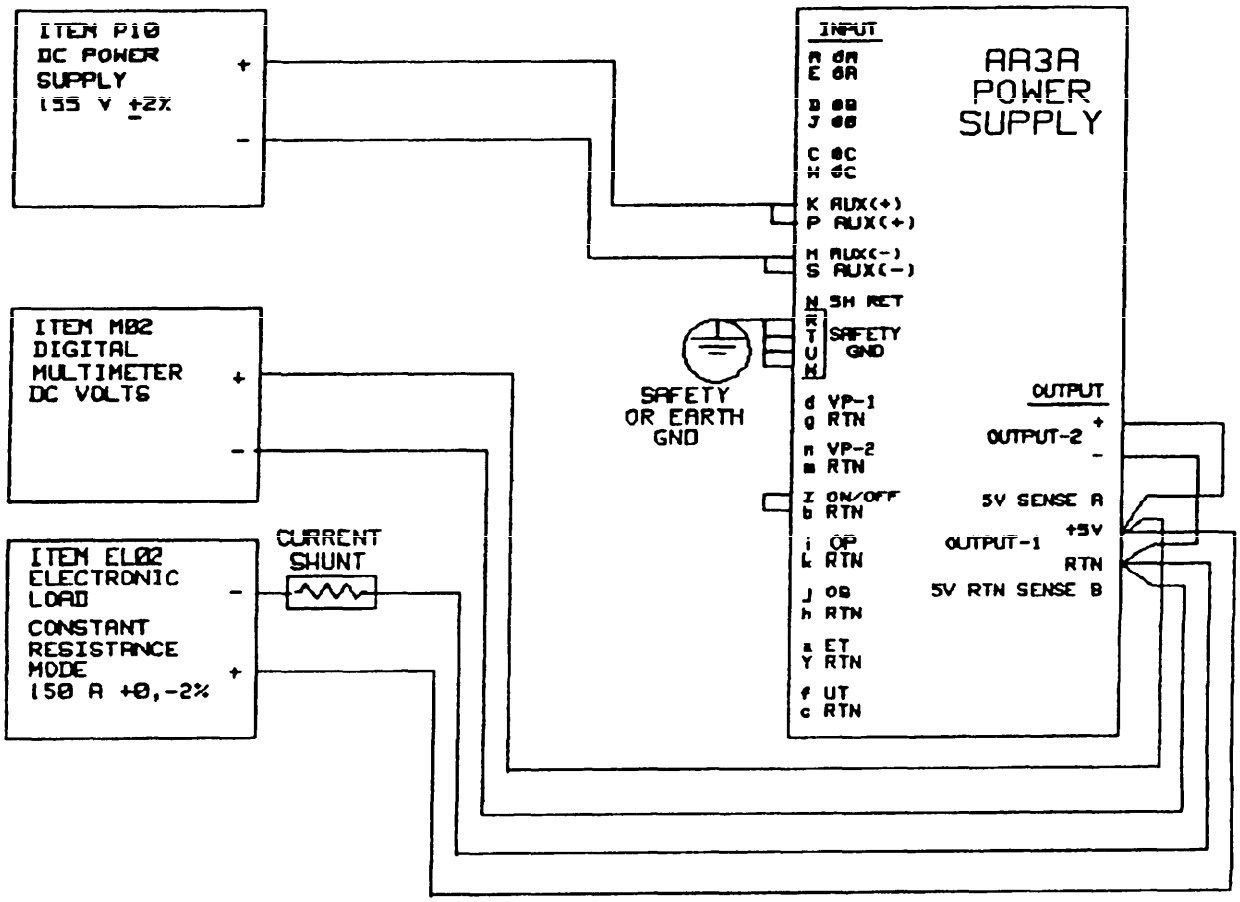


FIGURE 51. Reverse voltage protection test circuit for table XXVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. For test condition CC, connect as shown, then reverse the inputs polarity connection for a minimum of 5 seconds. Then reconnect to the circuit as shown and read the voltage on the item M02 multimeter. This voltage shall be within table XXVI limits. Item P10 shall be turned off when reversing the input leads and then turned back on.

FIGURE 51. Reverse voltage protection test circuit for table XXVI - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

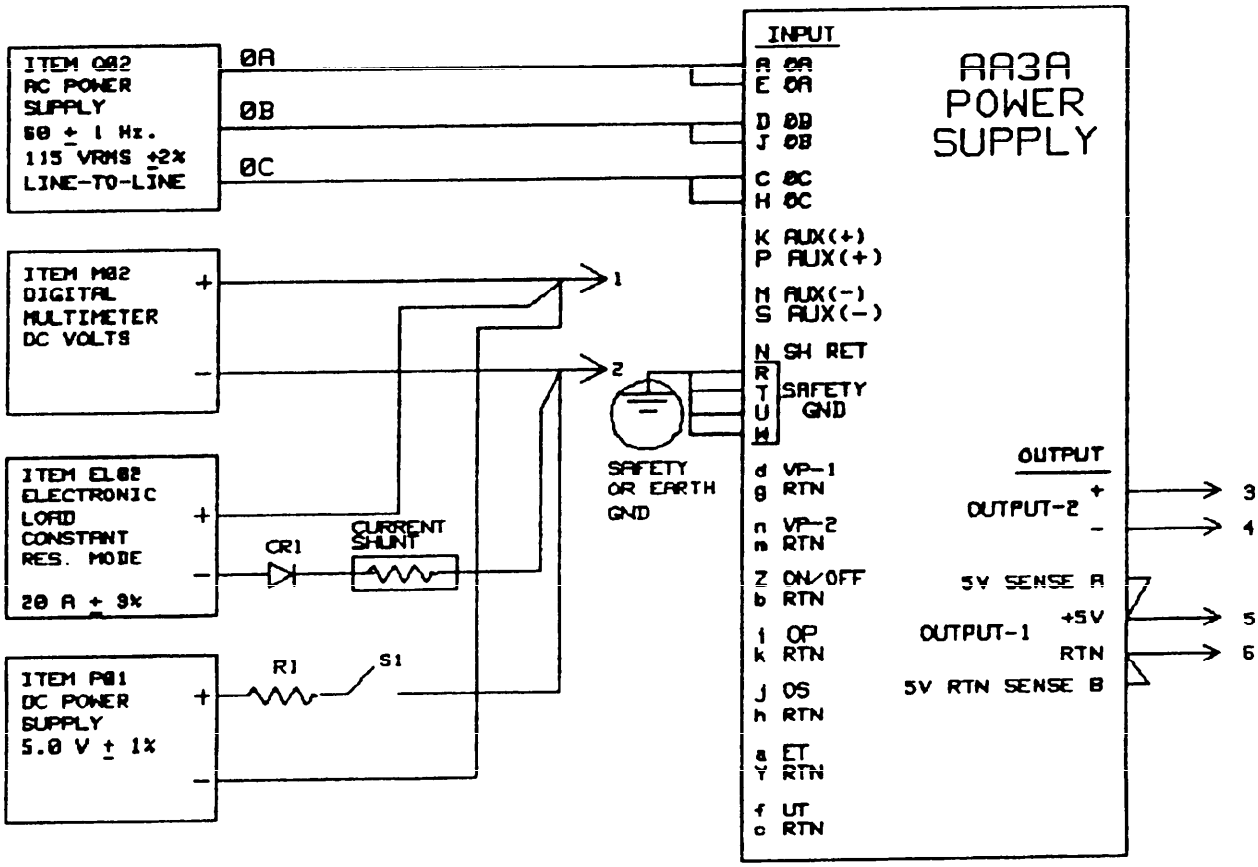


FIGURE 52. Reverse voltage protection/bias test circuit for table XXVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

(d) Repeat the above procedure for test conditions 22 and

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 50A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = spst switch.

6. R1 = 3 Ohm ± 10 percent, 50 watt resistor.

7. CR1 = 1N2132 or equivalent diode (require forward current = 20 amperes, and reverse voltage = 10 volts).

8. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

9. Test conditions DD1-DD2:

(a) Connect per table XXVI and this figure with S1 open.

(b) With item Q02 on, adjust item EL02 for 15 amperes and turn item Q02 off.

(c) Close S1 and read item M02 for test condition DD1 measurement.

(d) Turn item Q02 on and read item M02 for test condition DD2 measurement.

10. Test conditions DD3-DD6:

(a) Short input contacts i and k.

(b) Repeat note 8 procedure with connections according to table XXVI for test conditions DD3 and DD4 and for test conditions DD5 and DD6.

FIGURE 52. Reverse voltage protection/bias test circuit for table XXVI - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

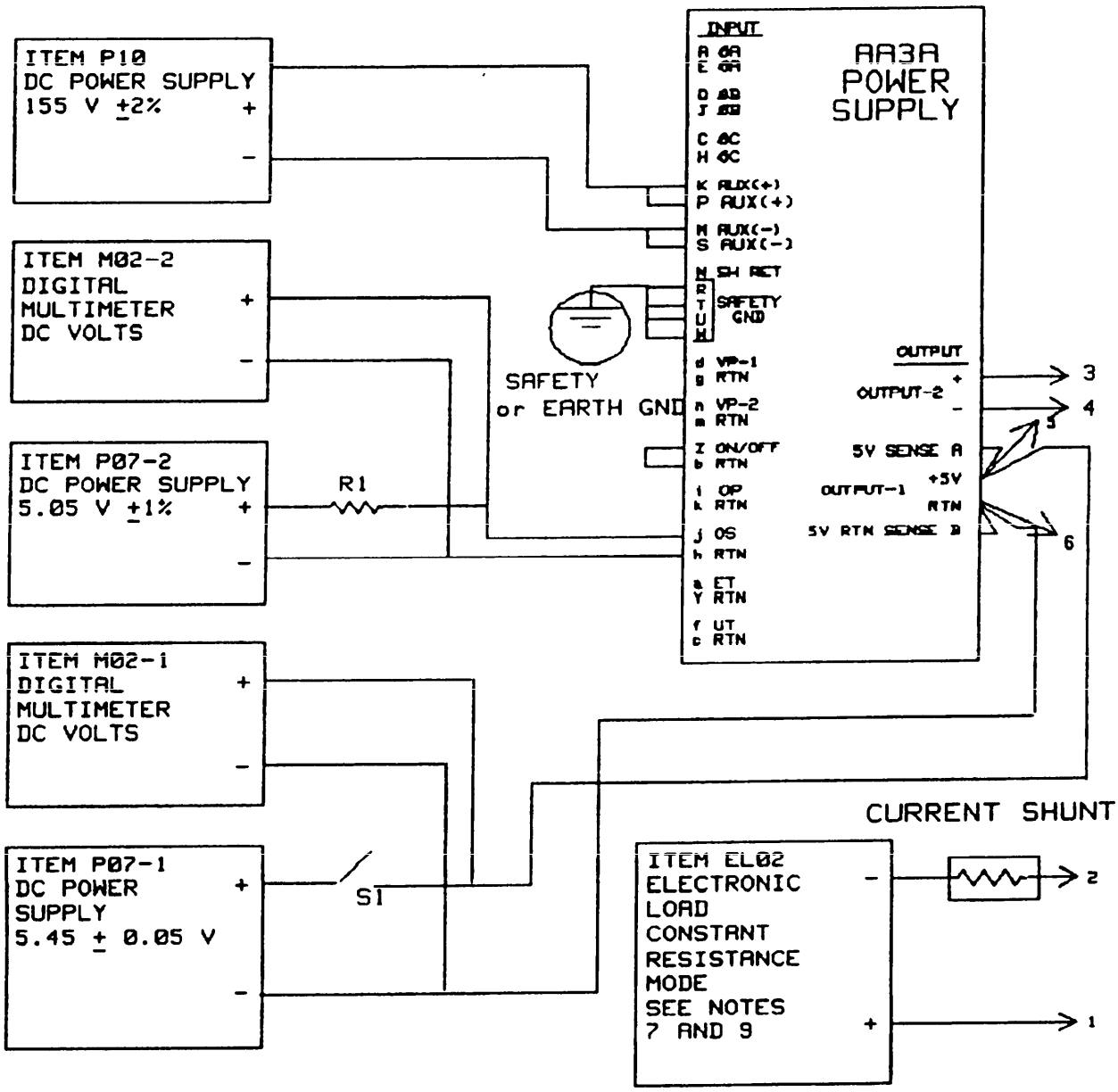


FIGURE 53. ON/OFF parallel outputs test circuit for table XXVI.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. R1 = 1.69 kOhm ± 1 percent, 1/8 watt resistor.

6. S1 = spst switch.

7. For test condition DD7, use the following procedure (see note 10):

(a) Connect according to table XXVI and this figure with S1 open.

(b) With item P10 on, set item EL02 to 150 amperes +0, -2 percent.

(c) Turn off the power supply under test by opening input contacts Z and b. Disconnect item EL02 (keep 3 connected to 5, and 4 connected to 6).

(d) Adjust item P07-1 to 5.05 volts ± 1 percent, close switch S1 for a minimum of 5 seconds (see note 8), then open S1. Repeat this four more times.

(e) With switch S1 open, connect item EL02, short input contacts Z and b, and measure the output voltage with item M02-1 to verify that the AA3A power supply was not damaged.

8. For test condition DD8, verify that item M02-2 remains within the limits of test condition DD8 when switch S1 is closed in 7(d) above. For test condition DD9, verify that the output status LED is off when switch S1 is closed in 7(d) above (see note 10).

FIGURE 53. ON/OFF parallel outputs test circuit for table XXVI - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

9. For test condition DD10 (see note 10).

(a) Connect according to table XXVI and this figure with S1 open.

(b) Short input contacts i to k.

(c) With item P10 on, set item EL02 to 80 amperes + 0, -2 percent.

(d) Turn off the power supply under test by opening input pins Z and b. Disconnect item EL02.

(e) Adjust item P07-1 to 5.05 volts ± 1 percent, close switch S1 for a minimum of 5 seconds, then open S1. Repeat this four more times.

(f) With switch S1 open, connect item EL02, short input contacts Z and b, and measure the output voltage with item M02-1 to verify that the AA3A power supply was not damaged.

10. The purpose of test conditions DD7 through DD10 is to simulate the connection of two or more power supplies in parallel when only one power supply is to be on at any one time.

11. For test condition DD11:

(a) Connect according to this figure with item P10 off, S1 closed and item P07-1 set to zero. Short points 3 to 5 and points 4 to 6 (item EL02 is not required).

(b) Adjust item P07-1 until item M02-1 reads 5.30 VDC ± 1 percent.

(c) Turn item P10 on. Open input contacts Z and b (remote ON/OFF). Short input contacts Z and b. Open switch S1. The item M02-1 voltage reading shall meet the table XXVI limits.

12. For test condition DD12:

(a) Short contacts i to k. Connect according to this figure with item P10 off, S1 closed and item P07-1 set to zero (item EL02 is not required and points 3 and 4 are not connected).

FIGURE 53. ON/OFF parallel outputs test circuit for table XXVI - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

(b) Adjust item P07-1 until item M02-1 reads 5.30 VDC ± 1 percent.

(c) Turn item P10 on. Open input contacts Z and b (remote ON/OFF). Short input contacts Z and b. Open switch S1. The item M02-1 voltage reading shall meet the table XXVI limits.

13. For test condition DD13:

(a) Short contacts i to k. Connect according to this figure except connect item M02-1, item P07-1, and R1 to points 3 and 4, respectively, instead of to points 5 and 6. Initially have item P10 off, S1 closed and item P07-1 set to zero.

(b) Adjust item P07-1 until item M02-1 reads 5.30 VDC ± 1 percent.

(c) Turn item P10 on. Open input contacts Z and b (remote ON/OFF). Short input contacts Z and b. Open switch S1. The item M02-1 voltage reading shall meet the table XXVI limits.

FIGURE 53. ON/OFF parallel outputs test circuit for table XXVI - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XXVII. Common mode output current test.

Test cond	Parameter	Connect point 1 to point:	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
			Min	Max		
EE1	Common mode current broadband, 150 A	3	0	200	mAp-p	54
EE2	Common mode current broadband, 75 A					54
EE3	Common mode current broadband, 40 A, 35 A, output-1					55
EE4	Common mode current broadband, 80 A, 70 A, output-1					
EE5	Common mode current broadband, 40 A, 35 A, output-2					
EE6	Common mode current broadband, 80 A, 70 A, output-2	3	0	200	mAp-p	55
EE7	Common mode current spectral line, 50Hz - 10kHz 1/ 2/	2	-INF	-80 3/	dBV / √Hz	54,55
EE8	Common mode current spectral line, 10kHz - 10MHz 1/ 2/	2	-INF	-56.5 3/		54,55
EE9	Common mode current spectral line, largest amplitude signal 1/ 2/	2	-INF	-46 3/	dBV / √Hz	54,55

-See footnotes on next page.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

1/ Test conditions EE7-EE9 shall be run using the setup from test conditions EE1-EE6 which gives the largest magnitude broadband current.

2/ Do not apply bandwidth correction to the EE7-EE9 limits.

3/ Common mode current (dBA) = (item B05 reading in dBV) - 20 Log (Z) = (item B05 reading in dBV + 20 dB)

Where Z = current probe transfer function of mV/mA (shown calculated above for z = 0.1), dBA = decibel (ref 1 ampere), and dBV = decibel (ref 1 volt). The table XXVII limits for EE7, EE8 and EE9 will change in accordance with the above relationship if z is not equal to 0.1. Refer to table I for dBA/$\sqrt{Hz}$ limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

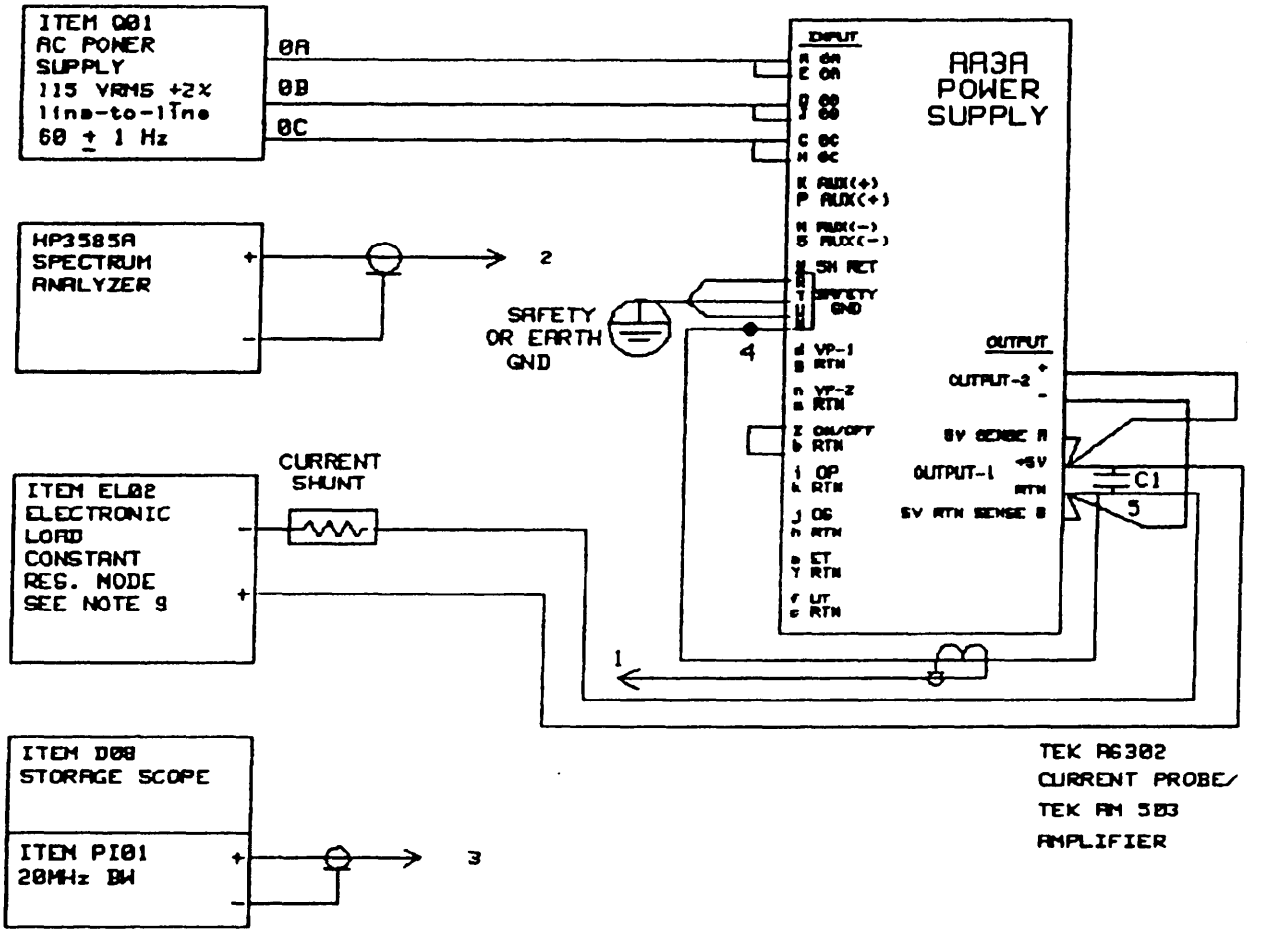


FIGURE 54. Common mode output current (single output) test circuit for table XXVII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. C1 = 10 uF ± 5 percent, 50 VDC, MIL-C-83421/01-359, or equivalent capacitor.

6. For test conditions EE1 and EE2, place the current probe around the wire going from point 4 to point 5. Measure the total peak-to-peak current observed on item D08 oscilloscope (including all spikes, hash, periodic and aperiodic components). Use the single sweep storage mode.

7. For test condition EE7, sweep the spectrum analyzer frequency between 50 Hz and 10 kHz. Use a 10 Hz bandwidth to make final measurements verifying the power supply meets the specified limits. The setup from test condition EE1-EE6 which gave the largest magnitude current shall be used.

8. For test conditions EE8 and EE9, sweep the spectrum analyzer frequency between 10 kHz and 10 MHz. Use a 30 Hz bandwidth to make final measurements verifying the power supply meets the specified limits. For test condition EE9, measure the spectral line of the largest amplitude observed on the spectrum analyzer. For test condition EE8, measure the spectral line of the second largest amplitude observed on the spectrum analyzer.

9. Item EL02 shall be set to 150 amperes +0, -2 percent for test condition EE1 and to 75 amperes ± 1 percent for test condition EE2. (The load used, if different than EL02, shall have its + and - terminals isolated from safety or chassis ground so the common mode current will flow to point 4).

FIGURE 54. Common mode output current (single output) test circuit for table XXVII - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

10. The connection from point 4 (safety gnd) to point 5 (output-1 RTN) shall be made using a 4 ± 0.125 inch piece of copper, stranded number 16 gauge wire (length measurements shall be made from the mating connector solder cups). The connection from the power supply output contacts to the C1 capacitor shall be kept as short as possible. The wires from input connector contacts R, T and U to the common tie point shall each be a 4.0 ± 0.5 inch piece of number 16 gauge wire. The output conections of output-2 + to output-1 +5V and output-2- to output-1 rtn shall be bussed together within 1.0 inch distance from the power supply mating connector.

11. When performing the tests of table XXVII, ensure that the current being measured is generated by the power supply under test and is not spurious noise from external sources.

FIGURE 54. Common mode output current (single output) test circuit for table XXVII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

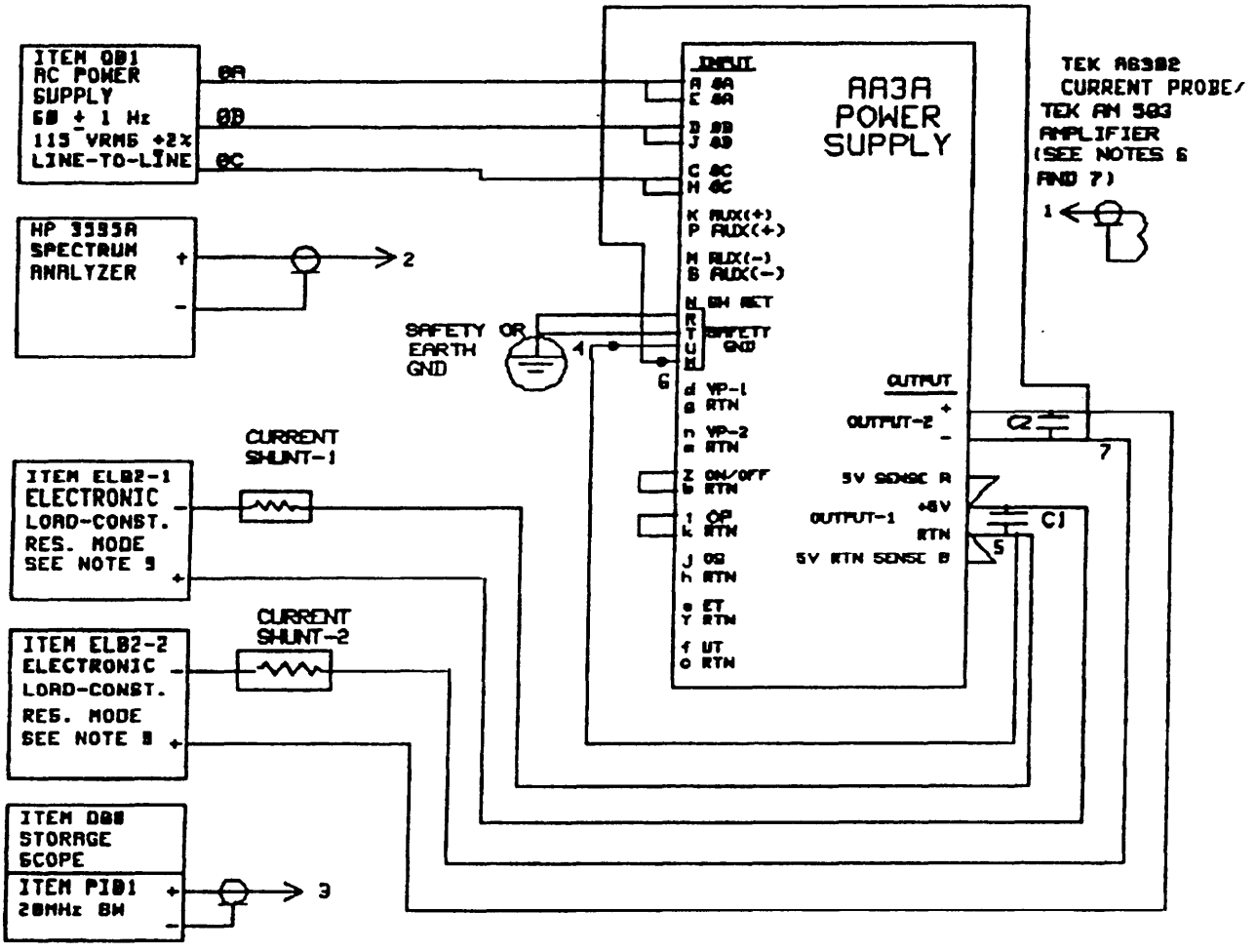


FIGURE 55. Common mode output current (dual output) test circuit for table XXVII.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. C1 = C2 = 10 uF ± 5 percent, 50 VDC, MIL-C-83421/01-359, or equivalent capacitor.

6. For test conditions EE3 and EE4, place the current probe around the wire going from point 4 to point 5. Measure the total peak-to-peak current observed on item D08 oscilloscope (including all spikes, hash, periodic and aperiodic components). Use the single sweep storage mode.

7. For test conditions EE5 and EE6, place the current probe around the wire going from point 6 to point 7. Measure the total peak-to-peak current observed on item D08 oscilloscope (including all spikes, hash, periodic and aperiodic components). Use the single sweep storage mode.

8. If test conditions EE7-EE9 are performed using this setup, then notes 7, 8 and 11 on figure 54 apply.

9. The electronic loads shall be set as follows:

TEST CONDITION	EL02-1(+0%,-2%)	EL02-2(+0%,-2%)
EE3	40 amperes	35 amperes
EE4	80 amperes	70 amperes
EE5	40 amperes	35 amperes
EE6	80 amperes	70 amperes

(The load used, if different then EL02, shall have its + and - terminals isolated from ground so the common mode current will flow to point 4 and point 6).

FIGURE 55. Common mode output current (dual output) test circuit for table XXVII - Continued.

Source: https://assist.dla.mil -- Downloaded: 20261-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES: - Continued.

10. The connections from safety ground to the output returns (point 4 to point 5 and point 6 to point 7) shall be made using a 4.0 ± 0.125 inch of copper, stranded, number 16 gauge wire (length measurements shall be made from the mating connector solder cups). The connection from the power supply output contacts to the C1-C2 capacitors and points 5 and 7 shall be kept as short as possible. The wires from input connector contacts R and T to the common tie point shall each be a 4.0 ± 0.5 inch of stranded, number 16 gauge wire.

FIGURE 55. Common mode output current (dual output) test circuit for table XXVII - Continued.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XXVIII. Voltage programming contacts signal rejection test.

Test cond	Parameter	Connect		Connect		Spectrum analyzer frequency ±1%	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		1 to input contact	2 to input contact	7 to point	8 to point		Min	Max		
FF1	VP-1 rejection at 100 Hz	d	g	5	6	100 Hz	15	-	dB	58
FF2	VP-1 rejection at 5 kHz	d	g	5	6	5 kHz				
FF3	VP-1 rejection at 50 kHz	d	g	5	6	50 kHz				
FF4	VP-2 rejection at 100 Hz	n	m	3	4	100 Hz				
FF5	VP-2 rejection at 5 kHz	n	m	3	4	5 kHz				
FF6	VP-2 rejection at 50 kHz	n	m	3	4	50 kHz	15	-	dB	58

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T01:52Z
Check the source to verify that this is the current version before use.

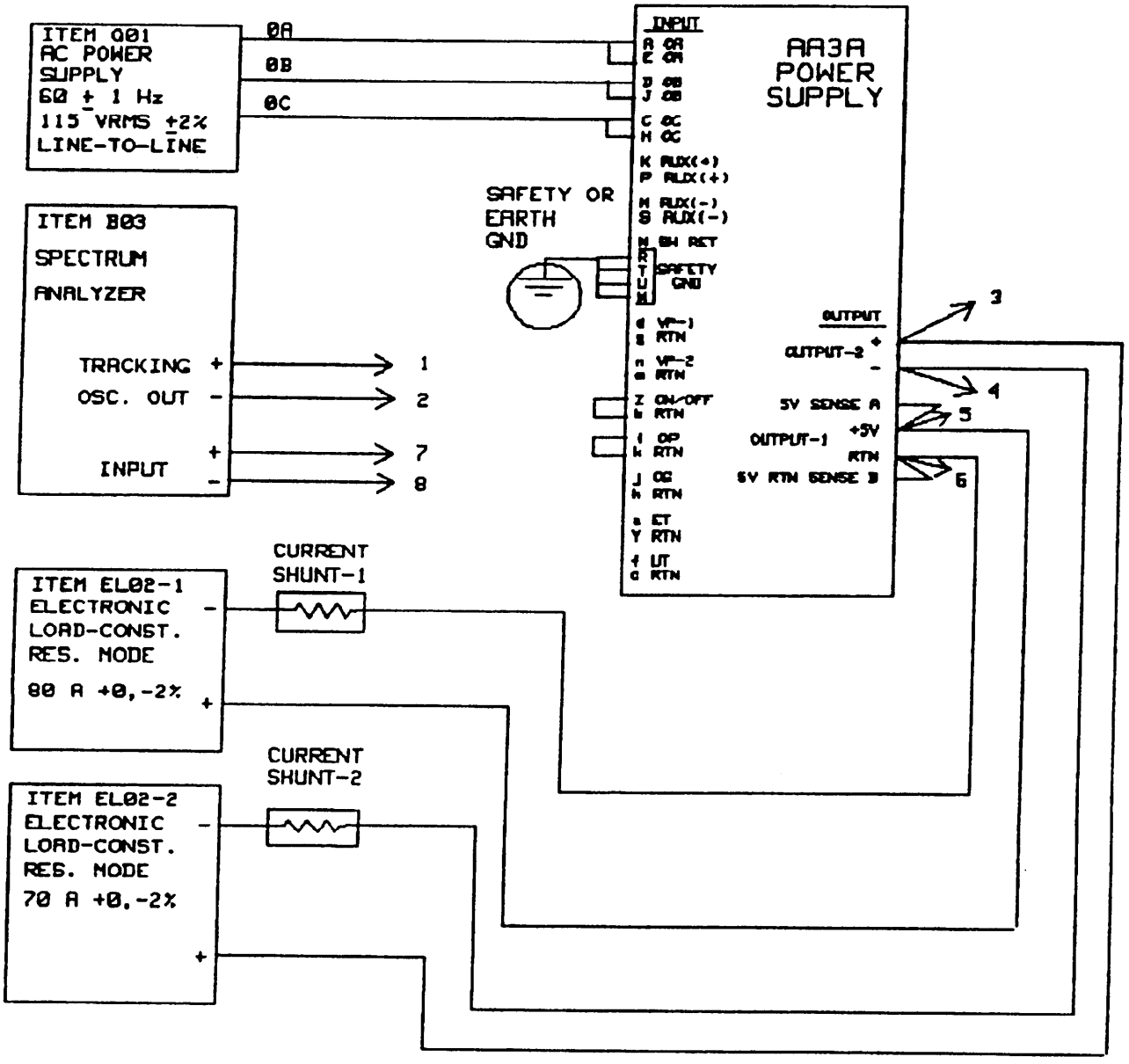


FIGURE 56. Voltage programming contacts signal rejection test circuit for table XXVIII.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 150A/50mV.

4. Monitor shunt voltage with an item VO4 voltmeter to ensure that the load current is within specification.

5. For each test condition, set the tracking oscillator output to 0.1 VRMS ± 5 percent (-20 ± 0.5 dBV), frequency per table XXVIII between points 1 and 2 with these points connected. Measure the AC voltage on the indicated power supply output with the spectrum analyzer. The spectrum analyzer measurement shall be 15 dB or greater below the -20 dB input signal (-20 dBV minus measurement level ≥ 15 dB). (If the tracking oscillator output cannot be adjusted as high as 0.1 VRMS, set it to the highest value obtainable. The spectrum analyzer measurement shall still be 15 dB or greater below the input signal.)

6. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

FIGURE 56. Voltage programming contacts signal rejection test circuit for table XXVIII - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XXIX. Input voltage and current characteristics test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
GG1	Input voltage turn-on threshold	100	135	VDC	57
GG2	Input voltage turn-off threshold	-	100		57
GG3	Input voltage turn-on/turn-off hysteresis 1/	10	-		-
GG4	Input voltage turn-on threshold, auxiliary input	100	135		57
GG5	Input voltage turn-off threshold, auxiliary input	-	100		57
GG6	Input voltage turn-on/turn-off hysteresis, aux. input 1/	10	-	VDC	-
GG7	Maximum steady state input current	-	14.7	ADC	57
GG8	Slowly rising input peak current		14.7	Apk	
GG9	Slowly rising input rate of current rise		11.4	A/ms	
GG10	Input peak current, remote turn-on		14.7	Apk	
GG11	Input rate of current rise, remote turn-on	-	11.4	A/ms	57

1/ Test conditions GG3 and GG6 are calculations: hysteresis = turn-on threshold minus turn-off threshold.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

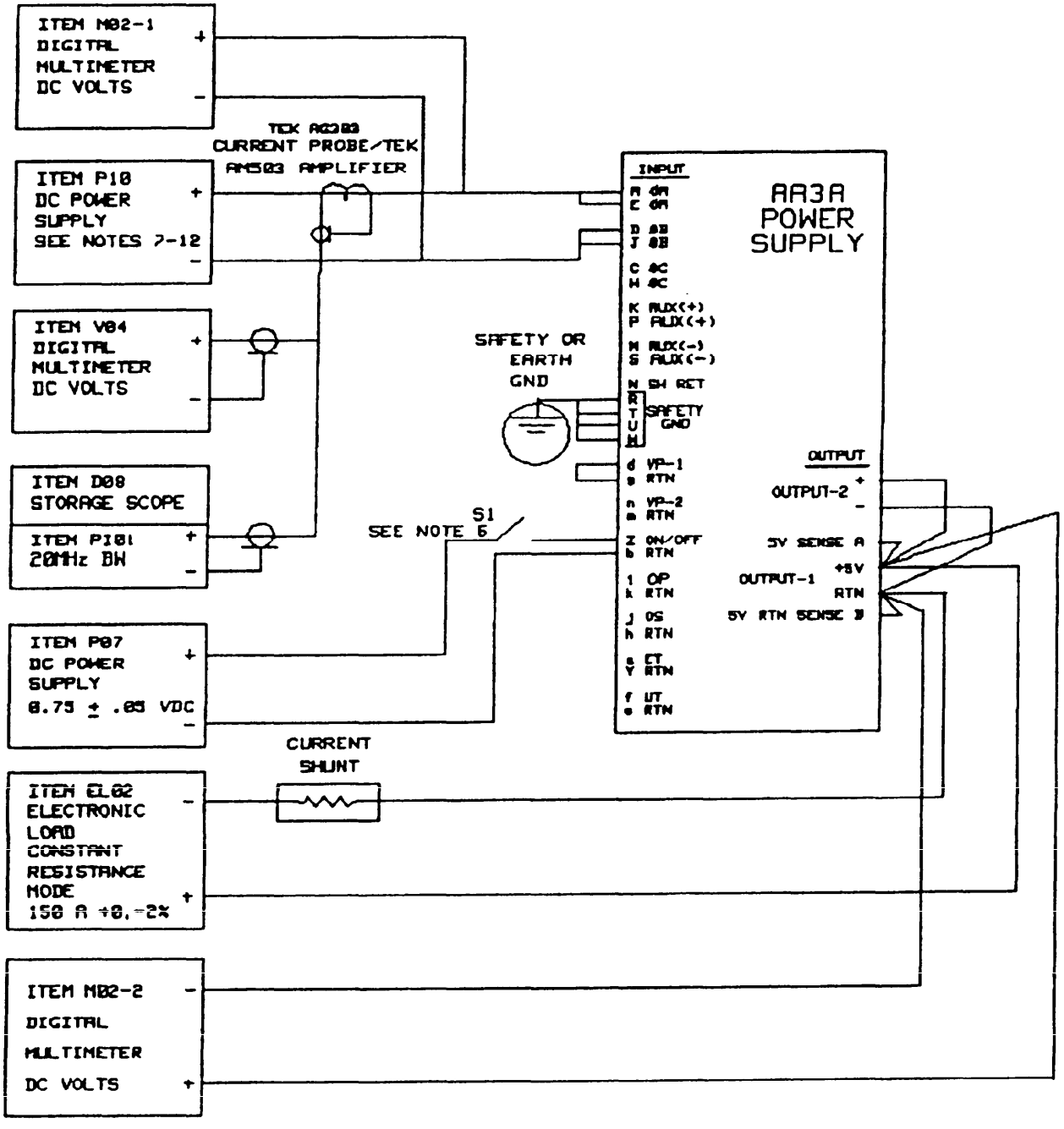


FIGURE 57. Input voltage and current characteristics test circuit for table XXIX.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. S1 = spst, 225 VDC, 15 ampere switch.

6. Switch S1 shall be closed except as indicated in note 12 for test conditions GG10 and GG11.

7. For test conditions GG1, GG2, and GG7 through GG11, connect item P10 power supply to the AC inputs as shown. For test conditions GG4 and GG5, connect item P10 and item M02-1 to the auxiliary inputs: + to input contacts K and P, - to input contacts M and S.

8. Test conditions GG1 and GG4 determine the AC/DC input and auxiliary input voltage threshold for power supply turn-on. Perform the test by slowly increasing the item P10 voltage (1 volt per second or less) from 0 volt until the output voltage, as measured with the item M02-2, just equals or exceeds 5.00 volts. The item M02-1 voltage at this point shall meet the test condition GG1 or GG4 limits.

9. Test conditions GG2 and GG5 determine the AC/DC input and auxiliary input voltage threshold for power supply turn-off. Perform the test by slowly decreasing the item P10 voltage (1 volt per second or less) from 155 volts until the output voltage, as measured with the item M02-2, just equals or drops below 5.00 volts. The item M02-1 voltage at this point shall meet the test condition GG2 or GG5 limits.

10. For test condition GG7, slowly (1 volt per second or less) sweep the item P10 voltage from 0 to 202 volts and back to 0 volts. The maximum steady state input current observed as measured with the current probe and item V04 shall meet the GG7 limits.

FIGURE 57. Input voltage and current characteristics test circuit for table XXIX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

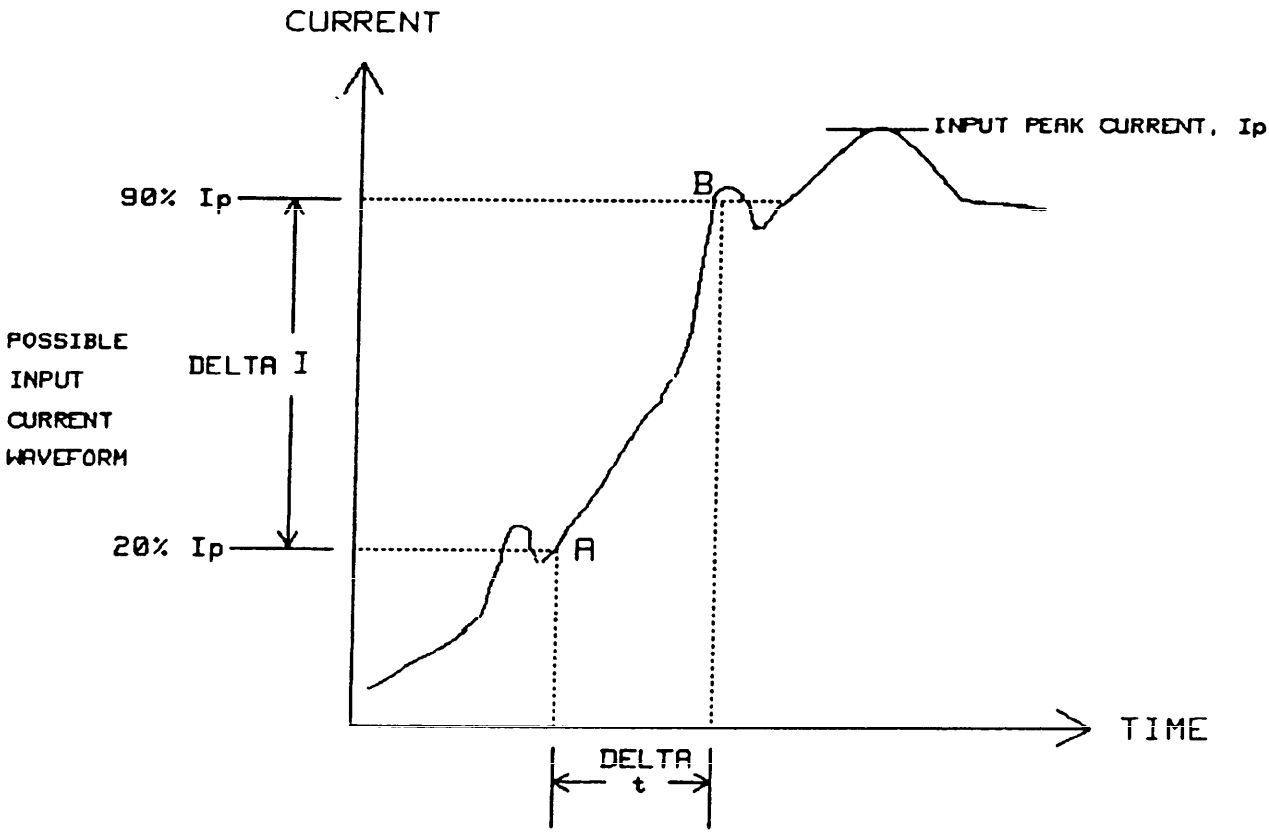
NOTES: - Continued.

11. For test conditions GG8 and GG9, increase the item P10 voltage, from 0 to 140 volts (above the turn-on threshold) at a rate of 1 volt per millisecond or less. Use the current probe and item D08 to store the input current waveform as the power supply turns on. The input current observed shall meet the GG8 and GG9 limits (see figure 58). The test shall be performed several times and the highest values recorded.

12. For test conditions GG10 and GG11, close switch S1 and adjust item P10 to 155 volts. Open switch S1. Use the current probe and item D08 to store the input current waveform as S1 is closed and the power supply turns on. The input current observed shall meet the GG10 and GG11 limits (see figure 58). The test shall be performed several times and the highest values recorded.

FIGURE 57. Input voltage and current characteristics test circuit for table XXIX - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.



NOTE:

1. The rate of current rise to be recorded for test condition GG9 or GG11 is the slope between points A and B on the rising input current vs time waveform. If the curve crosses the value of current represented by 20 percent Ip and/or 90 percent Ip more than once, the time between the last time it crosses 20 percent Ip and the first time it crosses 90 percent Ip shall be taken as the recorded delta t. The slope is defined as

rate of current rise = DELTA I in Amps/DELTA t in milliseconds.

(The figure is for illustrative purposes only and the input current waveform may look different.)

FIGURE 58. Input current waveform for table XXIX.

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XXX. Incorrect output programming protection.

Test cond	Parameter 1/	Item EL02-1 curr. +0,-2% 3/	Item EL02-2 curr. +0,-2% 3/	Connect i to point 2/	OP input contacts i to k	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
						Min	Max		
HH1	Output voltage	80A	0A	2	open	0.0	6.45	VDC	59
HH2		80A	0A	3					
HH3		0A	70A	2					
HH4		0A		3	open				
HH5		80A		2,3	shorted	0.0	6.45		
HH6		80A		2	shorted	5.00	5.10		
HH7	Output voltage	80A	70A	3	shorted	5.00	5.10	VDC	59

-See footnotes on next page.

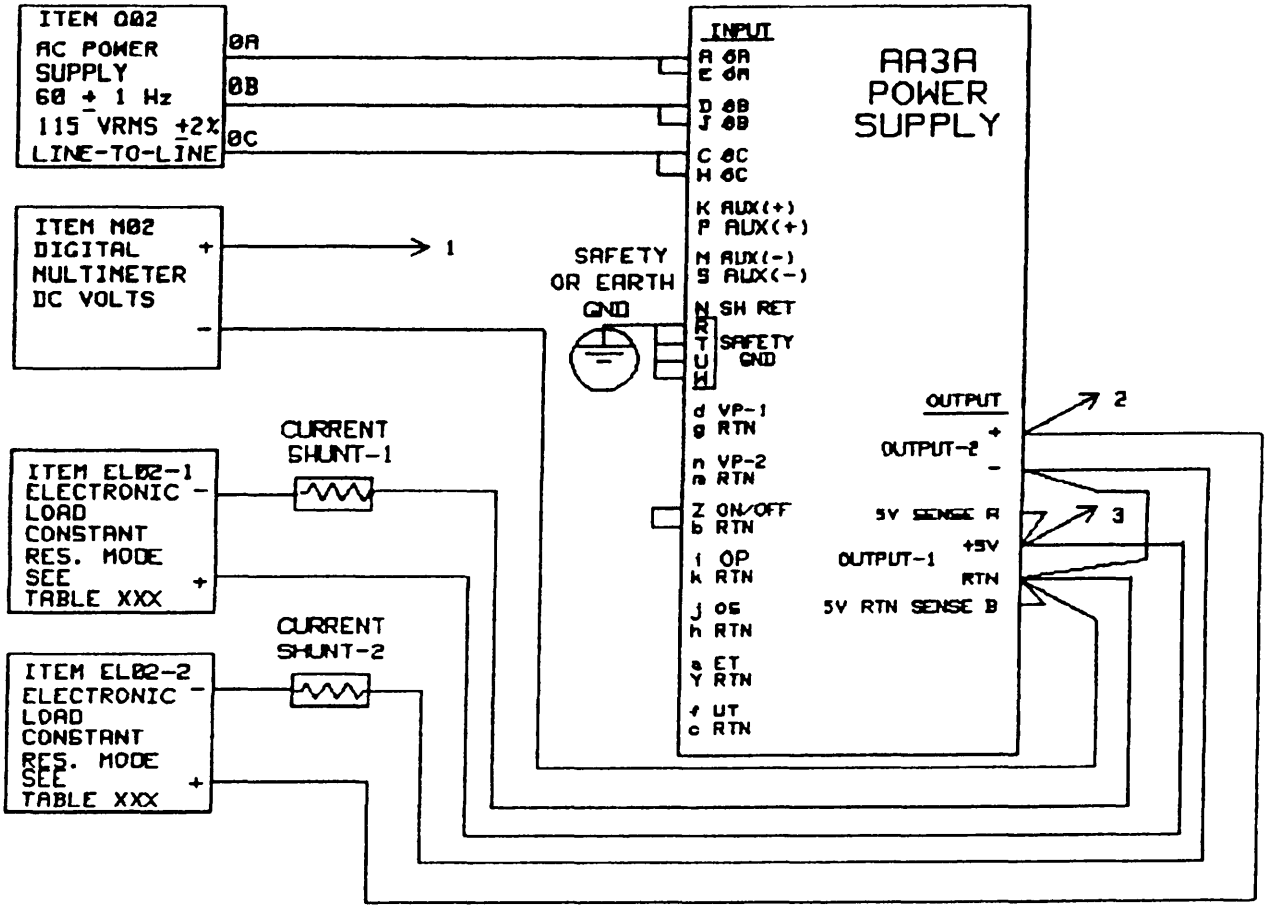
Source: https://assist.dla.mil -- Downloaded: 2026/10-20T01:52Z
Check the source to verify that this is the current version before use.

1/ Test conditions HH1-HH4 programs the power supply for a single output and operates as a dual output. Test condition HH5 programs the power supply for a dual output and operates as a single output. Test conditions HH6 and HH7 programs and operates the power supply as a dual output to verify no damage.

2/ Two numbers in one block (test condition HH5) indicate that points 2 and 3 shall be connected together in addition to being connected to point 1.

3/ For test conditions HH1 through HH5, the current specified for items EL02-1 and EL02-2 is what should be obtained if the output comes up to 5.05 volts. Should the output voltage be different, the current is allowed to vary accordingly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.



NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunts: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

FIGURE 59. Incorrect output programming protection test circuit for table XXX.

Source: https://assist.dla.mil -- Downloaded: 20261-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XXXI. Emergency condition input voltage protection test.

Test cond	Parameter 1/	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
JJ1	Emergency condition input voltage protection	4.99	5.51	VDC	60
JJ2	Emergency condition input voltage protection	5.20	5.30	VDC	60

1/ See figure 60, note 5 for further definition of these test conditions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

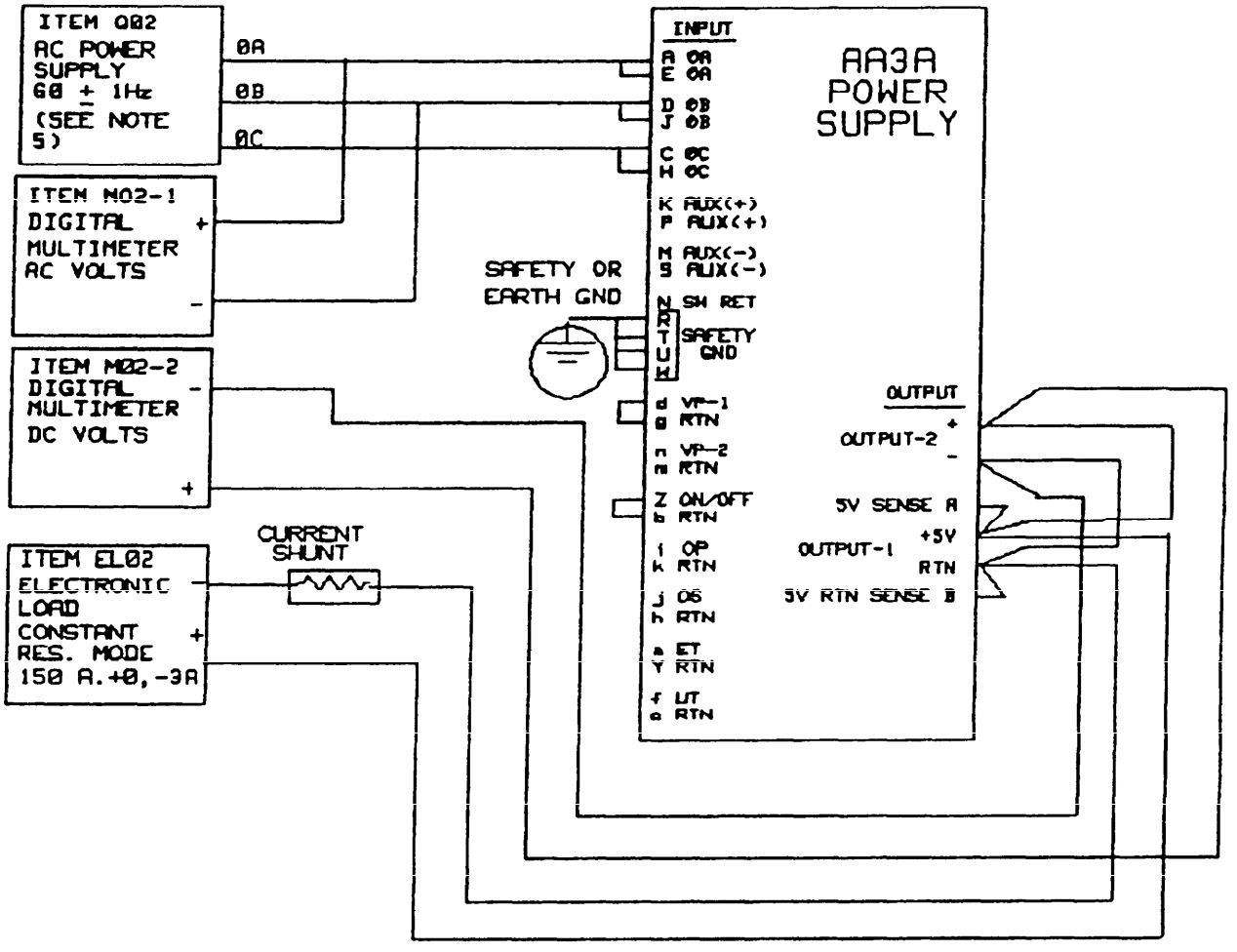


FIGURE 60. Emergency condition input voltage protection test circuit for table XXXI.

Source: https://assist.dla.mil -- Downloaded: 20261O-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. The tests shall be performed as follows:

(a) Connect as shown and set item Q02 to 115 VRMS ± 2 percent line-to-line.

(b) Increase the item Q02 voltage to 155 VRMS ± 1 percent line-to-line. Leave the 155 VRMS applied for a minimum of 2 minutes and then read item M02-2 to verify that the output voltage is within the test condition JJ1 limits.

(c) Reduce the item Q02 voltage to 115 VRMS ± 2 percent line-to-line. The item M02-2 voltage shall be within the test condition JJ2 limits.

FIGURE 60. Emergency condition input voltage protection test circuit for table XXXI - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T01:52Z
Check the source to verify that this is the current version before use.

TABLE XXXII. Output status noise test.

Test cond	Parameter	Initial limits 25 ± 5°C mounting surface temperature		Unit	Test fig
		Min	Max		
KK1	Output status noise, positive	0.0	500	mVp	61
KK2	Output status noise, negative	0.0	-500	mVp	61

Source: https://assist.dla.mil -- Downloaded: 20260-20T01:52Z
Check the source to verify that this is the current version before use.

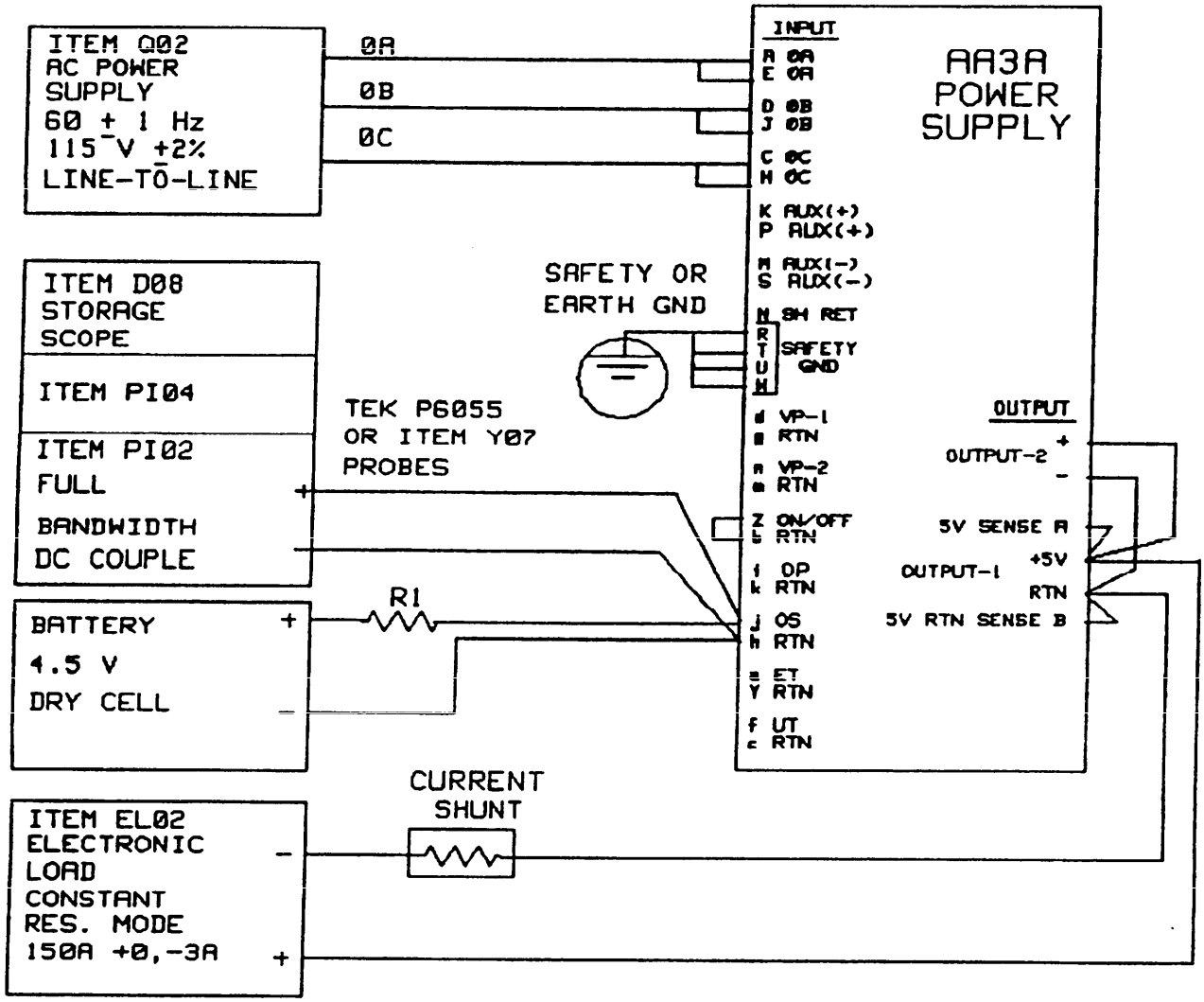


FIGURE 61. Output status noise test for table XXXII.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

NOTES:

1. See table II for output connector contact assignment.

2. For cooling requirements, refer to 3.5.6.

3. Current shunt: 150A/50mV.

4. Monitor shunt voltage with an item V04 voltmeter to ensure that the load current is within specification.

5. R1 = 5.9 kOhm, ± 1 percent, 1/8 watt resistor.

6. At the option of the testing facility, a DC input (item P10) of 155 VDC ± 2 percent connected between any two AC inputs may be used instead of the AC input specified.

7. The leads between the battery terminals and the output status contacts of the power supply under test shall be a tightly twisted pair, with a maximum length of 10 inches.

8. The millivolt peak (mVp) measurements for test conditions KK1 and KK2 shall be referenced to input contact h (OS RTN). The measurement shall include all spikes, hash, periodic and aperiodic components.

FIGURE 61. Output status noise test for table XXXII - Continued.

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - SH
Air Force - 99

Review activity:
DLA - GS

Preparing activity:
Navy - SH

Agent: NW

(Project 6130-0320)

Source: https://assist.dla.mil -- Downloaded: 202610-20T01:52Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-P-24764/1	2. DOCUMENT DATE *(YYMMDD)*

3. DOCUMENT TITLE
Power Supplies, Shipboard, Electronic, 5 Volts, Key Code AA3A

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)
	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*
Technical Point of Contact (TPOC): Mr. Randy Collins (Code 6023) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS	(1) Commercial: TPOC: 812-854-3826 (2) AUTOVON: 88-482-3826
c. ADDRESS *(Include Zip Code)* Commanding Officer Naval Weapons Support Center ATTN: Randy Collins, Bldg. 2917, Code 6023 Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-02-20T01:32Z
Check the source to verify that this is the current version before use.